EE361 Electromechanical Energy Conversion I — METU EEE — full-semester study notes

Chapter II — Magnetic Circuits

Manyetik Devreler · Weeks 4–6 of the 2026 syllabus

- Lesson 10 — Magnetic fields, flux, flux linkage and magnetic materials
- Lesson 11 — Magnetic materials: magnetization, B–H curves and saturation
- Lesson 12 — MMF, reluctance and magnetic equivalent circuits (DC excitation)
- Lesson 13 — Air gaps, fringing and leakage
- Lesson 14 — Flux linkage, self- and mutual inductance, coupled circuits and magnetic energy
- Lesson 15 — AC excitation: Faraday's and Lenz's laws, waveforms and saturation
- Lesson 16 — Hysteresis, eddy currents, core losses, lamination, skin effect and inductors

> Assistant-written teaching notes (English with Turkish terminology). Teacher sources LN3–LN6; textbook Sen Ch. 1. Additional archive problems for Chapters II–III are printed as an appendix to the Chapter III PDF. Every numerical result is checked by `assistant/calculations/full-semester/run_all.py`; source pages are cited in each lesson. Printed 2026-11-02 from the Markdown sources in `assistant/lessons/`.

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Lesson 10 — Magnetic fields, flux, flux linkage and magnetic materials

Chapter II: Magnetic Circuits (Manyetik Devreler) · Week 4 · Lesson 10 of 31

Assistant-written teaching note (2 Nov 2026). Teacher source: LN3 (all 26 slides) and LN4 slides 2–5. Book support: Sen, 3rd ed., §1.1.1–1.1.3, printed pp. 1–5 (PDF viewer pp. 25–29). Glossary: Part II §1 (Week 4). Exact page list at the end. Numbers are verified by `assistant/calculations/full-semester/lesson10_fields_materials.py`.

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1. Where this lesson sits and what you will be able to do

Chapter I treated electrical power as something that flows through wires and is described by voltages, currents and phasors. Every device that changes the *form* of electrical energy in this course — the inductor, the transformer, the DC machine — does it through a **magnetic field** (manyetik alan). Chapter II builds the tool that lets us calculate these fields with circuit-like arithmetic: the **magnetic circuit** (manyetik devre). This first lesson defines the physical quantities and the two field laws that the whole chapter rests on. Lesson 11 then studies how iron responds to a field, and Lesson 12 turns the results into equivalent circuits.

After this lesson you should be able to:

1. Define magnetic field intensity H, flux density B, flux Φ and flux linkage λ, with units, and say which one each law or formula uses.
2. State Ampère's circuital law, count the enclosed current with the correct signs, and use the right-hand rule to find field directions.
3. Use Gauss's law for magnetism to argue that flux is continuous ("what goes into a junction comes out").
4. Relate B and H through permeability $\mu = \mu_0\mu_r$, and classify materials as diamagnetic, paramagnetic or ferromagnetic.
5. Explain why air is a *circuit element* in magnetic circuits but practically an open circuit in electric circuits.
6. List the four modelling assumptions that turn a field problem into a lumped magnetic circuit, and recognise when each fails.

Prerequisites. Vector ideas (direction, dot product), the line integral as "sum of component along a path × length", and surface integrals as "sum of normal component × area". From Chapter I you only need the idea of a lumped circuit and Ohm's law.

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2. Physical picture: why we bother with magnetic fields

A current-carrying conductor is surrounded by a magnetic field. If we wind the conductor into a coil and put the coil on a piece of iron, the iron concentrates and guides that field. Three facts make this useful:

- **A changing magnetic field induces voltage** (Faraday's law, Lesson 15). That is how a transformer moves power from one winding to another with no electrical connection.
- **A magnetic field exerts force on currents and on iron** (Lorentz and reluctance forces, Lessons 22–24). That is how motors, relays and lifting magnets work.

- **A magnetic field stores energy** (Lesson 14). That is what an inductor does in a buck or boost converter (Lessons 28–29).

LN3 opens the topic by comparing electric and magnetic *materials*. In an electric circuit we choose a **conductor** (iletken) such as copper ($\rho = 1.68 \times 10^{-8}\ \Omega\,\mathrm{m}$) or aluminium ($2.82 \times 10^{-8}\ \Omega\,\mathrm{m}$) to carry current and an **insulator** (yalıtkan) such as plastic or air to block it. In a magnetic circuit the role of the "conductor" is played by a **ferromagnetic material** (ferromanyetik malzeme) such as iron, cobalt or electrical steel, because it carries flux easily. The role of the "insulator" would be a material that blocks flux — and the key lesson of LN3 slides 13–14 is that **no good magnetic insulator exists**. Copper is about 10^{22} times more conductive than air, so air really is an insulator for current. Iron, however, is only of the order of a few thousand times more permeable than air (LN3 uses "just 3000 times"). Air, copper and aluminium are all "non-magnetic" with $\mu_r \approx 1$, but flux still passes through them. Consequences that will follow us through the chapter:

- An **air gap** (hava aralığı) is a real circuit element with large but finite "resistance" to flux (Lesson 13).
- Some flux always escapes around the intended path — **leakage flux** (kaçak akı) — because the surrounding air is not a perfect insulator (Lessons 13, 18).

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3. The four quantities

3.1 Magnetic field intensity H (manyetik alan şiddeti)

H is a vector field measured in **amperes per metre** (A/m; in machine work often written At/m, "ampere-turns per metre"). It is the quantity tied directly to the currents that create the field: Ampère's law (Section 4) relates the circulation of H to current. A useful mental model is "magnetic push per metre of path".

3.2 Magnetic flux density B (manyetik akı yoğunluğu)

B is a vector field measured in **teslas**, $1\ \mathrm{T} = 1\ \mathrm{Wb/m^2}$. It describes how much magnetic flux crosses a unit area perpendicular to it and it is the quantity that produces forces and induced voltages. Typical values: Earth's field ≈ 50 µT; air gaps of machines 0.5–1 T; transformer cores 1.2–1.7 T; saturation of electrical steel about 1.8–2 T.

3.3 Magnetic flux Φ (manyetik akı)

The flux through a surface S is the surface integral of the normal component of B:

$$\Phi = \iint_S \mathbf{B} \cdot d\mathbf{A} \qquad [\mathrm{Wb}].$$

If B is uniform over a flat area A and perpendicular to it, $\Phi = BA$. In a core of cross-sectional area A_c with non-uniform B, we define the **average flux density** $B = \Phi / A_c$. This average is what all magnetic-circuit calculations use.

3.4 Flux linkage λ (akı kenetlenmesi)

A coil of N turns "links" the flux that passes through its turns. If every turn encloses the same flux Φ,

$$\lambda = N\Phi \qquad [\text{Wb-turn, or simply Wb}]$$

(LN4 slide 15: $\lambda = N_{turns}\Phi$, "turn·Weber"). If different turns enclose different flux — for example because some field lines leak out between turns — the correct definition is the sum over turns,

$$\lambda = \sum_{k=1}^{N} \Phi_k.$$

Flux linkage matters because Faraday's law says the coil voltage is $d\lambda/dt$ (Lesson 15) and inductance is flux linkage per ampere (Lesson 14). Keep the distinction sharp: Φ **is a property of a surface (or a core cross-section);** λ **is a property of a coil.**

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4. Ampère's circuital law and the right-hand rule

4.1 Statement

For any closed path C,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \sum I_{\text{enclosed}} = \iint_S \mathbf{J} \cdot d\mathbf{A},$$

where S is **any open surface bounded by** C and **J** is current density (A/m²). (LN3 slide 17 writes the right-hand side with a closed-surface symbol; it must be the open surface spanned by the contour — see errata E1.) This is the magnetostatic form; the displacement-current term of the full Maxwell equation (LN3 slide 16) is negligible at power frequencies inside machines.

Sen writes the same law as $\oint H\,dl\cos\theta = \sum i$ (Eq. 1.2), with θ the angle between **H** and the path element.

4.2 Signs: which currents count as positive

Choose a direction around C. Curl the fingers of your right hand in that direction; your thumb then points to the positive side of the surface S. A current crossing S in the thumb direction counts **positive**, one crossing the other way counts **negative**, and a current that does not cross S does not count at all. Sen's Fig. 1.2 example gives $\sum i = i_1 + i_2 - i_3$.

A coil of N turns, each carrying i, crosses the surface N times, so it contributes Ni — the **magnetomotive force** (manyetomotor kuvvet), defined formally in Lesson 12.

4.3 The right-hand rule for direction

Hold a straight conductor in your right hand with the thumb along the current; the fingers curl in the direction of **H** (and **B**). For a coil, curl the fingers along the current in the turns; the thumb points along the flux inside the coil. Figure 10.1 shows both views.

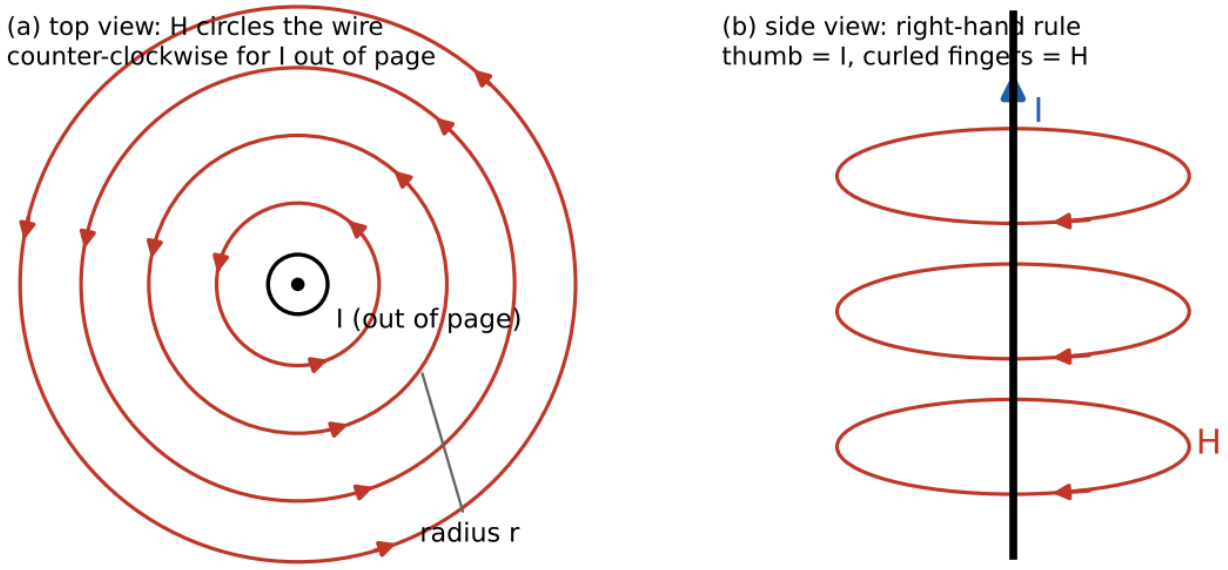


Figure 10.1. (a) Top view: for current out of the page, field lines are circles traversed counter-clockwise. (b) Side view of the right-hand rule. Redrawn after LN3 slide 17 and Sen Fig. 1.1.

4.4 Using the law: the long straight wire

When can Ampère's law give H directly, rather than only its circulation? When symmetry tells us the *shape* of the field. For an infinitely long straight wire carrying I:

1. **Symmetry:** rotating the problem about the wire or sliding along it changes nothing, so $|H|$ depends only on the distance r, and $\mathbf{H}$ is tangential (along circles) — a radial component would make the field lines start or end, which Gauss's law forbids (Section 5).
2. **Choose** C as a circle of radius r centred on the wire, traversed in the right-hand direction. Then $\mathbf{H} \parallel d\mathbf{l}$ everywhere and $|H|$ is constant on C.
3. **Evaluate:** $\oint H\, dl = H \cdot 2\pi r = I$, so

$$H(r) = \frac{I}{2\pi r} \qquad \text{(Sen Eq. 1.2a)}.$$

LN3 adds the remark "magnetic field intensity H is independent of the material properties". That statement is exact for the *circulation* $\oint \mathbf{H} \cdot d\mathbf{l}$, which depends only on the enclosed current. The *local* value of H is material-independent only when symmetry fixes the field shape regardless of the material, as for this wire with a uniform surrounding medium (or any arrangement of coaxial cylindrical layers). In a core with an air gap, the local H in iron and in air are very different — that is the subject of Lesson 13.

4.5 Using the law: the toroid (simirli/halka nüve)

A toroid is a ring-shaped core with a coil of N turns spread uniformly around it (Figure 10.2). By the same symmetry argument, $\mathbf{H}$ inside the core runs around the ring and depends only on the radius r. Take C as a circle of radius r inside the core: it is pierced by all N turns once, so

$$H(r) \cdot 2\pi r = Ni \quad \Longrightarrow \quad H(r) = \frac{Ni}{2\pi r} \qquad \text{(Sen Eq. 1.6)}.$$

Outside the toroid a circle encloses each turn twice in opposite directions (once going in, once coming back), so the net enclosed current is zero and $H = 0$ — an ideal toroid has no external field.

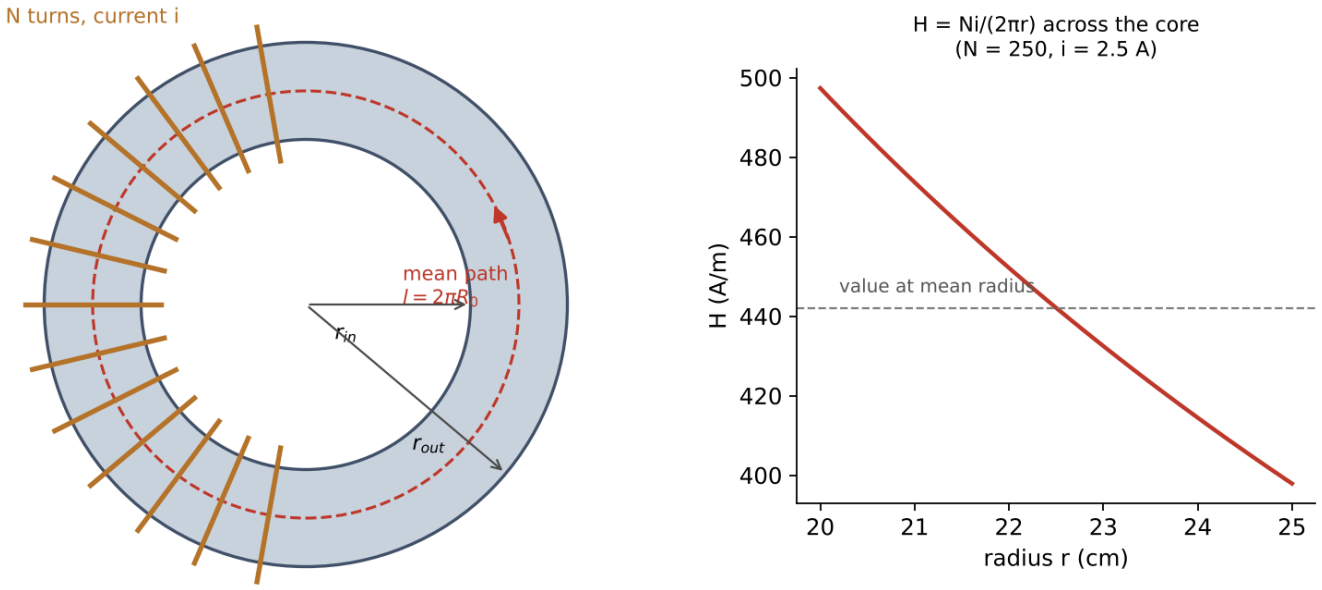


Figure 10.2. Left: toroid with distributed winding and the mean path of length $l = 2\pi R_0$. Right: $H(r)$ across the core for the numbers of Example 10.2. The field is strongest on the inner surface.

Two observations matter for every later calculation:

- H is *not* uniform across the core: it falls as $1/r$. For a thin core ($r_{out} - r_{in} \ll R_0$) the variation is small and we use the value at the **mean radius**, $H \approx Ni/l$ with $l = 2\pi R_0$ the **mean path length** (ortalama manyetik yol uzunluğu). Example 10.2 shows the error is about 0.3 % for a typical ring.
- The product $Hl = Ni$ is the prototype of every magnetic-circuit equation: “MMF = sum of $H \times$length around the loop”.

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5. Gauss’s law for magnetism: flux is continuous

The second field law is

$$\nabla \cdot \mathbf{B} = 0 \quad \Longleftrightarrow \quad \oiint_S \mathbf{B} \cdot d\mathbf{A} = 0 \ \text{ for every closed surface } S$$

(LN3 slides 18–19). Its practical meaning, as LN3 puts it: *there are no magnetic flux sources* and *no magnet has a single pole*. Every field line that enters a closed surface leaves it again.

For magnetic circuits this has an immediate consequence. Draw a closed surface around a **junction** where several limbs of a core meet (for example the top of the centre limb in Figure 12.3 of Lesson 12). With flux confined to the limbs, the surface integral reduces to the sum of the limb fluxes, so

$$\sum_k \Phi_k = 0 \qquad \text{(fluxes leaving the junction counted positive)}.$$

This is **Kirchhoff’s current law for flux**. Similarly, along an unbranched part of a core the same flux passes every cross-section, even if the area changes — only the flux *density* $B = \Phi/A$ changes.

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6. Permeability and the B–H relation

In vacuum, $\mathbf{B} = \mu_0 \mathbf{H}$ with the **permeability of free space** (boşluğun manyetik geçirgenliği)

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H/m} \qquad \text{(LN3 slide 22)}.$$

(The 2019 SI redefinition makes μ_0 a measured constant equal to this value within 10^{-9}; for EE361 use $4\pi \times 10^{-7}$ exactly.) In a material,

$$\mathbf{B} = \mu \mathbf{H} = \mu_0 \mu_r \mathbf{H},$$

where μ is the **permeability** (manyetik geçirgenlik) and $\mu_r = \mu/\mu_0$ the dimensionless **relative permeability** (bağıl manyetik geçirgenlik). LN3 slide 21 presents this as the magnetic twin of $\mathbf{J} = \sigma \mathbf{E}$: current density = conductivity × electric field, flux density = permeability × magnetic field.

Physically, μ_r measures how strongly the material's own atomic magnetic moments reinforce the applied field. LN3 slides 5–8 sort materials into three groups (Figure 10.3):

Class	μ_r	Behaviour	Examples (LN3)
Diamagnetic (diyamanyetik)	slightly < 1 (e.g. $1 - 10^{-5}$)	very slightly repelled from strong fields	copper, water, (famously) frogs
Paramagnetic (paramanyetik)	slightly > 1 (e.g. $1 + 2 \times 10^{-5}$)	very slightly attracted	aluminium
Ferromagnetic (ferromanyetik)	$\gg 1$, typically 10^3–10^5, **and not constant**	strongly attracted; can saturate and retain magnetization	iron, cobalt, nickel, electrical steels, ferrites

For engineering purposes the first two groups are "non-magnetic": take $\mu_r = 1$. Sen §1.1.2 gives μ_r of a few thousand (2000–6000) for machine steels; individual exam problems use values such as 1500 (ferrite, 2025 MT Q4), 4000 (LN3 exercise) and 25 000 (nanocrystalline toroid, 2021 MT Q2).

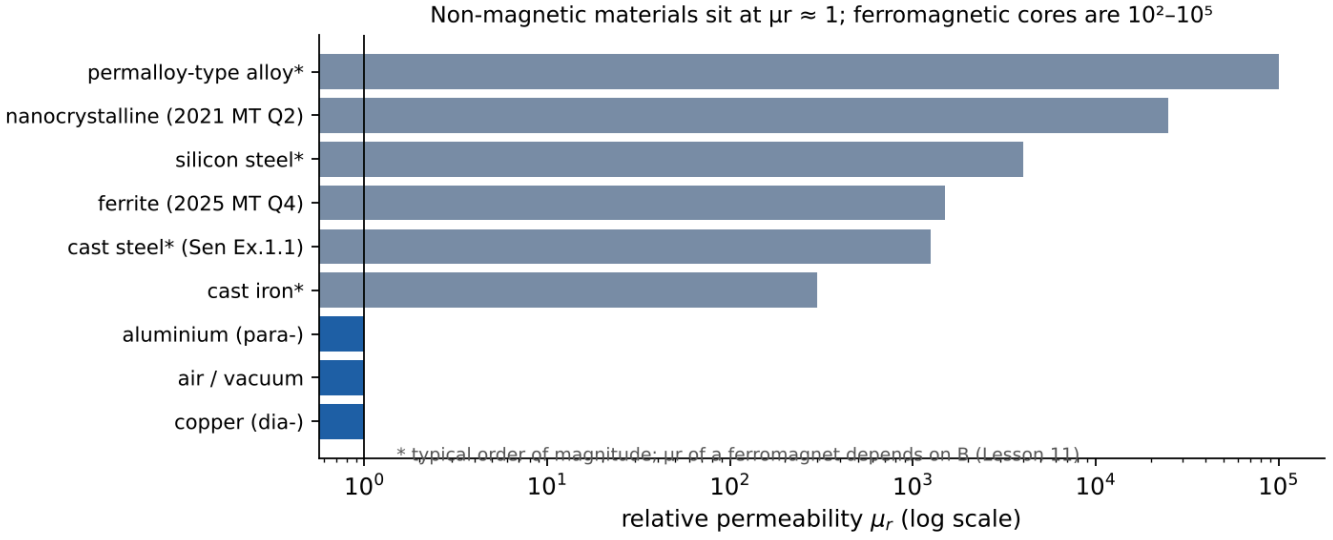


Figure 10.3. Orders of magnitude of μ_r (log scale). Values marked are typical; for ferromagnets μ_r depends strongly on the operating flux density, as Lesson 11 shows.*

Applications from LN3 slides 5 and 7 — scrap-lifting magnets, magnetic separators in iron-ore processing and car recycling — all exploit the enormous difference between ferromagnetic and non-magnetic materials: iron and steel are pulled into the field, aluminium and plastic are not.

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7. From fields to circuits: the analogy and its assumptions

7.1 The analogy table

LN3 slide 10 (repeated as LN4 slide 2) collects the correspondence that the rest of the chapter exploits:

Role	Hydraulic	Electric	Magnetic
Excitation	pump force (N)	EMF V (V)	MMF $\mathcal{F} = Ni$ (A or At)
"Push per metre"	pressure	electric field E (V/m)	magnetic field H (A/m)
Outcome (flow)	water flow (m³/s)	current I (A)	flux Φ (Wb)
Flow density	flow per area	current density J (A/m²)	flux density B (T = Wb/m²)
Flow limiter	pipe resistance	resistance $R = l/(\sigma A)$ (Ω)	reluctance $\mathcal{R} = l/(\mu A)$ (A/Wb)
Material property	—	conductivity σ (S/m); resistivity ρ (Ω·m)	permeability μ (H/m); reluctivity $1/\mu$ (m/H)

The analogy is mathematical, not physical: flux does not "flow" and nothing is dissipated by a constant flux in a reluctance. It works because both sets of quantities obey the same two laws: a loop law (KVL ↔ Ampère: $\oint H\,dl = Ni$) and a node law (KCL ↔ Gauss: $\sum \Phi = 0$), plus a linear constitutive relation ($J = \sigma E \leftrightarrow B = \mu H$).

7.2 The modelling assumptions (LN4 slide 5)

A real core is a three-dimensional field problem. We reduce it to a lumped circuit by assuming:

1. **No leakage:** all flux stays inside the magnetic material (and the intended gaps). Justified when μ_r is large and the windings are on the core; it is the *weakest* assumption because air is not a magnetic insulator.
2. **Uniform flux distribution** over each cross-section, so $B = \Phi/A$ and H is constant along each section of the mean path. Justified for limbs that are thin compared with their length; Example 10.2 quantifies the error for a toroid.
3. **No fringing:** flux crosses an air gap straight, so the gap area equals the core area. Justified for gaps much shorter than the pole width; Lesson 13 gives the usual correction.
4. **(Sometimes) infinite core permeability**, $\mu \to \infty$: the core needs no MMF and only gaps matter. Justified when the gap reluctance dominates (Lesson 13).

Each lesson will say which of these it uses. When a problem says "neglect leakage and fringing", it is invoking assumptions 1 and 3.

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8. Methods and common mistakes

Method — finding H with Ampère's law. (i) Identify the symmetry; (ii) choose a contour along which H is either constant and parallel to the path, or perpendicular to it; (iii) count the enclosed current with signs (right-hand rule); (iv) solve for H; (v) get B from $B = \mu H$ in each region; (vi) integrate B over the area for Φ; (vii) multiply by turns linking it for λ.

Common mistakes.

- *Using B in Ampère's law.* The circulation law is for H. Writing $\oint B\,dl = \mu_0 N i$ is only correct in a single uniform medium. In a core with an air gap you must use $\oint H\,dl$.
- *Forgetting the turns.* A coil of N turns crosses the contour N times: the enclosed current is Ni, not i.
- *Confusing Φ and λ.* Induced voltage and inductance involve $\lambda = N\Phi$. Mixing them gives errors by a factor of N.
- *Unit slips.* Convert mm and cm to metres before substituting; areas in mm² are 10^{-6} m² and in cm² are 10^{-4} m² (LN3 slide 23: "Don't forget to convert to SI units").
- *Assuming μ_r is a constant of iron.* It is constant only over a limited range (Lesson 11).

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9. Worked examples

Example 10.1 — Field of a long wire (conceptual + numbers)

Given. A long straight conductor carries $I = 100$ A. **Find** H and B at $r = 5$ cm (a) in air; (b) if the conductor is surrounded by a long, coaxial cylindrical iron sleeve with $\mu_r = 4000$ that contains the point.

Solution. The symmetry argument of Section 4.4 holds in both cases, because a coaxial cylindrical sleeve does not break the rotational symmetry. Hence

$$H = \frac{I}{2\pi r} = \frac{100}{2\pi(0.05)} = 318.3 \text{ A/m}$$

in both cases. Then (a) $B = \mu_0 H = 4\pi \times 10^{-7} \times 318.3 = 4.0 \times 10^{-4}$ T $= 0.4$ mT; (b) $B = 4000\,\mu_0 H = 1.6$ T.

Check. The code integrates H numerically around the circle and recovers 100 A.
Interpretation. Same H, 4000 times larger B: iron does not change the "push" set by the current, it multiplies the flux that the push produces. (1.6 T is near saturation for real steel, so in practice μ_r would be lower there — Lesson 11.)

Example 10.2 — Toroid: field across the core and the mean-radius approximation

Given. A toroid with $N = 250$ turns carries $i = 2.5$ A. The core has a circular cross-section, inner radius 20 cm and outer radius 25 cm (dimensions of Sen Example 1.5), and is assumed linear with $\mu_r = 2000$ (our assumption; Sen uses silicon steel's curve). **Find** (a) H and B at the inner, mean and outer radii; (b) the core flux exactly and with the mean-radius approximation; (c) the inductance $L = \lambda/i$.

Solution. (a) From $H = Ni/(2\pi r)$ with $Ni = 625$ At:

radius	H (A/m)	$B = \mu H$ (T)
$r_{in} = 0.20$ m	497.4	1.250
$R_0 = 0.225$ m	442.1	1.111
$r_{out} = 0.25$ m	397.9	1.000

b. *Exact:* integrate $B(r) = \mu N i/(2\pi r)$ over the circular section of radius $a = 2.5$ cm centred at R_0. Using $\iint dA/r = 2\pi\left(R_0 - \sqrt{R_0^2 - a^2}\right)$ for a circle,

$$\Phi = \mu N i\left(R_0 - \sqrt{R_0^2 - a^2}\right) = (2000 \cdot 4\pi \times 10^{-7})(625)(0.225 - 0.223607) = 2.1884\ \text{mWb}.$$

Mean-radius approximation: $\Phi \approx B(R_0)\,\pi a^2 = 1.1111 \times \pi (0.025)^2 = 2.1817$ mWb.

The approximation underestimates by 0.31 %. (c) $L = N\Phi/i = 0.2188$ H exactly, 0.2182 H approximately; the latter equals $N^2/\mathcal{R}$ with $\mathcal{R} = l/(\mu A)$ — the reluctance formula of Lesson 12.

Check. The code computes the flux both by a numerical double integral and by the closed form (agreement to 10^{-7}). **Interpretation.** The flux density is 25 % higher on the inside of the ring than on the outside; a real core would start saturating on the inner surface first. For a "fat" core this matters; for normal limbs the uniform-flux assumption is good to a fraction of a percent.

Example 10.3 — Flux and flux linkage, including imperfect linkage

Given. A uniform flux density $B = 1.2$ T crosses a core section of 20 cm² at right angles. A 200-turn coil is wound on that section. **Find** (a) Φ; (b) λ; (c) λ if 50 of the turns sit at the end of the coil where only 80 % of the core flux passes through them.

Solution. (a) $\Phi = BA = 1.2 \times 20 \times 10^{-4} = 2.4 \times 10^{-3}$ Wb. (b) $\lambda = N\Phi = 200 \times 2.4 \times 10^{-3} = 0.48$ Wb·turn. (c) $\lambda = \sum \Phi_k = 150(2.4 \times 10^{-3}) + 50(0.8)(2.4 \times 10^{-3}) = 0.456$ Wb·turn, equivalent to an "effective" 190 turns linking the full flux.

Interpretation. Part (c) is exactly what leakage does: some turns link less than the full core flux. Lessons 13 and 18 model the missing linkage with a separate leakage inductance.

Example 10.4 — Counting enclosed current with signs

Given. Three long parallel conductors pass through a circular contour of radius 0.1 m: $i_1 = 30$ A and $i_2 = 20$ A out of the page, $i_3 = 15$ A into the page (the configuration of Sen Fig. 1.2). The contour is traversed counter-clockwise when viewed from the side where currents come out.

Find $\oint \mathbf{H} \cdot d\mathbf{l}$, and say when you could also state H on the contour.

Solution. Counter-clockwise traversal puts the right-hand thumb out of the page, so out-of-page currents are positive: $\oint \mathbf{H} \cdot d\mathbf{l} = i_1 + i_2 - i_3 = 35$ A. The *local* H cannot be obtained from this alone, because three off-centre wires do not produce a field of constant magnitude along the circle. Only if the net current were a single wire on the axis would $H = 35/(2\pi \cdot 0.1) = 55.7$ A/m everywhere on the contour.

Interpretation. Ampère's law always gives the circulation; it gives the field only with symmetry. Magnetic circuits supply that "symmetry" artificially by assuming flux and H follow the core uniformly (Section 7.2).

Example 10.5 — Direction of flux in a core (right-hand rule)

Given. A rectangular ring core like LN3 slide 23, with the winding on the left limb. Assume that, viewed from the front, current in the front segments of the turns flows from left to right (and therefore right to left in the segments behind the limb).

Find the direction of the flux in the left limb and in the top limb.

Solution. Curl the right-hand fingers along the current: across the front from left to right, then back behind the limb. The thumb points **upward** along the left limb. Flux therefore goes up the left limb, right along the top limb, down the right limb and back left along the bottom — clockwise as seen from the front. This is the circulation sense drawn by the "magnetic flux lines" arrow on LN3's figure (top limb, pointing right); the slide's sketch does not make the winding sense unambiguous, which is exactly why exam figures add dots or arrows.

Interpretation. Reversing either the current or the winding sense reverses the flux. In multi-winding problems (Lesson 12) you must fix each winding's flux direction this way *before* writing the equivalent circuit; this is what dot marks encode (Lesson 14).

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10. Optional practice (original problems unless marked)

P10.1 (coaxial cable). A coaxial cable has a solid inner conductor of radius 2 mm carrying 50 A uniformly distributed, and a return outer conductor occupying 6–7 mm, also uniformly distributed. Find H at r = 1 mm, 3 mm, 6.5 mm and 10 mm.

▼ **Solution**

Symmetry gives circular field lines; take circles of radius r.

- $r < a$: enclosed current Ir^2/a^2, so $H = Ir/(2\pi a^2) = 50(0.001)/(2\pi \cdot 4 \times 10^{-6}) = 1989$ A/m.
- $a < r < b$: enclosed I, $H = I/(2\pi r) = 2653$ A/m at 3 mm.
- $b < r < c$: enclosed $I\,(c^2 - r^2)/(c^2 - b^2)$, so at 6.5 mm $H = \dfrac{50}{2\pi(0.0065)} \cdot \dfrac{49 - 42.25}{49 - 36} = 636$ A/m.
- $r > c$: enclosed current zero, $H = 0$ — a coax has no external field, like the toroid. Check: at $r = a$ both formulas give 3979 A/m (continuity). *Verified in* `lesson10_fields_materials.py`.

P10.2 (concept). Using the numbers on LN3 slides 13–14, explain why an air gap in a transformer core changes the magnetizing current dramatically, while an air gap of the same proportions in a copper bus bar would simply break the circuit.

▼ **Solution**

In the bus bar, copper is ~10^{22} times more conductive than air, so a gap is effectively an infinite resistance: current stops. In the core, iron is only ~10^3 times more permeable than air, so a gap is a *finite* reluctance in series with the iron. A gap of length g has the same reluctance as $\mu_r g$ of iron; with $\mu_r = 3000$, a 1 mm gap "looks like" 3 m of iron. Flux still flows, but the MMF (current) needed for the same flux rises enormously. Quantified in Lesson 13.

P10.3 (long solenoid; original variant of Sen Problem 1.1). An air-cored solenoid is 50 cm long and 5 cm in diameter with 250 turns and carries 100 A. Treating the inside field as uniform and neglecting the field outside, find H, B, Φ and L.

▼ **Solution**

Take a rectangular contour with one long side inside the coil (length l) and the other far outside where $H \approx 0$; the short sides are perpendicular to H. Ampère: $Hl = Ni$, so $H = 250 \times 100/0.5 = 5.0 \times 10^4$ A/m. $B = \mu_0 H = 0.0628$ T. $\Phi = B\pi d^2/4 = 1.234 \times 10^{-4}$ Wb. $L = N\Phi/i = 0.308$ mH, which equals $\mu_0 N^2 A/l$. *Verified in code.*

P10.4 (signs). A contour encloses a 4-turn coil carrying 2 A in one sense and a 3-turn coil carrying 5 A in the opposite sense. What is the net enclosed current, and what does a negative answer mean?

▼ **Solution**

$\sum I = 4(2) - 3(5) = -7$ A (with the first coil's sense taken as positive). The negative sign means the net circulation of H is opposite to the chosen traversal direction: the second coil "wins" and sets the flux direction. Two windings on one core with opposing MMFs behave exactly like this (Lesson 12).

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11. Summary of key relations

Relation	Meaning	Validity
$\oint_C \mathbf{H} \cdot d\mathbf{l} = \sum I_{enc}$	Ampère's circuital law	Magnetostatics/low frequency; open surface bounded by C; signs by right-hand rule
$\oiint_S \mathbf{B} \cdot d\mathbf{A} = 0$	Gauss's law for magnetism; flux continuity	Always
$\mathbf{B} = \mu_0 \mu_r \mathbf{H}$, $\mu_0 = 4\pi \times 10^{-7}$ H/m	Constitutive relation	μ_r constant only for linear materials
$\Phi = \iint \mathbf{B} \cdot d\mathbf{A} = BA$	Flux	BA form for uniform normal B
$\lambda = N\Phi$ (or $\sum \Phi_k$)	Flux linkage	$N\Phi$ when every turn links the same flux
$H = I/(2\pi r)$	Long straight wire	Uniform medium, infinite wire
$H = Ni/(2\pi r)$; $H \approx Ni/l$	Toroid	Distributed winding; mean-radius form for thin cores

12. Vocabulary (EN – TR)

English	Türkçe	Symbol / unit
electromagnetic system	elektromanyetik sistem	—
magnetic field	manyetik alan	—
magnetic field intensity	manyetik alan şiddeti	H, A/m
magnetic flux density	manyetik akı yoğunluğu	B, T
magnetic flux	manyetik akı	Φ, Wb
flux linkage	akı kenetlenmesi	λ, Wb·turn

English	Türkçe	Symbol / unit
magnetic field lines	manyetik alan çizgileri	—
permeability / relative permeability	manyetik geçirgenlik / bağıl manyetik geçirgenlik	μ, H/m / μ_r
permeability of free space	boşluğun manyetik geçirgenliği	μ_0
ferromagnetic, paramagnetic, diamagnetic	ferromanyetik, paramanyetik, diyamanyetik	—
magnetic core	manyetik nüve	—
magnetomotive force	manyetomotor kuvvet	$\mathcal{F}$, A·turn
leakage flux	kaçak akı	—
right-hand rule	sağ el kuralı	—

13. Sources

- LN3 = `sources/lectures/361 LN3_rotated.pdf`: viewer pp. 5–60 (slides 2–22) for materials, analogy, Ampère and Gauss laws, permeability; pp. 61–64 (slides 23–25) exercises used in Lesson 12. Review notes: LN03.
- LN4 = `sources/lectures/361 LN4_rotated.pdf`: viewer pp. 2–10 (slides 2–5): review table, Ampère, MMF, course assumptions; p. 38 (slide 15) flux linkage. Notes: LN04.
- Sen 3rd ed.: §1.1.1 i–H relation, printed pp. 1–2 (viewer 25–26), Eqs. 1.1–1.2a; §1.1.2 B–H relation, p. 3 (viewer 27), Eqs. 1.3–1.4; §1.1.3 toroid, pp. 3–4 (viewer 27–28), Eqs. 1.5–1.11; Example 1.5 dimensions, p. 14 (viewer 38).
- Glossary: `sources/glossaries/EE361_Magnetic_Circuits_Glossary_Weeks_4-6.pdf`, p. 1–2 (Week 4 terms).
- Verification: `assistant/calculations/full-semester/lesson10_fields_materials.py` → `records/lesson10_fields_materials.json`.

Lesson 11 — Magnetic materials: magnetization, B–H curves and saturation

Chapter II: Magnetic Circuits · Week 4 · Lesson 11 of 31

Assistant-written teaching note. Teacher source: LN4 slides 8–10; LN5 slides 2–4. Book support: Sen 3rd ed. §1.1.2 and §1.1.4 (printed pp. 3–5, viewer 27–29), Fig. 1.7 (p. 6, viewer 30), Examples 1.1 and 1.5 (pp. 8–9, 14–15), §1.2 opening (p. 15–17), §1.4 permanent magnets (pp. 26–30, optional extension). Numbers: `lesson11_bh_saturation.py`*.*

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1. Place in the course and goals

Lesson 10 treated permeability as a number. For air that is exactly right. For iron it is a convenient fiction that holds only over a limited range. This lesson explains *why* iron is so permeable, why its permeability falls as you push it harder (**saturation**, manyetik doyma), and what that means for any calculation that uses $B = \mu H$ or $L = N^2/\mathcal{R}$. The ideas reappear in the magnetizing current of transformers (Lesson 15), in the magnetization curve of DC machines (Lesson 26), and in the decision to put an air gap in an inductor (Lesson 13).

You should be able to:

1. Explain magnetization in terms of magnetic domains and why it produces $\mu_r \gg 1$.
2. Read a **B–H characteristic** (magnetization curve, mıknatıslanma eğrisi), find H for a given B and vice versa, and compute secant and incremental permeability.
3. State precisely what “linear” and “nonlinear” mean for a magnetic material, and quantify the error of a constant-μ model.
4. Distinguish soft and hard magnetic materials by their loops, and know where each is used.
5. Explain why a saturating core makes inductance depend on current, and distinguish secant from incremental inductance.

Prerequisites: Lesson 10 (B, H, μ, Ampère’s law for a toroid).

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2. Why iron is so permeable: domains and magnetization

Every atom carries a tiny magnetic moment from the spin and orbital motion of its electrons. In most materials these moments are random or cancel, and they respond only feebly to an applied field — the diamagnetic and paramagnetic cases with $\mu_r \approx 1$ (Lesson 10).

In **ferromagnetic** materials (iron, nickel, cobalt and their alloys) a quantum-mechanical exchange interaction makes neighbouring moments line up spontaneously. A crystal therefore breaks up into **magnetic domains** (manyetik bölgeler) — regions perhaps tens of micrometres across, each fully magnetized in one direction. Without an external field, domains point in different directions and the material shows no net magnetization (Figure 11.1a).

When we apply a field H (for example by passing current through a coil on the core):

1. **Low H — wall motion.** Domains already aligned roughly with H grow at the expense of the others by moving the boundaries (domain walls) between them (Figure 11.1b). A small H produces a large net magnetization: this is the steep part of the curve, where μ_r is in the thousands.

2. **Higher H — rotation.** Once wall motion is exhausted, the remaining moments must be rotated toward H, which takes much more field per unit of extra magnetization.
3. **Very high H — saturation.** All moments are aligned (Figure 11.1c). The material can contribute no more; any further increase of B comes only from the vacuum term $\mu_0 H$, so the slope of the curve tends to μ_0.

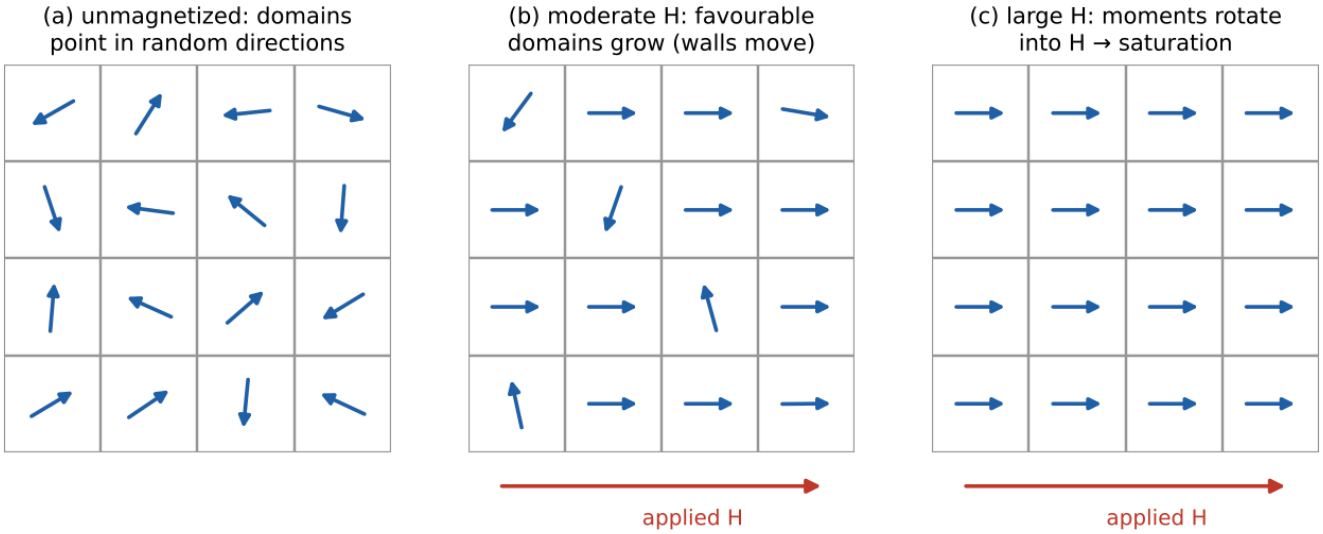


Figure 11.1. Schematic of magnetic domains: (a) random, no net magnetization; (b) favourable domains grow; (c) saturation. Assistant-drawn schematic.

In formula terms, the flux density inside a material is the vacuum part plus the material's **magnetization** $\mathbf{M}$ (A/m, the magnetic moment per unit volume, mıknatıslanma):

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}).$$

Writing $\mathbf{M} = \chi_m \mathbf{H}$ defines the susceptibility χ_m and gives $\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}$, i.e. $\mu_r = 1 + \chi_m$. For a ferromagnet χ_m is huge at low fields and collapses towards zero at saturation — which is exactly why μ_r is not a constant. You will not be asked to compute M in EE361; it is introduced so the shape of the B–H curve makes sense.

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3. The B–H characteristic

3.1 How it is measured and what it shows

Wind N turns on a toroid of mean length l and area A, increase the current slowly from zero on an unmagnetized core, and measure the flux (from the induced voltage, Lesson 15). Then $H = Ni/l$ (Lesson 10) and $B = \Phi/A$. Plotting B against H gives the **magnetization curve** (LN4 slide 10, LN5 slide 3, Sen Fig. 1.6).

Its regions:

- **Initial region** near the origin: in some steels (cast steel in Figure 11.2) the curve starts with a moderate slope before becoming steep — the initial permeability is lower than the maximum.
- **Approximately linear region** (LN5: "approximately linear"): steep and roughly straight. Most devices are designed to work near the top of this region.
- **Knee** (dirsek): the slope drops quickly.

- **Saturation region:** nearly flat; large increases in H give little extra B (LN4: “B limit ⇒ saturation”).

Air and vacuum, by contrast, have a straight-line characteristic $B = \mu_0 H$ through the origin (LN4 slide 9: the plot reaches only 0.0025 T at $H = 2000$ A/m). Linear materials have **constant slope**; LN4 states “slope gives permeability $\mu = B/H$; constant slope = linear material”.

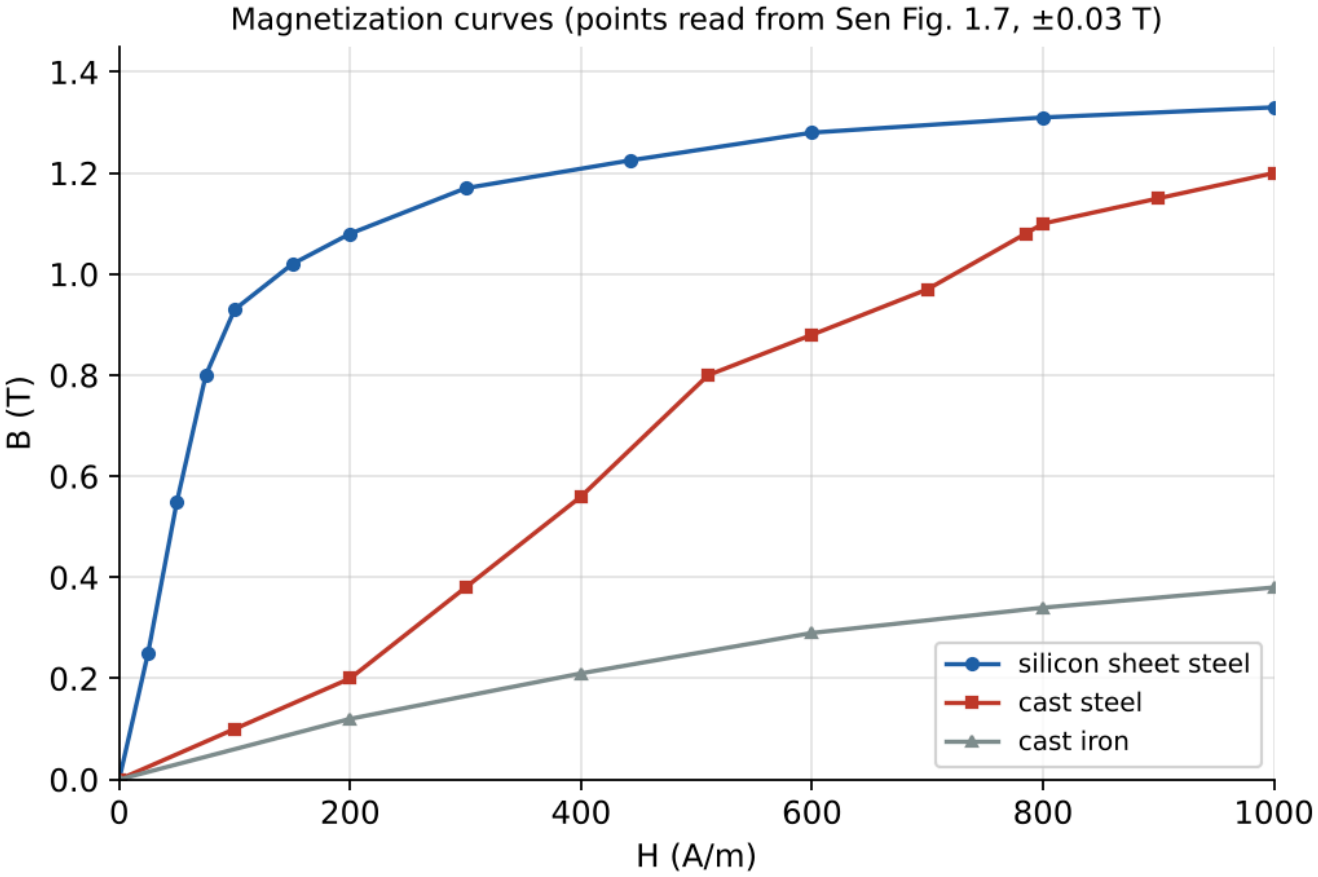


Figure 11.2. Magnetization curves of silicon sheet steel, cast steel and cast iron. Data points read by eye from Sen Fig. 1.7 (estimated uncertainty ±0.03 T), with the points Sen states numerically (cast steel 0.8 T at 510 A/m, 1.08 T at 785 A/m, 1.1 T at 800 A/m; silicon steel 1.225 T at 442.3 A/m) inserted exactly. The data table is in `assistant/calculations/full-semester/bh_data.py`.

Two practical readings of Figure 11.2:

- To reach 0.35 T, silicon steel needs about 33 A/m, cast steel about 283 A/m, cast iron about 850 A/m (Practice P11.2). Better material means less MMF, hence less magnetizing current and smaller copper loss.
- Sen notes (§1.1.4) that the reluctance of a ferromagnetic path “is low when B is low, and high when B is high”. Unlike a resistor, a reluctance depends on the flux it carries.

3.2 Two kinds of permeability

Because the curve is not a straight line through the origin, “the permeability” is ambiguous. Two definitions are used (Figure 11.3):

- **Secant (amplitude, “apparent”) permeability**

$$\mu = \frac{B}{H} \quad \text{(slope of the line from the origin to the operating point).}$$

This is what Sen uses in Example 1.1(b) and Example 1.5, and what you need when you want *total* flux from *total* MMF, e.g. $\mathcal{R} = l/(\mu A)$ at a given operating point.

- **Incremental (differential) permeability**

$$\mu_{inc} = \frac{dB}{dH} \quad \text{(slope of the tangent at the operating point).}$$

This governs *small changes* about an operating point, e.g. the response to a small AC ripple superimposed on a DC current (the inductor of a buck converter, Lesson 28).

Above the knee $\mu_{inc} < \mu$: the tangent is flatter than the secant.

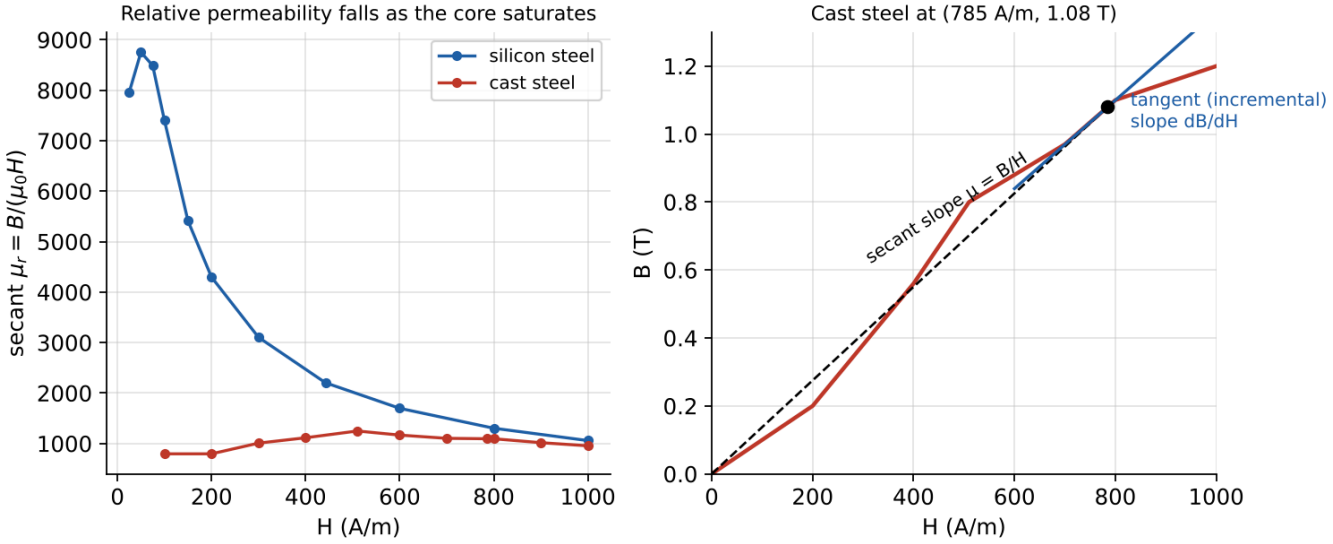


Figure 11.3. Left: secant relative permeability $B/(\mu_0 H)$ versus H for the two steels — it falls by a factor of several as the core saturates. Right: secant and tangent lines on the cast-steel curve at (785 A/m, 1.08 T).

3.3 Linear versus nonlinear behaviour

A material is **linear** (doğrusal) over a range if $B = \mu H$ with a single constant μ throughout that range — then reluctances are constants, fluxes add (superposition holds), inductance does not depend on current, and doubling the current doubles the flux (LN4 slide 7). A material is **nonlinear** (doğrusal olmayan) when μ depends on H; none of those statements then hold. Real iron is linear only approximately and only below the knee. The usual engineering compromises are:

1. **Constant-μ model** fitted near the intended operating point (all of Lessons 12–14 when μ_r is "given").
2. **Piecewise-linear model:** two or more straight segments (Sen Problems 1.15–1.16).
3. **Graphical/numerical solution with the actual curve:** load line or iteration (Lesson 12, Sen Example 1.2).
4. **Ignore the iron altogether ($\mu \to \infty$)** when an air gap dominates (Lesson 13).

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4. Soft and hard magnetic materials

If we reduce H back to zero after magnetizing, B does not retrace the curve: the material keeps some flux density, the **remanence** (kalıcı mıknatıslanma, B_r). A reverse field of magnitude H_c, the **coercivity** (koersivite), is needed to bring B to zero. LN5 slide 4 illustrates remanence with a cassette tape and a magnetic-stripe transit card — both store information as remanent magnetization. The full closed B–H cycle is the **hysteresis loop**; Lesson 16 studies it and the energy it costs.

The width of that loop sorts materials into two families (glossary Part II §3; Sen §1.4):

	Soft magnetic material (yumuşak manyetik malzeme)	**Hard magnetic material (sert manyetik malzeme)**
Loop	narrow; low H_c (typically < 100 A/m for electrical steels)	wide; high H_c (hundreds of kA/m for ferrite and NdFeB magnets)
Remanence	may be large, but is easily removed	large and hard to remove
Purpose	carry changing flux with little loss and little MMF	provide a permanent field without excitation
Examples	silicon (electrical) steel, ferrites, nanocrystalline and amorphous alloys	alnico, ferrite magnets, samarium–cobalt, neodymium–iron–boron
Used in	transformer and machine cores, inductors	permanent-magnet motors, loudspeakers, holding magnets

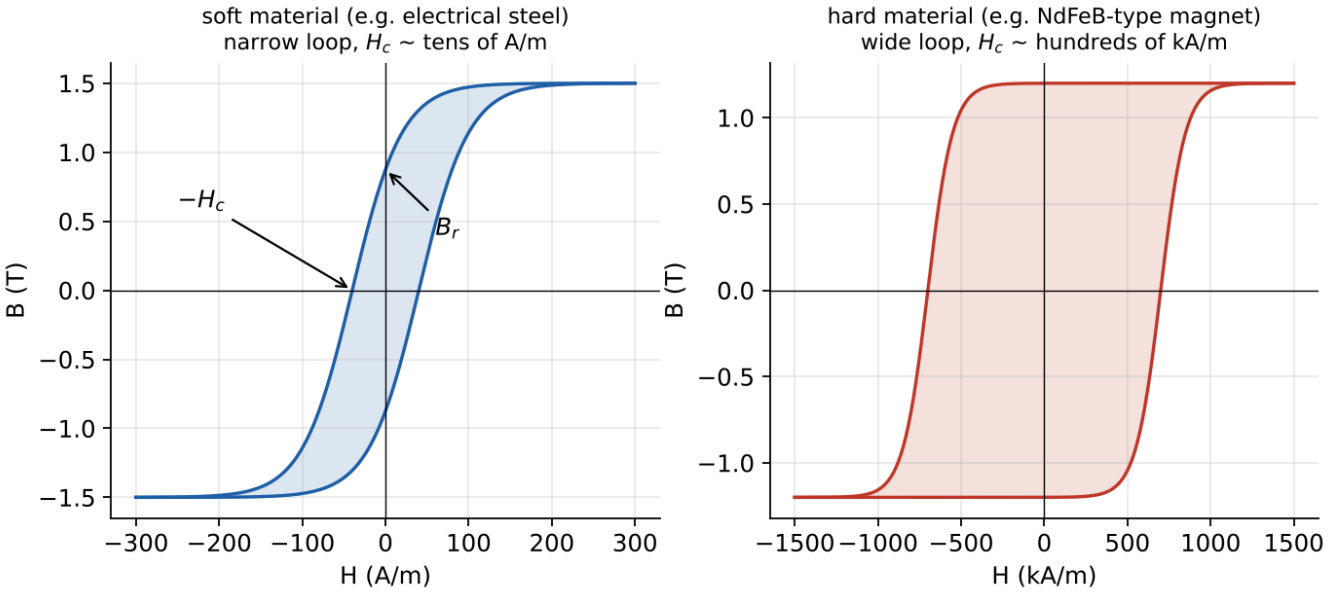


Figure 11.4. Illustrative hysteresis loops (assistant-generated shapes, not data of a specific grade): a soft material has a narrow loop; a hard material a wide loop with large H_c. Note the very different H scales.

Everything in Chapters II and III assumes soft materials. Permanent magnets enter EE361 only through the constant-flux DC machine (LN15 slide 28, “permanent magnet DC machine”) — Example 11.6 shows the basic magnet design calculation from Sen as an optional extension.

5. Consequences for inductance: secant and incremental inductance

For a coil of N turns on a core of length l and area A with no gap, $\lambda = N\Phi = NBA$ and $i = Hl/N$. So the λ–i curve of the coil is the B–H curve with both axes rescaled. Two inductances follow:

$$L_{sec} = \frac{\lambda}{i} = \frac{N^2 A}{l}\,\frac{B}{H}, \qquad L_{inc} = \frac{d\lambda}{di} = \frac{N^2 A}{l}\,\frac{dB}{dH}.$$

LN6 slide 4 defines inductance incrementally as $L = N\,d\Phi/dI = d\lambda/dI$, and slide 5 notes that in a *linear* (air-cored) system $L = N\Phi/I = d\lambda/dI$ — the two coincide only when the λ–i curve is a straight line through the origin. With a saturating core both depend on the current, and the one that matters depends on the question:

- The *voltage* across the coil for a changing current is $v = d\lambda/dt = L_{inc}\,di/dt$.
- The *stored energy* is the area $\int i\,d\lambda$ (Lesson 14), not $\tfrac{1}{2}L_{sec}i^2$ unless the coil is linear.

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6. Methods and common mistakes

Method — magnetic quantity from a B–H curve.

1. Compute what is fixed by the problem: if flux is given, $B = \Phi/A$ in each section; if current is given, you need the nonlinear methods of Lesson 12.
2. Read H from the curve for each section's B (interpolate linearly between tabulated points and say so).
3. MMF of that section $= Hl$; total MMF = sum (Ampère).
4. If asked for permeability or reluctance, use the **secant** value at that operating point.

Common mistakes.

- *Using a single "μr of iron" everywhere.* Example 11.5 shows a constant-μ model fitted at 1.08 T is wrong by −42 % at 0.93 T and +79 % at the higher field.
- *Expecting superposition with saturation.* Two MMFs acting on a saturated core do not give the sum of their separate fluxes.
- *Reading the curve with wrong units.* Sen's axis is At/m; some data sheets use Oe (1 Oe = 79.6 A/m) or gauss (1 G = 10^{-4} T).
- *Confusing B_r with saturation.* B_r is the flux density at $H = 0$ after magnetization; saturation is the high-H limit.

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7. Worked examples

Example 11.1 — Permeability at given operating points (Sen Examples 1.1 and 1.5)

Given. Cast steel at $B = 0.8$ T needs $H = 510$ A/m (Sen Ex. 1.1); at 1.08 T it needs 785 A/m (Sen Ex. 1.2). Silicon steel at $H = 442.3$ A/m gives 1.225 T (Sen Ex. 1.5). **Find** the secant μ and μ_r at each point.

Solution.

point	$\mu = B/H$ (H/m)	$\mu_r = \mu/\mu_0$
cast steel, 0.8 T	$0.8/510 = 1.569 \times 10^{-3}$	1248 (Sen rounds to 1250)
cast steel, 1.08 T	$1.08/785 = 1.376 \times 10^{-3}$	1095

point	$\mu = B/H$ (H/m)	$\mu_r = \mu/\mu_0$
silicon steel, 1.225 T	$1.225/442.3 = 2.770 \times 10^{-3}$	2204

Interpretation. The same cast steel has μ_r 12 % lower at 1.08 T than at 0.8 T — the knee is beginning. “The” relative permeability of a steel is meaningful only together with an operating point.

Example 11.2 — Doubling the current does not double the flux in saturation

Given. A silicon-steel toroid: mean length $l = 0.5$ m, area $A = 10$ cm², $N = 200$ turns. **Find** H, B and $L_{sec} = N\Phi/I$ for I = 0.125, 0.25, 0.5, 1.0 and 2.0 A, using the Figure 11.2 data.

Solution. $H = NI/l = 400I$ A/m; read B; $L_{sec} = NBA/I$.

I (A)	H (A/m)	B (T)	L_{sec} (H)
0.125	50	0.55	0.880
0.25	100	0.93	0.744
0.5	200	1.08	0.432
1.0	400	1.21	0.242
2.0	800	1.31	0.131

From 0.125 A to 0.25 A, B rises by a factor 1.7 (still roughly proportional); from 0.5 A to 1 A, only by 12 %. The “inductance” drops by a factor of almost 7 over the range.

Interpretation. LN4 slide 7 (“current doubled → MMF doubled → flux doubled”) is the *linear* statement. MMF always doubles; flux doubles only below the knee.

Example 11.3 — Secant versus incremental inductance

Given. The toroid of Example 11.2 operating at $I = 0.5$ A ($H = 200$ A/m, $B = 1.08$ T). **Find** L_{sec} and L_{inc}, estimating dB/dH by a central difference over ±50 A/m.

Solution. $L_{sec} = \dfrac{N^2 A}{l}\dfrac{B}{H} = \dfrac{200^2(10^{-3})}{0.5}\cdot\dfrac{1.08}{200} = 0.432$ H.

$dB/dH \approx [B(250) - B(150)]/100 = (1.125 - 1.02)/100 = 1.05 \times 10^{-3}$ H/m, so $L_{inc} = \dfrac{N^2 A}{l}\dfrac{dB}{dH} = 80 \times 1.05 \times 10^{-3} = 0.084$ H.

Interpretation. A small 50 Hz ripple on top of 0.5 A DC would “see” only 84 mH, five times less than λ/I. This is why converter inductors are specified at their DC bias current, and why they are often gapped (Lesson 13) to keep L nearly constant.

Example 11.4 — Sen Example 1.5: toroid on silicon sheet steel

Given. $N = 250$, $i = 2.5$ A, inner and outer radii 20 cm and 25 cm, circular cross-section, silicon sheet steel. **Find** (a) B at the mean radius; (b) L, assuming uniform B equal to the mean-radius value.

Solution. (a) Mean radius $R_0 = 22.5$ cm, $H = Ni/(2\pi R_0) = 625/(1.4137) = 442.1$ A/m (Sen: 442.3). Reading the silicon-steel curve, $B = 1.225$ T. (b) $A = \pi(0.025)^2 = 1.963 \times 10^{-3}$ m²; $\Phi = BA = 2.405 \times 10^{-3}$ Wb; $\lambda = N\Phi = 0.601$ Wb·turn; $L = \lambda/i = 0.2405$ H ≈ 240.4 mH (Sen).

Same answer by reluctance: $\mu = B/H = 2.77 \times 10^{-3}$ H/m, $\mathcal{R} = 2\pi R_0/(\mu A) = 2.60 \times 10^5$ A/Wb, $L = N^2/\mathcal{R} = 0.2405$ H.

Interpretation. Sen's "inductance" here is the *secant* inductance at 2.5 A. At another current it would differ (Example 11.2).

Example 11.5 — How wrong is a constant-μ model?

Given. Model silicon steel with a constant μ_r fitted at $H = 200$ A/m, $B = 1.08$ T. **Find** the model's B at 100 and 400 A/m and compare with the curve.

Solution. $\mu_r = 1.08/(\mu_0 \cdot 200) = 4297$. Then

H (A/m)	model $B = \mu_0\mu_r H$	curve B	model error
100	0.54 T	0.93 T	−42 %
200	1.08 T	1.08 T	0 (fit point)
400	2.16 T	1.21 T	+79 %

Interpretation. A linear model is a local approximation. Above the knee it can predict physically impossible flux densities (2.16 T in silicon steel). When an exam gives μ_r, it is telling you to assume operation in the region where that value holds.

Example 11.6 — (Optional extension) Sen Example 1.8: sizing a permanent magnet

Given. An alnico-5 magnet with a soft-iron keeper removed must give $B_g = 0.8$ T in an air gap with $A_g = 2.5$ cm², $l_g = 0.4$ cm. Choose the maximum-energy-product operating point $B_m = 0.95$ T, $H_m = -42$ kA/m. Neglect leakage, fringing and the soft-iron MMF. **Find** l_m and A_m.

Solution. With no coil, Ampère's law around magnet + gap gives $H_m l_m + H_g l_g = 0$ (Sen Eq. 1.43); with $H_g = B_g/\mu_0$:

$$l_m = \frac{B_g l_g}{\mu_0 |H_m|} = \frac{0.8(0.004)}{4\pi \times 10^{-7}(42\,000)} = 0.0606 \text{ m} = 6.06 \text{ cm}.$$

Flux continuity $B_m A_m = B_g A_g$ (Eq. 1.45): $A_m = 0.8(2.5)/0.95 = 2.105$ cm². Magnet volume $= 12.76$ cm³, which matches Sen Eq. 1.48, $V_m = B_g^2 V_g/(\mu_0 B_m H_m)$.

Interpretation. Inside the magnet H is *negative* (opposite to B): the magnet works in the second quadrant of its loop, and its large coercivity is what lets it push flux across the gap. A soft material would simply demagnetize.

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8. Optional practice

P11.1. Using Figure 11.2, find the current needed in a 400-turn coil on a cast-steel ring of mean length 40 cm to produce 1.0 T. Then find the secant μ_r.

▼ **Solution**

From the cast-steel table, $B = 1.0$ T lies between (700 A/m, 0.97 T) and (785 A/m, 1.08 T); linear interpolation gives $H \approx 700 + 85(0.03/0.11) = 723$ A/m. $NI = Hl = 723(0.4) = 289$ At, so $I = 0.72$ A. $\mu_r = 1.0/(\mu_0 \cdot 723) = 1100$. (Reading uncertainty ±0.03 T on the curve gives roughly ±25 A/m in H.)

P11.2. Which material needs the least H for 0.35 T: cast iron, cast steel or silicon steel? Give approximate values.

▼ **Solution**
Cast iron ≈ 850 A/m, cast steel ≈ 283 A/m, silicon steel ≈ 33 A/m (Figure 11.2 data). Silicon steel needs about 25 times less MMF than cast iron for the same flux density. *Values recorded in* `lesson11_bh_saturation.py`.

P11.3 (concept). A coil on an ungapped ferrite core is tested with a small AC signal at zero DC current and then at a large DC bias. Its measured inductance drops sharply in the second test. Explain using Section 5.

▼ **Solution**
A small-signal test measures the incremental inductance $L_{inc} \propto dB/dH$ at the bias point. At zero bias the operating point is on the steep part of the curve (large slope). With a large DC bias the operating point moves above the knee where dB/dH is small, so L_{inc} falls even though the coil and core are unchanged.

P11.4 (concept). Why are transformer cores made of soft material while the rotor of a PMDC motor carries hard material?

▼ **Solution**
A transformer core carries flux that reverses every half-cycle; a narrow loop (low H_c) means small energy lost per cycle (Lesson 16) and little MMF needed. A PMDC motor needs a constant field without excitation; a wide loop (high H_c, high B_r) keeps the magnet magnetized against the demagnetizing field of the air gap and the armature current.

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9. Summary

Idea	Statement	Limits
Magnetization	$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$; domains align with H	Microscopic picture; qualitative in EE361
Secant permeability	$\mu = B/H$ at the operating point	Use for total flux ↔ total MMF
Incremental permeability	$\mu_{inc} = dB/dH$	Use for small changes about a bias
Linear material	μ constant ⇒ constant $\mathcal{R}$ and L, superposition	Iron only below the knee
Saturation	slope → μ_0; μ_r falls	Electrical steel ≈ 1.6–2 T
Remanence, coercivity	B_r at $H = 0$; H_c to remove it	Loop details in Lesson 16
Soft vs hard	narrow vs wide loop	Cores vs permanent magnets
Inductance with iron	$L_{sec} = \lambda/i$, $L_{inc} = d\lambda/di$	Equal only if λ–i is a line through 0

10. Vocabulary (EN – TR)

English	Türkçe
magnetic material	manyetik malzeme

English	Türkçe
magnetic domain	manyetik bölge (domain)
magnetization	mıknatıslanma
B–H characteristic; magnetization curve	B–H karakteristiği; mıknatıslanma eğrisi
magnetic saturation	manyetik doyma
linear / nonlinear magnetic material	doğrusal / doğrusal olmayan manyetik malzeme
remanence (residual flux density)	kalıcı mıknatıslanma (artık akı yoğunluğu)
coercivity	koersivite (zorlayıcı alan şiddeti)
soft / hard magnetic material	yumuşak / sert manyetik malzeme
electrical steel	elektrik çeliği
permanent magnet	sürekli (kalıcı) mıknatıs
knee of the curve	eğrinin dirseği

11. Sources

- LN4 viewer pp. 20–26 (slides 8–10): B = μH, linear air curve, nonlinear iron with saturation. Notes.
- LN5 viewer pp. 2–5 (slides 2–4): linear vs nonlinear materials, saturation curve, residual magnetism examples. Notes.
- LN6 viewer pp. 7–9 (slides 4–5): incremental and linear definitions of inductance.
- Sen 3rd ed.: §1.1.2 p. 3; §1.1.4 p. 5 and Fig. 1.6; Fig. 1.7 p. 6 (data source for Figure 11.2); Example 1.1 pp. 8–9; Example 1.5 pp. 14–15; §1.2 hysteresis pp. 15–17; §1.4 permanent magnets and Example 1.8 pp. 26–30 (viewer pp. 50–54).
- Glossary Part II, p. 1–2 and p. 4 (soft/hard material).
- Verification: `lesson11_bh_saturation.py` → `records/lesson11_bh_saturation.json`.

Lesson 12 — MMF, reluctance and magnetic equivalent circuits (DC excitation)

Chapter II: Magnetic Circuits · Week 5 · Lesson 12 of 31

Assistant-written teaching note. Teacher source: LN3 slides 20–25, LN4 slides 3–7. Book: Sen 3rd ed. §1.1.3–1.1.5, Examples 1.1–1.4 (printed pp. 3–14, viewer 27–38). Exams: MT I Fall 2014 Q1; 2025 MT1 Q4(a); 2021 MT Q2(a,b). Numbers: `lesson12_mmf_reluctance.py`.

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1. Place and goals

Lessons 10 and 11 gave us the field laws and the material behaviour. This lesson turns them into the working method of the whole chapter: replace a core by a **magnetic equivalent circuit** (manyetik eşdeğer devre) made of MMF sources and reluctances, then solve it with the same KVL/KCL arithmetic you know from Chapter I. With **DC excitation** (constant current), nothing changes in time, so there is no induced voltage yet — that comes in Lesson 15.

After this lesson you should be able to:

1. Define MMF, reluctance and permeance, compute reluctance of any core section in SI units, and explain each step of the derivation $\Phi = \mathcal{F}/\mathcal{R}$.
2. Draw the equivalent circuit of series, parallel and multi-winding cores, including correct MMF polarities.
3. Solve linear magnetic circuits for flux given current, or current given flux, with mesh or nodal equations.
4. Solve nonlinear circuits with a real B–H curve by the forward method (flux given), the load-line method and trial-and-error/iteration (current given).
5. Check answers with Ampère’s law and flux continuity.

Prerequisites: Lessons 10–11; Chapter I circuit analysis (series/parallel resistors, mesh equations).

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2. From Ampère’s law to “magnetic Ohm’s law”

2.1 Derivation for a uniform core

Take a closed core of mean length l and uniform cross-section A made of a linear material of permeability μ, wound with N turns carrying a constant current i (Figure 12.1). Assume (Lesson 10, Section 7.2) no leakage and uniform flux.

1. **Ampère’s law** along the mean path: H is parallel to the path and constant on it, so $\oint H\,dl = Hl$. The path is threaded by all N turns, so
$$Hl = Ni. \tag{12.1}$$
2. **Material law:** $B = \mu H = \mu N i / l$.
3. **Flux:** uniform B over A, so
$$\Phi = BA = \frac{\mu A}{l}\,Ni = \frac{Ni}{l/(\mu A)}. \tag{12.2}$$
4. **Name the pieces:**

$$\boxed{\mathcal{F} = Ni \ \ [\text{A or At}], \qquad \mathcal{R} = \frac{l}{\mu A} \ \ [\text{A/Wb} = \text{H}^{-1}], \qquad \Phi = \frac{\mathcal{F}}{\mathcal{R}}} \tag{12.3}$$

$\mathcal{F}$ is the **magnetomotive force** (manyetomotor kuvvet), measured in ampere-turns (amper-sarım); $\mathcal{R}$ is the **reluctance** (relüktans); its reciprocal $\mathcal{P} = 1/\mathcal{R} = \mu A/l$ is the **permeance** (permeans), in henry (Wb/A). LN3 slide 20 sets these side by side with the electric case: $V = IR$ $\leftrightarrow$ $\mathcal{F} = \Phi\mathcal{R}$, $R = \rho l/A = l/(\sigma A)$ $\leftrightarrow$ $\mathcal{R} = l/(\mu A)$ (Sen Eqs. 1.12–1.14).

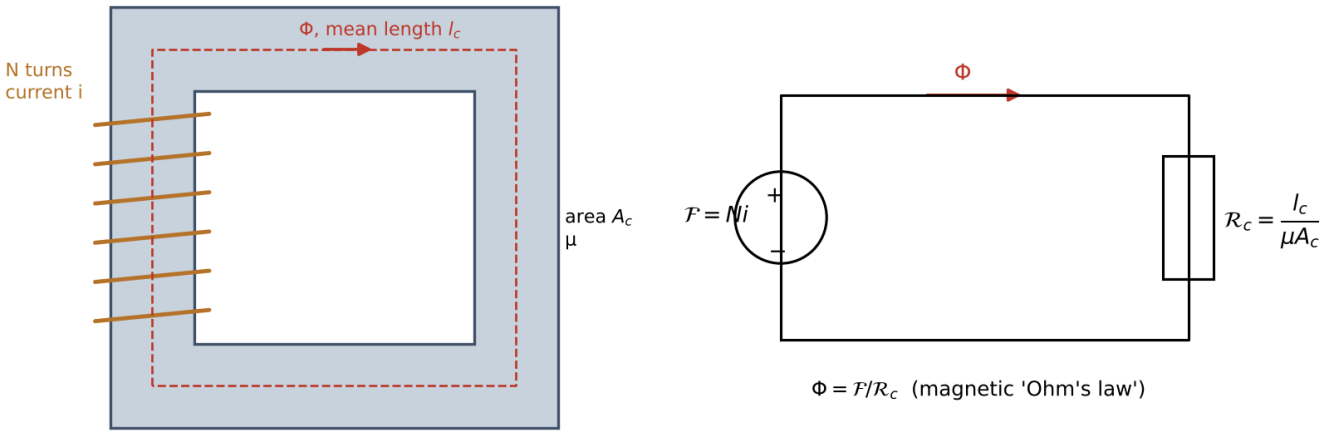


Figure 12.1. Left: closed core with an N-turn winding; dashed line = mean path of length l_c. Right: the magnetic equivalent circuit, an MMF source driving flux through one reluctance. After LN3 slide 25(b) and Sen Fig. 1.5.

2.2 What the analogy does and does not mean

- **Same algebra:** Ampère's law around any closed loop of the core is "sum of MMF sources = sum of $\mathcal{R}\Phi$ drops" (KVL), and Gauss's law at any junction is "sum of fluxes = 0" (KCL). So every Chapter I technique — series/parallel combination, voltage-divider-like splitting, mesh and nodal equations, superposition (only if linear) — applies.
- **Different physics:** a constant flux dissipates no power, so a reluctance is *not* a lossy element; it is closer to an inverse "spring constant" that stores energy (Lesson 14). And a reluctance depends on flux if the iron saturates (Lesson 11), which no ordinary resistor does.
- **Reluctance drop** $\mathcal{R}\Phi$ equals Hl for that section: the MMF "used up" by that section. The phrase "MMF drop across the gap" means exactly $H_g l_g$.

2.3 Units checklist

$\mu_0 = 4\pi \times 10^{-7}$ H/m, lengths in m, areas in m². A typical iron section has $\mathcal{R} \sim 10^4$–10^5 A/Wb; a millimetre air gap of a few cm² has $\sim 10^6$ A/Wb. If your gap reluctance comes out tiny, you probably used mm instead of m.

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3. Building equivalent circuits

3.1 Recipe

1. **Identify sections** with constant area and material along the mean path: each limb, each yoke, each air gap. Use the centre lines of the limbs for lengths.
2. **Compute each reluctance** $\mathcal{R}_k = l_k/(\mu_k A_k)$. For air $\mu = \mu_0$.
3. **Place MMF sources** $N_k i_k$ in the limbs that carry the windings. **Polarity**: use the right-hand rule (Lesson 10, Example 10.5) to find the direction in which each winding alone would drive flux, and orient the source so it pushes flux that way.
4. **Connect** sections exactly as flux can flow: sections in series if the same flux passes through them, in parallel if flux divides between them.
5. **Choose flux directions** (arbitrary; a negative answer means the opposite direction) and write KVL (Ampère) around independent loops and KCL (Gauss) at junctions.
6. **Solve, then check** with an independent equation (e.g. a different loop, or nodal analysis).

3.2 Series circuits

Same flux through every section: $\mathcal{R}_{tot} = \sum \mathcal{R}_k$, $\Phi = \mathcal{F}/\mathcal{R}_{tot}$, and $B_k = \Phi/A_k$ in each part. Where the area halves, B doubles: **narrow sections saturate first**.

3.3 Parallel (branched) circuits

When a core has three or more limbs, flux divides. For the common “shell” geometry with the winding on one leg (like the 2025 Q4 core seen from coil 1), the source sees its own leg in series with the parallel combination of the others, and the flux divides like current in a current divider:

$$\Phi_a = \Phi_{tot}\frac{\mathcal{R}_b}{\mathcal{R}_a + \mathcal{R}_b} \qquad \text{(flux takes the low-reluctance path preferentially)}.$$

3.4 Several windings

Each winding is an MMF source. If two windings on the same loop drive flux the same way their MMFs **aid** (add); if opposite ways they **oppose** (subtract) — Lesson 10, P10.4. With windings on different limbs, write one KVL per independent loop. Sen Example 1.3 (worked below) is the standard case.

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4. Nonlinear circuits: using the real B–H curve

With saturable iron, $\mathcal{R}_{iron}$ depends on its own flux, so we cannot simply add reluctances. Two problem types arise.

4.1 Type A — flux (or B) given, find the current (“forward” problem)

This is always direct:

1. From the given flux, compute $B_k = \Phi_k/A_k$ in every section (use KCL for branched cores).
2. Iron sections: read H_k from the B–H curve. Gaps: $H_g = B_g/\mu_0$.
3. MMF needed $= \sum H_k l_k$ around the loop containing the winding.
4. $i = \mathcal{F}/N$.

4.2 Type B — current given, find the flux (“inverse” problem)

Now H_{iron} and B are both unknown and related by the curve. For a core with one iron path and an air gap, Ampère’s law gives one more equation:

$$Ni = H_c l_c + \frac{B}{\mu_0} l_g \quad \Longrightarrow \quad B = -\mu_0 \frac{l_c}{l_g} H_c + \frac{\mu_0 N i}{l_g}. \tag{12.4}$$

This is a straight line in the B–H plane — the **load line** (yük doğrusu, Sen Eq. 1.21). Its intercepts are $B = \mu_0 N i / l_g$ (all MMF on the gap, $H_c = 0$) and $H_c = Ni/l_c$ (all MMF on the iron, $B = 0$); its slope is $-\mu_0 l_c / l_g$. The operating point is where the load line crosses the iron’s B–H curve.

The picture shows *why* an air gap linearizes a magnetic circuit. A **longer gap** lowers the B-intercept and makes the line flatter (smaller |slope|), so the crossing moves down into the straight part of the iron curve and the flux becomes nearly proportional to the current. A **shorter gap** raises the intercept and steepens the line, pushing the crossing toward the knee and into saturation.

The alternatives to drawing are:

- **Trial and error** (Sen): assume B, compute i by the Type A method, adjust B until i matches.
- **Bisection/iteration** (what the verification code does): define $f(B) = H_c(B) l_c + B l_g / \mu_0 - Ni$ and find the root; f increases monotonically with B, so the root is unique.

Superposition fails in nonlinear circuits: never compute fluxes from two windings separately and add them unless the iron stays in its linear region.

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5. Methods summary and common mistakes

Situation	Method
Linear, one loop	$\Phi = \mathcal{F} / \sum \mathcal{R}$
Linear, branched	series–parallel reduction or mesh/nodal equations
Nonlinear, flux known	sections’ $B \to H$ from curve $\to \sum Hl$
Nonlinear, current known, one iron path + gap	load line or iteration
Nonlinear, branched, current known	iterate on one unknown flux (Sen P1.15 “trial and error”)

Common mistakes.

- Forgetting that the **same flux** gives **different** B in sections of different area.
- Using $\mathcal{R}_{iron}$ computed at one flux density for another flux density.
- Getting an MMF polarity wrong in a two-winding problem; always draw the flux each winding would produce alone first.
- Subtracting the gap length from the iron length (correct, $l_c - g$) and then forgetting it — usually negligible, but do it consistently (see Example 12.8 and errata E8).
- Treating “magnetic Ohm’s law” as permission to compute power $\Phi^2 \mathcal{R}$ — there is no such loss.

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6. Worked examples

Example 12.1 — Reluctance of a core (LN3 slide 23 exercise)

Given. Iron core, $\mu_r = 4000$, mean length $l = 500$ mm, area $A = 2500$ mm². **Find** $\mathcal{R}$ "in terms of μ_0" and numerically.

Solution. Convert: $l = 0.5$ m, $A = 2.5 \times 10^{-3}$ m².

$$\mathcal{R} = \frac{l}{\mu_r \mu_0 A} = \frac{0.5}{4000 \times 2.5 \times 10^{-3}\,\mu_0} = \frac{0.05}{\mu_0} = \frac{0.05}{4\pi \times 10^{-7}} = 3.979 \times 10^4 \text{ A/Wb}.$$

Interpretation. The same core with a 3 mm air gap is Example 13.1; the gap will multiply the reluctance by about 25.

Example 12.2 — Flux, B and H in the ungapped core (LN4 slide 6)

Given. The core of Example 12.1 with $N = 5$ turns and $I = 10$ A (DC). **Find** B and H in the core.

Solution. $\mathcal{F} = NI = 50$ At. $\Phi = \mathcal{F}/\mathcal{R} = 50/(3.979 \times 10^4) = 1.257 \times 10^{-3}$ Wb. $B = \Phi/A = 0.503$ T. $H = B/\mu = 0.503/(4000\mu_0) = 100$ A/m. **Check.** Ampère directly: $H = NI/l = 50/0.5 = 100$ A/m ✓. (LN4 says "same dimensions of the previous example"; if the gapped core of slide 24 is meant, see Example 13.2.)

Example 12.3 — Series core with two different cross-sections (original; cf. Sen P1.2)

Given. A rectangular core: two limbs, each 0.30 m long with area 50 cm²; two yokes, each 0.20 m long with area 100 cm² (thicker). $\mu_r = 2000$ (linear), $N = 300$, $I = 1$ A. **Find** (a) Φ; (b) B in limbs and yokes; (c) L; (d) the current that brings the limbs to 1.2 T (if still linear).

Solution. (a) $\mathcal{R}_{limbs} = \dfrac{2(0.30)}{2000\mu_0(50 \times 10^{-4})} = 4.775 \times 10^4$ A/Wb; $\mathcal{R}_{yokes} = \dfrac{2(0.20)}{2000\mu_0(100 \times 10^{-4})} = 1.592 \times 10^4$ A/Wb. Series: $\mathcal{R}_{tot} = 6.366 \times 10^4$ A/Wb, $\Phi = 300/6.366 \times 10^4 = 4.712 \times 10^{-3}$ Wb.

b. $B_{limb} = 4.712 \times 10^{-3}/0.005 = 0.942$ T; $B_{yoke} = 0.471$ T.
c. $L = N^2/\mathcal{R}_{tot} = 90\,000/63\,662 = 1.414$ H.
d. $I = 1.2(0.005)\mathcal{R}_{tot}/300 = 1.273$ A. **Check.** $\sum Hl = (0.942/\mu)(0.6) + (0.471/\mu)(0.4) = 300$ At ✓. **Interpretation.** The thin limbs carry twice the flux density of the yokes; they would saturate first, so a designer sizes the *thinnest* section for the allowed B.

Example 12.4 — Flux given, find current, with nonlinear iron (Sen Example 1.1)

Given. A primitive relay: $N = 500$, cast-steel core with mean length $l_c = 360$ mm, two air gaps of 1.5 mm each; $B = 0.8$ T is needed in the gaps. Neglect fringing. **Find** (a) the coil current; (b) μ and μ_r of the core; (c) the current with zero air gap.

Solution (Type A). No fringing ⇒ same area ⇒ $B_c = B_g = 0.8$ T.

- Core: from the cast-steel curve $H_c = 510$ A/m, so $\mathcal{F}_c = H_c l_c = 510(0.36) = 184$ At.
- Gaps: $\mathcal{F}_g = \dfrac{B_g}{\mu_0}(2l_g) = \dfrac{0.8}{4\pi \times 10^{-7}}(3 \times 10^{-3}) = 1910$ At.
- Total $\mathcal{F} = 2094$ At, so $i = 2094/500 = 4.19$ A.

b. $\mu = 0.8/510 = 1.57 \times 10^{-3}$ H/m, $\mu_r = 1250$. (c) Without gaps, $i = 184/500 = 0.368$ A.
Interpretation. Two gaps totalling 3 mm need ten times the MMF of 360 mm of steel. This is the first hint of Lesson 13's theme.

Example 12.5 — Current given, find flux: load line and iteration (Sen Example 1.2)

Given. The same relay with 1 mm gaps (2 mm total) and $i = 4$ A. **Find** B.

Solution (Type B). Load line (12.4): slope $-\mu_0 l_c/l_g = -4\pi \times 10^{-7}(0.36/0.002) = -2.26 \times 10^{-4}$ T/(A/m); B-intercept $\mu_0 N i/l_g = 1.257$ T; H-intercept $Ni/l_c = 5556$ A/m. Plotting on the cast-steel curve (Figure 12.2) gives the crossing at about $B = 1.08$ **T,** $H_c = 784$ **A/m** (Sen: 1.08 T).

Trial and error (Sen's table): $B = 1.1$ T → $H_c = 800$, $\mathcal{F} = 288 + 1751 = 2039$ At, $i = 4.08$ A (too high); $B = 1.08$ T → $H_c = 785$, $\mathcal{F} = 283 + 1719 = 2002$ At, $i = 4.00$ A ✓.

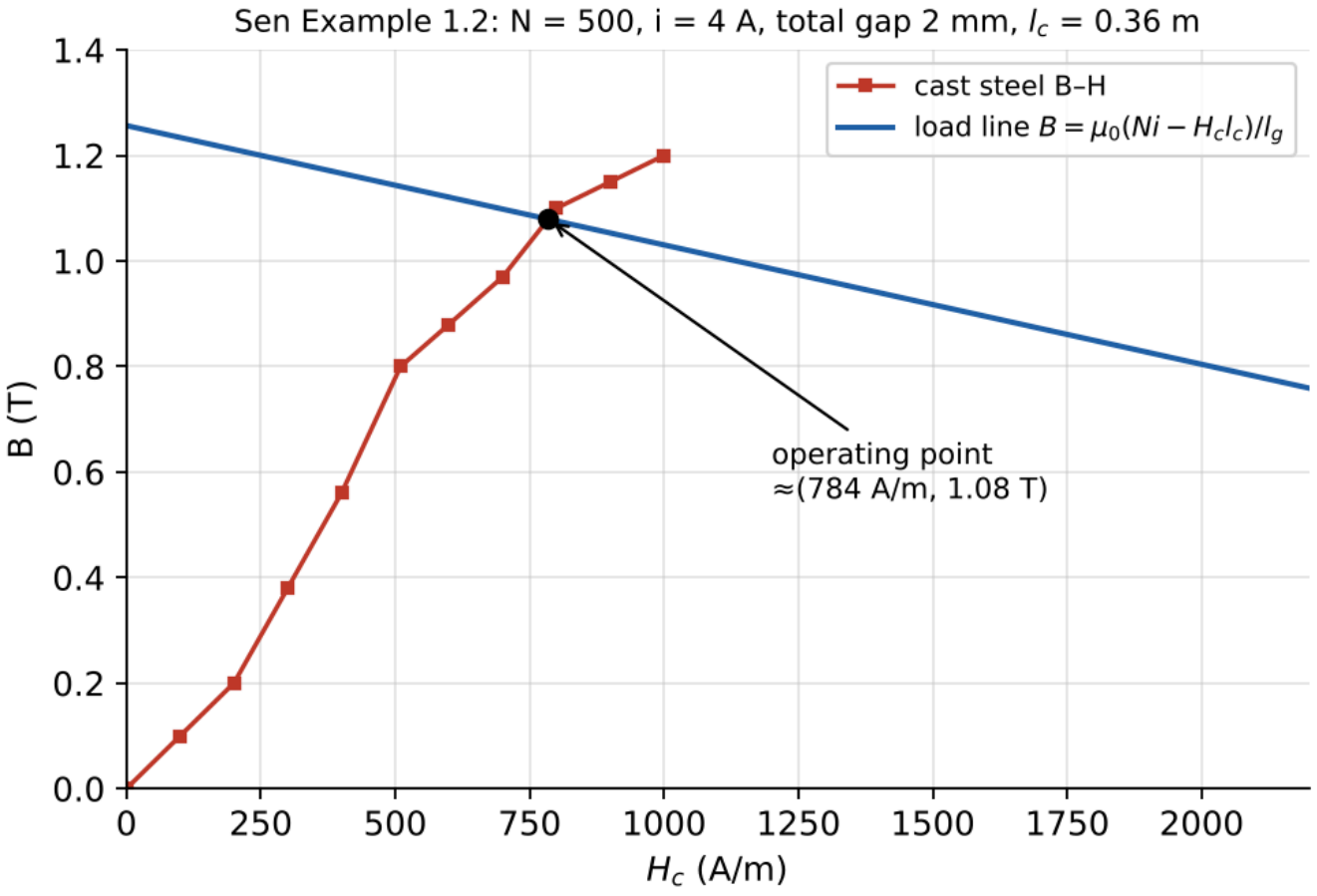


Figure 12.2. Load line $B = \mu_0(Ni - H_c l_c)/l_g$ on the cast-steel curve. The code finds the root at 1.079 T by bisection.

Interpretation. With 2 mm of gap the gap absorbs 86 % of the MMF (1719 of 2002 At) and the iron sits just at its knee. Halving the gap would double the B-intercept and the slope of the load line, pushing the iron further into saturation (Practice P12.3 shows the diminishing return of extra current at the knee).

Example 12.6 — Two windings, parallel paths (Sen Example 1.3)

Given. The three-limb core of Figure 12.3: all limbs 2 cm × 2 cm ($A = 4$ cm²), $\mu_r = 1200$; outer paths b-a-f-e and b-c-d-e each 3×52 cm; centre limb b-e 52 cm long containing a 0.5 cm gap (so 51.5 cm of iron); $N_1 = N_2 = 500$, $I_1 = I_2 = 10$ A, both windings driving flux *down* the centre limb.

Neglect leakage and fringing. **Find** the gap flux, B_g and H_g.

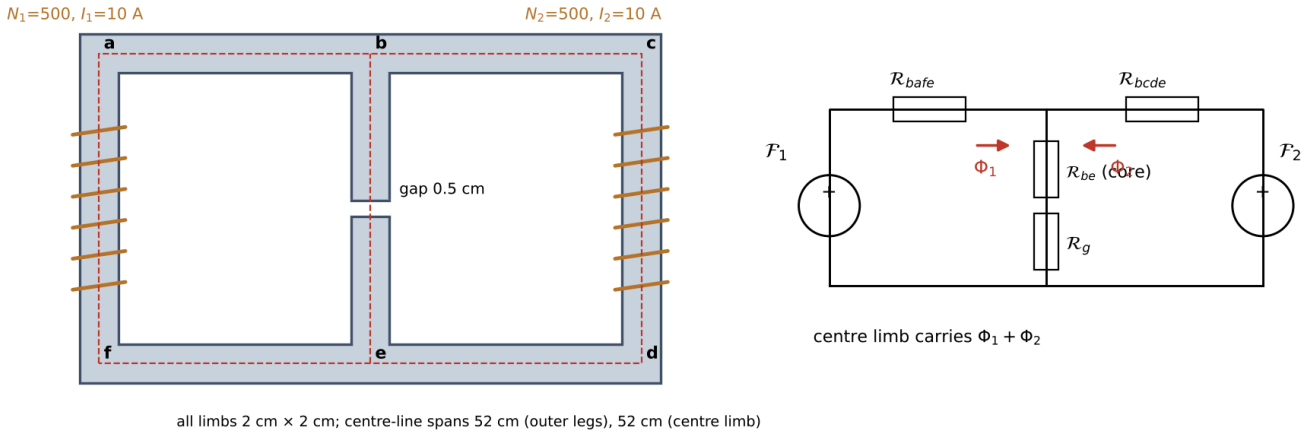


Figure 12.3. Redrawn from Sen Fig. E1.3: (left) geometry with path letters; (right) equivalent circuit.

Solution. Reluctances: $\mathcal{R}_{bafe} = \mathcal{R}_{bcde} = \dfrac{1.56}{1200\mu_0(4\times10^{-4})} = 2.586\times10^{6}$; $\mathcal{R}_{be,core} = \dfrac{0.515}{1200\mu_0(4\times10^{-4})} = 0.854\times10^{6}$; $\mathcal{R}_g = \dfrac{0.005}{\mu_0(4\times10^{-4})} = 9.947\times10^{6}$ A/Wb. $\mathcal{F}_1 = \mathcal{F}_2 = 5000$ At.

Mesh equations with Φ_1, Φ_2 both flowing down the centre limb:

$$\Phi_1(\mathcal{R}_{bafe}+\mathcal{R}_{be}+\mathcal{R}_g) + \Phi_2(\mathcal{R}_{be}+\mathcal{R}_g) = 5000,$$

$$\Phi_1(\mathcal{R}_{be}+\mathcal{R}_g) + \Phi_2(\mathcal{R}_{bcde}+\mathcal{R}_{be}+\mathcal{R}_g) = 5000.$$

By symmetry $\Phi_1 = \Phi_2 = 5000/(\mathcal{R}_{bafe}+2\mathcal{R}_{be}+2\mathcal{R}_g) = 2.067\times10^{-4}$ Wb. Gap flux $\Phi_g = 4.134\times10^{-4}$ Wb, $B_g = 1.034$ T, $H_g = B_g/\mu_0 = 8.22\times10^{5}$ A/m.

Check (nodal method). Let u be the magnetic potential of node b relative to e. KCL: $(5000-u)/\mathcal{R}_{bafe} + (5000-u)/\mathcal{R}_{bcde} = u/(\mathcal{R}_{be}+\mathcal{R}_g)$ gives the same Φ_g to 10^{-10}. *(Sen prints $\mathcal{R}_{be} = 0.82\times10^{6}$; the stated length and area give 0.854×10^{6}, and Sen's final fluxes agree with the latter — errata E12.)*

Variant. Reverse winding 2. Now the centre-limb flux is exactly zero by symmetry, and the two MMFs act in series around the outer loop a-b-c-d-e-f: $\Phi = (\mathcal{F}_1+\mathcal{F}_2)/(\mathcal{R}_{bafe}+\mathcal{R}_{bcde}) = 10\,000/(2\times2.586\times10^{6}) = 1.933\times10^{-3}$ Wb, i.e. 4.8 T if the material stayed linear — impossible; the iron would saturate. The linear model's answer is a warning, not a prediction.

Example 12.7 — Nonlinear core with six gaps (MT I Fall 2014 Q1)

Given. (exams/archive/MT/MT I - Fall2014.pdf pp. 2–3.) A rectangular sheet-steel core, area $A = 10$ cm² everywhere, steel mean length $l_s = 80$ cm, six identical gaps of 0.3 mm in one limb (total 1.8 mm), a 240-turn coil with $I = 10$ A DC. The exam's B–H curve (Fig. 2) is a straight line from the origin to about (500 A/m, 1.40 T), then a knee to about 1.8 T at 1000 A/m and 2.0 T at 3000 A/m (read from the figure, ±0.03 T). Neglect leakage and fringing. **Find** (a) the operating point; (b) energy stored in the steel; (c) reluctances of steel and gaps; (d) L; (e) L if the steel were infinitely permeable.

Solution. (a) Load line: $2400 = H_s(0.8) + B_s(1.8\times10^{-3})/\mu_0$. Intercepts: $B_s = 2400\mu_0/0.0018 = 1.676$ T; $H_s = 3000$ A/m. Try the straight segment $B_s = (1.40/500)H_s$: $2400 = H_s(0.8 + 2.8\times10^{-3}\times1.8\times10^{-3}/\mu_0) = 4.811H_s$, so $H_s = 498.9$ A/m and $B_s = 1.397$ T — just inside the linear segment, so this is

the operating point.

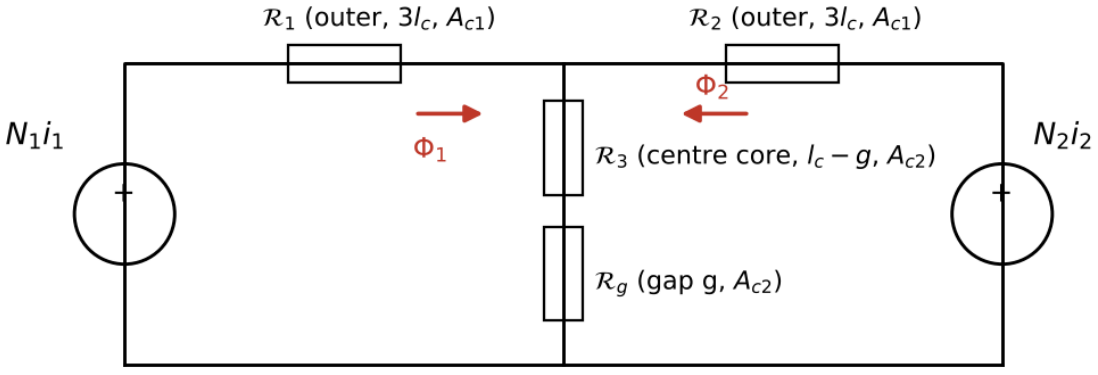
b. On the straight segment $\int_0^B H\,dB = \frac{1}{2}BH$: $W_s = \frac{1}{2}(1.397)(498.9)(10^{-3}\times 0.8) = 0.279$ J.
c. $\Phi = B_s A = 1.397\times 10^{-3}$ Wb; $\mathcal{R}_s = H_s l_s/\Phi = 2.857\times 10^5$ A/Wb; $\mathcal{R}_g = 1.8\times 10^{-3}/(\mu_0\times 10^{-3}) = 1.432\times 10^6$ A/Wb.
d. $L = N\Phi/I = 240(1.397\times 10^{-3})/10 = 33.5$ mH.
e. $L = N^2/\mathcal{R}_g = 57\,600/1.432\times 10^6 = 40.2$ mH. **Check.** $\Phi(\mathcal{R}_s+\mathcal{R}_g) = 2400$ At ✓; total energy $\frac{1}{2}LI^2 = 1.676$ J $= W_s + W_g$ with $W_g = 1.398$ J ✓. **Interpretation.** The steel is 444 times longer than the gaps, yet the gaps carry 83 % of the reluctance and 83 % of the stored energy. Assuming $\mu\to\infty$ overestimates L by 20 % here because the steel is close to its knee. (The exam's remark that splitting the gap into six "makes the field uniform" refers to reduced fringing per gap — Lesson 13.)

Example 12.8 — Reluctances and flux split of the 2025 MT1 Q4 core, part (a)

Given. (exams/2025/questions/Midterm_1_2025_v2.pdf p. 9.) Two ferrite E-cores form a core with outer paths of length $3l_c$ and area $A_{c1} = 1.5$ cm², a centre limb of length l_c and area $A_{c2} = 2.5$ cm² containing a gap $g = 0.5$ mm; $l_c = 0.1$ m, $\mu_r = 1500$; coil 1 ($N_1 = 400$) on the left leg, coil 2 ($N_2 = 800$) on the right leg. **Find** the equivalent circuit and reluctances; then (beyond the question) the flux split when $i_1 = 1.2$ A and coil 2 is open.

Solution. Outer paths: $\mathcal{R}_1 = \mathcal{R}_2 = \dfrac{3(0.1)}{1500\mu_0(1.5\times 10^{-4})} = 1.061\times 10^6$ A/Wb. Centre iron: $\mathcal{R}_3 = \dfrac{0.1-0.0005}{1500\mu_0(2.5\times 10^{-4})} = 0.211\times 10^6$ A/Wb (the handwritten key uses 0.095 m and gets 0.202×10⁶ — errata E8, effect < 0.1 % on the final answers). Gap: $\mathcal{R}_g = \dfrac{5\times 10^{-4}}{\mu_0(2.5\times 10^{-4})} = 1.592\times 10^6$ A/Wb. Circuit: Figure 12.4.

From coil 1, $\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 \parallel (\mathcal{R}_3+\mathcal{R}_g) = 1.061\times 10^6 + 0.668\times 10^6 = 1.729\times 10^6$ A/Wb. At $i_1 = 1.2$ A: $\Phi_1 = 480/\mathcal{R}_{eq} = 2.776\times 10^{-4}$ Wb, which divides into $\Phi_{centre} = \Phi_1\mathcal{R}_2/(\mathcal{R}_2+\mathcal{R}_3+\mathcal{R}_g) = 1.029\times 10^{-4}$ Wb and $\Phi_{right} = 1.748\times 10^{-4}$ Wb.



Both MMFs drawn so that their flux enters the centre limb from the top (assumed winding senses).

Interpretation. Even though the centre limb is wider, the 0.5 mm gap makes it the *higher*-reluctance branch, so most of coil 1's flux returns through the far outer leg — good for coupling to coil 2 (Lesson 14). Note also B in the left leg: $\Phi_1/A_{c1} = 1.85$ T, far above what real ferrite can

carry (ferrites saturate at roughly 0.3–0.5 T). The exam instructs “assume operation in the linear region”, which is a modelling instruction, not a statement about real ferrite — exactly the issue raised in its part (e).

Example 12.9 — Toroidal inductor of a GaN charger (2021 MT Q2 Part I)

Given. (exams/archive/MT/2021_Midterm_Q2_solution.pdf p. 5.) Nanocrystalline toroid: outer diameter 24 mm, inner 12 mm, height 6 mm; $\mu_r = 25\,000$, $B_{sat} = 1.2$ T, $N = 25$. **Find** (a) L; (b) the maximum current before saturation.

Solution. Mean path $l_e = \pi(24+12)/2$ mm $= 18\pi$ mm $= 0.05655$ m; area $A_e = \frac{24-12}{2} \times 6 = 36$ mm^2 $= 36 \times 10^{-6}$ m^2 (the key writes “mm²” — a unit slip, errata E10). $\mathcal{R} = l_e/(\mu_r \mu_0 A_e) = 5.0 \times 10^4$ A/Wb. (a) $L = N^2/\mathcal{R} = 625/5 \times 10^4 = 12.5$ mH. (b) $NI_{max} = \Phi_{sat}\mathcal{R} = B_{sat}A_e\mathcal{R} \Rightarrow I_{max} = 1.2(36 \times 10^{-6})(5 \times 10^4)/25 = 86.4$ mA. **Interpretation.** A tiny current saturates a high-μ ungapped core. Power inductors that must carry amperes are therefore gapped or use low-μ powder cores (Lesson 13, Example 13.4).

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7. Optional practice

P12.1. For the core of Example 12.3, the yoke area is reduced to 50 cm² as well. Find the new Φ for 1 A and the new current for 1.2 T in the limbs.

▼ **Solution**
All four sides now have $A = 50$ cm²; total length 1.0 m. $\mathcal{R} = 1.0/(2000\mu_0 \times 0.005) = 7.958 \times 10^4$ A/Wb, $\Phi = 300/7.958 \times 10^4 = 3.770 \times 10^{-3}$ Wb ($B = 0.754$ T). For 1.2 T, $\Phi = 6 \times 10^{-3}$ Wb and $I = 6 \times 10^{-3} \times 7.958 \times 10^4/300 = 1.59$ A. (Equivalent: $I = Hl/N$ with $H = 1.2/\mu$.)

P12.2. In Sen Example 1.3 (both windings aiding, linear core), what equal current in both windings gives $B_g = 1.2$ T?

▼ **Solution**
The circuit is linear, so B_g scales with current: $I = 10 \times 1.2/1.0336 = 11.61$ A. *Code: 11.610 A by root-finding.*

P12.3. The relay of Sen Example 1.2 must reach 1.15 T with 2 mm total gap. Find the current (Type A problem).

▼ **Solution**
From the cast-steel data, $H_c(1.15\text{ T}) = 900$ A/m: $\mathcal{F}_c = 900(0.36) = 324$ At. $\mathcal{F}_g = (1.15/\mu_0)(0.002) = 1830$ At. $i = 2154/500 = 4.31$ A. Note that 7.7 % more current than Example 12.5 buys only 6.5 % more flux: the iron is entering its knee.

P12.4 (concept). Why does the 2025 Q4 key compute L_{22} with the same $\mathcal{R}_{eq}$ as L_{11}?

▼ **Solution**
$L_{22} = N_2^2/\mathcal{R}_{eq,2}$ where $\mathcal{R}_{eq,2}$ is the reluctance seen by coil 2: $\mathcal{R}_2 + \mathcal{R}_1 \parallel (\mathcal{R}_3 + \mathcal{R}_g)$. Because the core is symmetric ($\mathcal{R}_1 = \mathcal{R}_2$), this equals $\mathcal{R}_{eq,1}$. For an unsymmetric core you must recompute it.

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8. Summary

Relation	Use
$\mathcal{F} = Ni$; $\mathcal{R} = l/(\mu A)$; $\mathcal{P} = 1/\mathcal{R}$	Definitions (SI units)
$\sum \mathcal{F} = \sum \mathcal{R}_k \Phi_k = \sum H_k l_k$ around a loop	Ampère/KVL
$\sum \Phi = 0$ at a junction	Gauss/KCL
$B = -\mu_0 \frac{l_c}{l_g} H_c + \frac{\mu_0 N i}{l_g}$	Load line (iron + gap, current given)
$L = N^2/\mathcal{R}_{eq}$	Inductance from the circuit (linear)

9. Vocabulary

English	Türkçe
magnetomotive force (MMF)	manyetomotor kuvvet (MMK)
ampere-turn	amper-sarım
reluctance / permeance	relüktans / permeans
magnetic equivalent circuit	manyetik eşdeğer devre
series / parallel magnetic circuit	seri / paralel manyetik devre
magnetic flux path; mean path length	manyetik akı yolu; ortalama yol uzunluğu
excitation current	uyartım akımı
load line	yük doğrusu
trial and error	deneme-yanılma
composite structure	bileşik yapı
winding / coil; number of turns	sargı / bobin; sarım sayısı

10. Sources

- LN3 viewer pp. 53–64 (slides 20–25): magnetic Ohm’s law, reluctance, J = σE vs B = μH, exercises with and without gap, equivalent circuits. LN4 viewer pp. 3–17 (slides 3–7): Ampère, MMF, assumptions, exercise N = 5, I = 10 A, doubling current.
- Sen §1.1.3–1.1.5 pp. 3–7 (viewer 27–31), Table 1.1, Eqs. 1.12–1.21; Examples 1.1–1.3 pp. 8–12 (viewer 32–36); Fig. 1.7 p. 6.
- MT I Fall 2014 Q1: exams/archive/MT/MT I - Fall2014.pdf pp. 2–3 (no key supplied; solution here is ours).
- 2025 MT1 Q4(a): exams/2025/questions/Midterm_1_2025_v2.pdf p. 9; key EE361_2025f_MT1_Q4_Sol.pdf p. 1.
- 2021 MT Q2(a,b): exams/archive/MT/2021_Midterm_Q2_solution.pdf p. 5 (typed prompt + handwritten key).
- Verification: `lesson12_mmf_reluctance.py`.
- More solved archive problems: Ch II–III archive collection §7.4 (Homework 1, three-leg nonlinear core).

Lesson 13 — Air gaps, fringing and leakage

Chapter II: Magnetic Circuits · Week 5 · Lesson 13 of 31

Assistant-written teaching note. Teacher source: LN3 slides 14 and 24; LN4 slides 5–6; LN8 slides 9–12 (leakage, preview of transformers). Book: Sen §1.1.5 (printed pp. 5–7, viewer 29–31), Example 1.4 (p. 13), Problems 1.12–1.13. Exams: MT I Fall 2015 Q1, Fall 2016 Q1, Fall 2017 Q1. Numbers: `lesson13_air_gaps.py`.

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1. Place and goals

Lesson 12 showed in passing that a few millimetres of air can need more MMF than half a metre of iron. This lesson makes that the centre of attention, because almost every real magnetic device has an air gap, either because it must move (relays, machines: the rotor is separated from the stator by an air gap) or because a gap is deliberately added (inductors). It also deals honestly with the two "air paths" the simple model ignores: **fringing** at the edges of a gap and **leakage** around the windings.

After this lesson you should be able to:

1. Explain physically and numerically why a short gap dominates the reluctance of a core, and when the iron can be treated as infinitely permeable.
2. Apply the usual fringing correction for gap area and judge when it matters.
3. Explain leakage flux, model it with a parallel air reluctance, and relate it to leakage inductance.
4. Use gaps deliberately: design an inductor for a given L, compare gapped and ungapped inductors for saturation current and stored energy, and evaluate sensitivity to gap tolerance.

Prerequisites: Lessons 10–12.

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2. Why the air gap dominates

2.1 The "equivalent iron length" of a gap

Compare a gap of length g with a piece of iron of length l_{fe}, both of area A:

$$\mathcal{R}_g = \frac{g}{\mu_0 A}, \qquad \mathcal{R}_{fe} = \frac{l_{fe}}{\mu_r \mu_0 A} \qquad \Longrightarrow \qquad \mathcal{R}_g = \mathcal{R}_{fe} \ \text{ when } \ l_{fe} = \mu_r g.$$

A gap "looks like" μ_r times its length of iron. With $\mu_r = 4000$, the 3 mm gap of LN3 slide 24 has the reluctance of **12 m** of iron, while the whole core path is only 0.5 m. This is LN3's message on slide 14 ("iron is just ~3000 times more permeable than air; therefore air is a circuit element").

2.2 Where the MMF goes

For a core of iron length l_c and gap g in series (same flux, same area, no fringing), Ampère's law reads

$$Ni = H_c l_c + H_g g = \frac{B}{\mu_r \mu_0} l_c + \frac{B}{\mu_0} g = \frac{B}{\mu_0}\left(\frac{l_c}{\mu_r} + g\right). \tag{13.1}$$

The two terms are the MMF drops across iron and gap. Their ratio is $(l_c/\mu_r) : g$. Whenever $g \gg l_c/\mu_r$ — that is, whenever the gap is longer than the "iron equivalent air length" l_c/μ_r — the gap takes nearly all of the MMF and

$$B \approx \frac{\mu_0 N i}{g}, \qquad L \approx \frac{\mu_0 N^2 A}{g} \qquad (\mu \to \infty \text{ approximation}). \tag{13.2}$$

Figure 13.1 plots the gap's share of the total reluctance and the resulting inductance for the LN3 core. Already at g = 0.5 mm the gap holds 80 % of the reluctance.

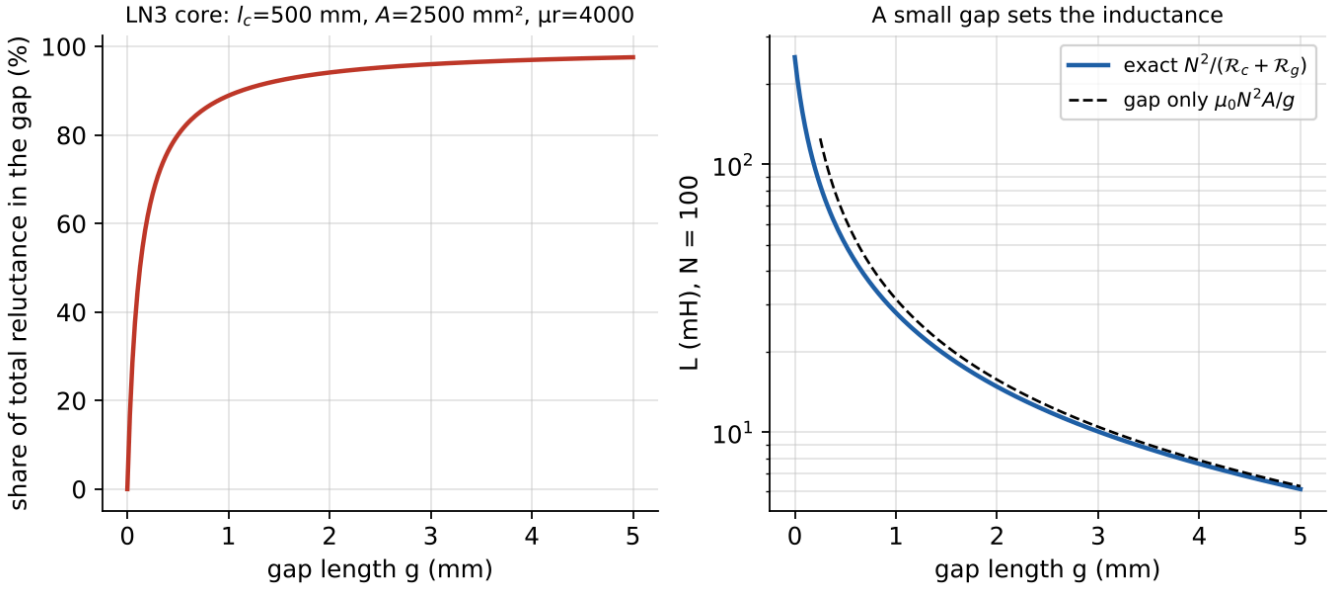


Figure 13.1. LN3 core (l_c = 500 mm, A = 2500 mm², μ_r = 4000). Left: share of total reluctance in the gap. Right: inductance for N = 100, exact (solid) and with the iron neglected (dashed); the two merge for gaps above about 1 mm.

2.3 Consequences you must be able to explain

1. **Linearization.** The gap's B–H relation is exactly linear. If the gap carries most of the MMF, the circuit's overall Φ–i relation is nearly linear even though the iron is not (load-line picture of Lesson 12, Section 4.2). Inductors that must have a stable L are gapped for this reason.
2. **Less sensitivity to the iron.** $L \approx \mu_0 N^2 A/g$ does not contain μ_r, so temperature and batch variations of the iron hardly matter. (They do matter in an ungapped core: Example 13.7 shows a factor of 7 between ungapped and gapped.)
3. **More current for the same flux.** The price of a gap is a larger magnetizing current. This is why **transformers are built without gaps** (Lesson 18; MT I Fall 2019 Q2(g) asks exactly this), while inductors and machines live with or need them.
4. **Energy is stored in the gap.** Energy density is $B^2/(2\mu)$ (Lesson 14); for equal B the gap stores μ_r times more energy per unit volume than the iron. In a gapped inductor most of the energy sits in a few cubic millimetres of air.
5. **A gap delays saturation.** For the same L, a gapped core needs more turns but saturates at a higher current (Example 13.4).

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3. Fringing

3.1 What it is

At the edges of a gap, field lines bulge outward (Figure 13.2): the flux spreads into the surrounding air before it re-enters the other pole face. The flux crosses an effective area larger than the pole face, so the gap's true reluctance is *lower* than $g/(\mu_0 A_c)$ (Sen Fig. 1.10).

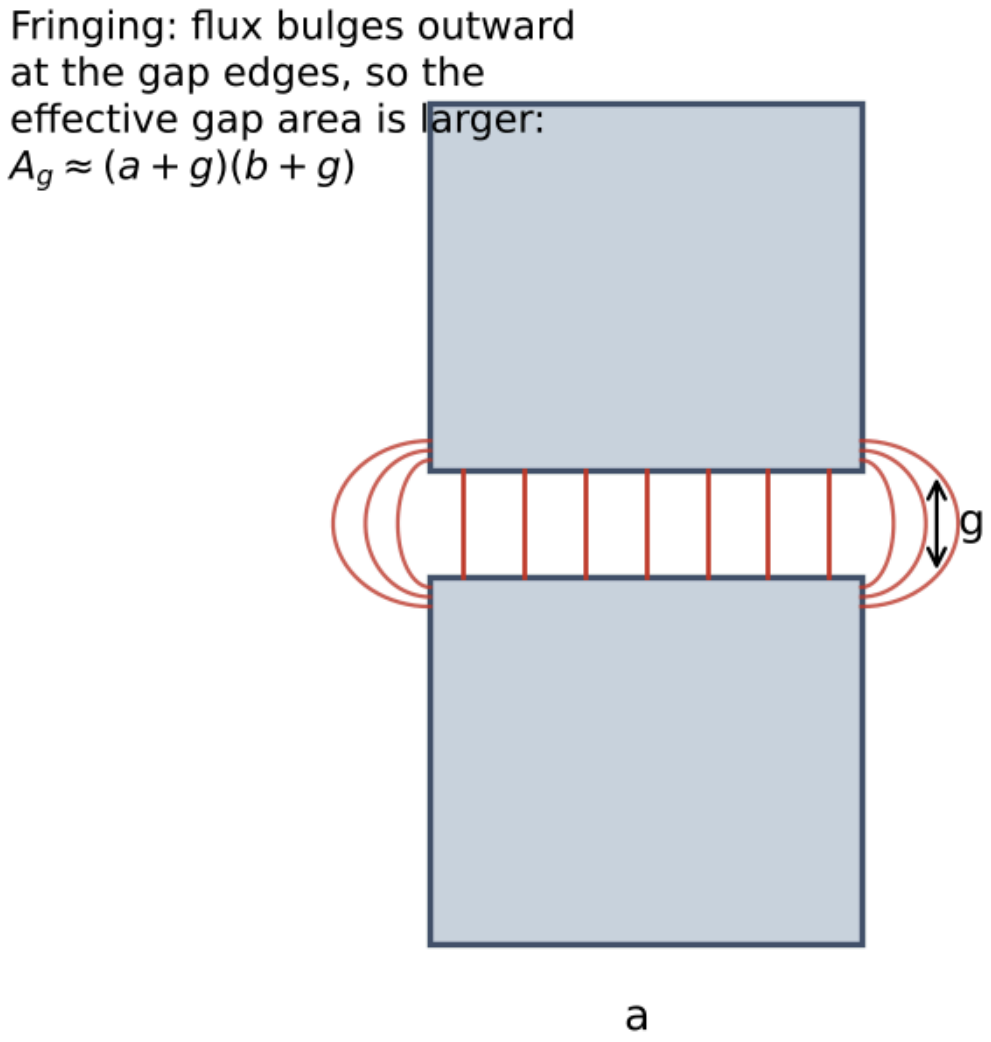


Figure 13.2. Fringing flux at an air gap between pole faces of width a (and depth b). Assistant schematic.

3.2 The empirical correction

For a rectangular pole face $a \times b$ and a gap g small compared with a and b, the usual first correction (used in Fitzgerald & Kingsley and many course texts) adds the gap length to each dimension:

$$A_g \approx (a+g)(b+g). \tag{13.3}$$

Some problems instead state a percentage increase (Sen Problem 1.12: "assume a 10 % increase in the effective cross-sectional area of the air gap"). Three remarks:

- The correction is **empirical**. It is reasonable for $g \lesssim a/10$; for gaps comparable to the pole width the field must be computed (finite elements) and no simple formula is accurate.
- Fringing **increases** flux for a given MMF, and **decreases** the gap flux density relative to the core (the same flux spread over a larger area).
- Splitting one long gap into several short gaps (MT I Fall 2014 Q1's "six identical air-gap portions") keeps each gap small relative to the pole face, which reduces fringing and the stray field near the winding — the exam's phrase "makes the magnetic field distribution within the circuit uniform".

Unless a problem asks for it, **neglect fringing** (LN4 slide 5 assumption 3) and say so.

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4. Leakage flux

4.1 Physical picture

Because air is not a magnetic insulator, part of the flux produced by a coil closes through the air around the coil instead of following the core (Figure 13.3). This **leakage flux** (kaçak akı) links the coil that produced it but not (all of) the other windings or the gap. The flux that follows the intended path and links all windings is the **main** or **mutual flux** (ana/ortak akı). LN8 slide 9 uses the same picture to introduce the transformer's leakage reactance: "leakage through air (linear B–H, no hysteresis)".

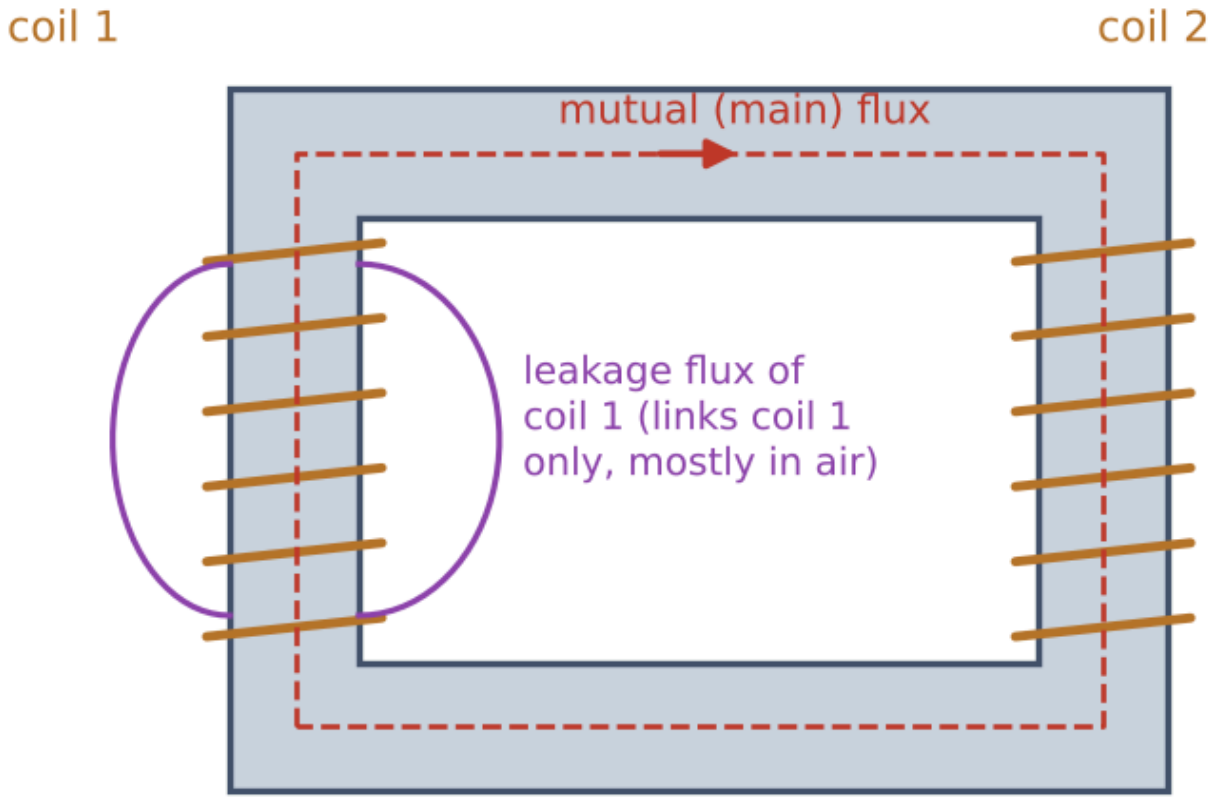


Figure 13.3. Mutual flux follows the core; leakage flux of coil 1 closes through the air near the coil and links coil 1 only.

4.2 A circuit model of leakage

Leakage can be represented by an air reluctance $\mathcal{R}_\ell$ placed **in parallel** across the MMF source of the coil (the leakage path starts and ends at the coil). If the rest of the circuit (iron + gap) has reluctance $\mathcal{R}_m$, the coil's MMF drives

$$\Phi_m = \frac{Ni}{\mathcal{R}_m}, \qquad \Phi_\ell = \frac{Ni}{\mathcal{R}_\ell}, \qquad \frac{\Phi_\ell}{\Phi_m + \Phi_\ell} = \frac{\mathcal{R}_m}{\mathcal{R}_m + \mathcal{R}_\ell}. \tag{13.4}$$

Two lessons follow from (13.4):

- **Leakage grows with the gap.** A larger gap raises $\mathcal{R}_m$, so a larger fraction of the coil's flux leaks (Example 13.8). Machines and gapped inductors have much more leakage than transformers.
- **Leakage is linear.** It flows mostly in air, so its reluctance is constant even when the core saturates. In circuit terms leakage becomes a constant series inductance $L_\ell = N^2/\mathcal{R}_\ell$ (Lessons 14 and 18).

How to reduce leakage (LN8 slides 10–11): put windings close together on the same limb (concentric windings, one over the other), use high-permeability cores, and avoid unnecessary gaps.

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5. Designing with gaps

Inductance target. With $\mu \to \infty$: $L = \mu_0 N^2 A/g$, so for a required L either $g = \mu_0 N^2 A/L$ (choose N) or $N = \sqrt{Lg/(\mu_0 A)}$ (choose g). With finite μ_r replace g by $g + l_c/\mu_r$.

Saturation current. The core saturates when $B = B_{sat}$: $NI_{sat} = B_{sat} A \mathcal{R}_{tot}$. For a fixed $L = N^2/\mathcal{R}_{tot}$, multiplying $\mathcal{R}_{tot}$ by k requires $N \to \sqrt{k}\,N$ and gives $I_{sat} \to \sqrt{k}\,I_{sat}$ and maximum energy $\tfrac{1}{2}LI_{sat}^2 \to k\times$. **A gap lets the same core store more energy before saturating** (Example 13.4).

Sensitivity. $L \propto 1/(g + l_c/\mu_r)$, so $\dfrac{\Delta L}{L} \approx -\dfrac{\Delta g}{g + l_c/\mu_r}$. A ±0.05 mm tolerance on a 0.63 mm gap changes L by about ∓6–7 % (Example 13.7).

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6. Common mistakes

- Using the core area for the gap when the problem gives fringing information (or applying fringing when told to neglect it).
- Forgetting there are **two** gaps when a C-core and a bar (crane magnet, relay plunger) are separated: flux crosses air twice.
- Subtracting g from the iron length but also using l_c elsewhere inconsistently; for $g \ll l_c$ either choice is fine, but be consistent.
- Treating leakage as "lost" flux that disappears: it still links its own coil and still stores energy.
- Assuming a bigger gap always means less inductance *and* more energy: at fixed N and fixed current, a bigger gap reduces L and stored energy; at fixed flux (fixed B) a bigger gap increases stored energy. State what is held constant (MT I Fall 2016 Q1 Part II).

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7. Worked examples

Example 13.1 — The LN3 core with a 3 mm air gap (LN3 slide 24)

Given. Core of Example 12.1 (μ_r = 4000, l = 500 mm, A = 2500 mm²) with a 3 mm gap (μ_r = 1 in the gap). **Find** the total reluctance and the gap's share.

Solution. Iron length $0.5 - 0.003 = 0.497$ m: $\mathcal{R}_c = 0.497/(4000\mu_0 \cdot 2.5 \times 10^{-3}) = 3.955 \times 10^4$ A/Wb. Gap: $\mathcal{R}_g = 0.003/(\mu_0 \cdot 2.5 \times 10^{-3}) = 9.549 \times 10^5$ A/Wb. Total 9.945×10^5 A/Wb, i.e. **25 times** the ungapped core; the gap holds **96 %** of it. **Interpretation.** The gap's equivalent iron length is $\mu_r g = 12$ m. If we neglected the iron entirely, $\mathcal{R}$ would be only 4 % low.

Example 13.2 — B and H with the gap (LN4 slide 6, gapped reading)

Given. The gapped core of Example 13.1 with $N = 5$, $I = 10$ A. **Find** Φ, B, H_c and H_g.

Solution. $\Phi = 50/9.945 \times 10^5 = 5.028 \times 10^{-5}$ Wb; $B = \Phi/A = 0.0201$ T (in both iron and gap, no fringing). $H_c = B/(4000\mu_0) = 4.0$ A/m; $H_g = B/\mu_0 = 1.60 \times 10^4$ A/m. MMF drops: gap $H_g g = 48.0$ At, iron $H_c l_c = 2.0$ At (sum 50 ✓). **Interpretation.** Compared with Example 12.2 (0.503 T, 100 A/m), the same 50 At now produce 25 times less flux. H in the gap is 4000 times H in the iron — the "local H is material-independent" idea of Lesson 10 clearly fails here, as noted there.

Example 13.3 — Effect of fringing on the same core

Given. Treat the core as a 50 mm × 50 mm square section (2500 mm²) and use $A_g = (a + g)(b + g)$. **Find** the change in flux.

Solution. $A_g = 53 \times 53 = 2809$ mm² (+12.4 %). $\mathcal{R}_g = 0.003/(\mu_0 \cdot 2.809 \times 10^{-3}) = 8.499 \times 10^5$; total 8.894×10^5 A/Wb. $\Phi = 50/8.894 \times 10^5 = 5.62 \times 10^{-5}$ Wb — **11.8 % more** than without fringing. Gap flux density $\Phi/A_g = 0.0200$ T, core flux density $\Phi/A_c = 0.0225$ T. **Interpretation.** Because the gap dominates, a 12 % change in gap area changes flux and inductance by almost 12 %. With a 3 mm gap on a 50 mm pole the condition $g \ll a$ is reasonably met; for larger gaps the correction becomes unreliable.

Example 13.4 — Gapped versus ungapped toroidal inductor (MT I Fall 2015 Q1)

Given. (exams/archive/MT/MT I - Fall2015.pdf pp. 1–2.) Toroid OD 20 mm, ID 10 mm, height 5 mm; $\mu_r = 250$ (linear region), $B_{sat} = 0.6$ T. Part I: ungapped inductor of 37.5 µH. Part II: a gap is cut so that the total reluctance is four times that of Part I; keep L = 37.5 µH. **Find** (a) the core reluctance; (b) N; (c) I_{max}; (d) the new N; (e) the new I_{max}; (f) compare maximum stored energies; (g, bonus) the gap length.

Solution. Mean path $l_e = \pi(15\text{ mm}) = 47.12$ mm; area $A_e = 5 \times 5 = 25$ mm².

a. $\mathcal{R} = 0.04712/(250\mu_0 \cdot 25 \times 10^{-6}) = 6.00 \times 10^6$ A/Wb.
b. $N = \sqrt{L\mathcal{R}} = \sqrt{37.5 \times 10^{-6} \times 6 \times 10^6} = 15$ turns.
c. $I_{max} = B_{sat} A_e \mathcal{R}/N = 0.6(25 \times 10^{-6})(6 \times 10^6)/15 = 6$ A.
d. $\mathcal{R}' = 24 \times 10^6 \Rightarrow N' = \sqrt{37.5 \times 10^{-6} \times 24 \times 10^6} = 30$ turns.
e. $I'_{max} = 0.6(25 \times 10^{-6})(24 \times 10^6)/30 = 12$ A.
f. $W = \frac{1}{2} L I_{max}^2$: 0.675 mJ ungapped, 2.70 mJ gapped — **four times more**.
g. Require $\dfrac{g}{\mu_0 A_e} + \dfrac{l_e - g}{250\mu_0 A_e} = 4\mathcal{R} \Rightarrow g\left(1 - \frac{1}{250}\right) = \dfrac{3 l_e}{250} \Rightarrow g = 0.568$ mm (0.565 mm if the iron length is not reduced). **Check.** In the gapped design the gap holds 75 % of the reluctance and hence 75 % of the stored energy. **Interpretation.** Same core, same inductance: the

gap doubles the usable current and quadruples the energy capacity, at the cost of twice the turns (twice the copper length, so more resistance).

Example 13.5 — Junkyard crane electromagnet (MT I Fall 2016 Q1)

Given. (exams/archive/MT/MT I - Fall2016.pdf pp. 2–4.) A C-core with N_t turns lifts an iron part across two air gaps of length g (flux crosses down and back up). Neglect leakage and fringing. Part I: $\mu \to \infty$; N_t = 500, I = 20 A DC, g = 0.5 cm, A_c = 100 cm², l_c = 90 cm, l_i = 60 cm. **Find** (a, b) total reluctance, Φ and B; (c) L; (d) energy densities. Part II: (a) gap halved at constant current — stored energy? (b) μ_r = 1000 instead of ∞ — how do B_{core} and H_{core} change? (c) the core saturates at 1.5 T; sketch B and H for 20 A rms at 50 Hz.

Solution. (a) Only the two gaps count: $\mathcal{R} = 2g/(\mu_0 A_c) = 0.01/(\mu_0 \cdot 0.01) = 7.958 \times 10^5$ A/Wb. (b) $\Phi = N_t I/\mathcal{R} = 10\,000/7.958 \times 10^5 = 1.257 \times 10^{-2}$ Wb; $B = \Phi/A_c = 1.257$ T. (c) $L = N_t^2/\mathcal{R} = 0.314$ H. (d) Air: $w = B^2/(2\mu_0) = 6.28 \times 10^5$ J/m³; core and iron part: $w = B^2/(2\mu) = 0$ for $\mu \to \infty$. Total energy = $w \times (2gA_c) = 62.8$ J $= \frac{1}{2}LI^2$ ✓.

Part II. (a) Halving g halves $\mathcal{R}$, doubles L; at constant current $W = \frac{1}{2}LI^2$ **doubles** (and B doubles to 2.5 T — physically the iron would saturate first; with $\mu \to \infty$ assumed, the arithmetic answer is "doubles"). (b) With μ_r = 1000 the iron adds $\mathcal{R}_{fe} = (0.9 + 0.6)/(1000\mu_0 \cdot 0.01) = 1.194 \times 10^5$ A/Wb; B_{core} falls from 1.257 T to 1.093 T, while H_{core} rises from 0 to $B/\mu = 870$ A/m. (c) Peak current $20\sqrt{2} = 28.3$ A would give 1.545 T in a linear core: above 1.5 T. The core saturates only near the current peaks: B follows the sinusoid until $|i| = 27.45$ A (from about 76° to 104° of each half-cycle) and is flattened at ≈1.5 T there, while H_{core} rises steeply in those intervals (a peaky H). Lesson 15 develops such waveforms.

Example 13.6 — E-core inductor with gaps in the outer legs (MT I Fall 2017 Q1)

Given. (exams/archive/MT/MT I - Fall2017.pdf pp. 2–4.) Core depth 5 cm; outer legs 5 cm wide (area 25 cm²), each with a 0.1 cm gap; centre leg 10 cm wide (area 50 cm²) with 300 turns; window 20 cm × 40 cm; $\mu \to \infty$. I = 1 A DC. **Find** (a) the equivalent circuit and total reluctance; (b) fluxes in each leg and gap flux density; (c) L; (d) energy densities; (e) the current if both gaps are halved at constant total flux; (f, bonus) N for the same L if μ_r = 2000.

Solution. (a) Each gap: $\mathcal{R}_g = 0.001/(\mu_0 \cdot 25 \times 10^{-4}) = 3.183 \times 10^5$ A/Wb. The centre-leg MMF source sees the two outer paths in parallel: $\mathcal{R}_{tot} = \mathcal{R}_g/2 = 1.592 \times 10^5$ A/Wb.

b. $\Phi_{centre} = 300/1.592 \times 10^5 = 1.885 \times 10^{-3}$ Wb; each outer leg 9.42×10^{-4} Wb; $B_g = 9.42 \times 10^{-4}/25 \times 10^{-4} = 0.377$ T.
c. $L = N^2/\mathcal{R}_{tot} = 0.565$ H.
d. Each gap $w = B_g^2/(2\mu_0) = 5.65 \times 10^4$ J/m³; iron legs 0 ($\mu \to \infty$).
e. Halving each gap halves each $\mathcal{R}_g$, so $\mathcal{R}_{tot}$ halves. With Φ constant, $I = \Phi\mathcal{R}_{tot}/N$ halves: I_{new} **= 0.5 A**. The handwritten key writes 0.25 A by comparing the new total $\mathcal{R}_g/4$ with a quarter of the old total — an arithmetic slip (errata E11).
f. Mean paths: each outer loop path (yoke + outer leg + yoke) ≈ 100 cm at 25 cm², centre leg 45 cm at 50 cm²: $\mathcal{R}_{outer,fe} = 1.592 \times 10^5$, $\mathcal{R}_{centre,fe} = 0.358 \times 10^5$ A/Wb. $\mathcal{R}_{tot} = 0.358 \times 10^5 + (1.592 \times 10^5 + 3.183 \times 10^5)/2 = 2.745 \times 10^5$ A/Wb. For $L = 0.565$ H: $N = \sqrt{0.565 \times 2.745 \times 10^5} = 394$ turns (key 393.9). With finite μ and the original current, flux falls, so B_{core} decreases while H_{core} becomes non-zero.

Example 13.7 — Designing the gap for a target inductance (Sen Problem 1.13) and sensitivity

Given. Core with $A_c = A_g = 5$ cm², $\mu \to \infty$, $N = 100$. **Find** g for L = 10 mH. Then, for an iron path of l_c = 0.2 m with μ_r = 2000, find the effect of a ±0.05 mm gap tolerance and compare with no gap at all.

Solution. $g = \mu_0 N^2 A/L = 4\pi \times 10^{-7}(10^4)(5 \times 10^{-4})/0.01 = 0.628$ mm. With finite iron, $L \propto 1/(g + l_c/\mu_r)$ where $l_c/\mu_r = 0.1$ mm. A gap 0.05 mm shorter raises L by 7.4 %; 0.05 mm longer lowers it by 6.4 %. Removing the gap entirely would raise L by a factor $(0.628 + 0.1)/0.1 = 7.3$ — and make it proportional to μ_r, which varies with temperature, flux level and batch. **Interpretation.** Gapped inductors trade some inductance per turn for predictability.

Example 13.8 — How the gap increases leakage (original)

Given. A coil sees a main path of reluctance $\mathcal{R}_m$ (iron + gap) and a leakage path $\mathcal{R}_\ell = 8 \times 10^6$ A/Wb in parallel. **Find** the leakage fraction $\Phi_\ell/(\Phi_m + \Phi_\ell)$ for $\mathcal{R}_m$ = 0.4, 1.59 and 6.4 ×10⁶ A/Wb (gap quadrupled each step).

Solution. From (13.4): 4.8 %, 16.6 %, 44.4 %. **Interpretation.** The leakage path does not change, but as the gap grows, the main path becomes comparable to it and a growing share of the coil's flux bypasses the gap. This is why machine windings (large gaps) have much larger leakage reactances than transformer windings.

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8. Optional practice

P13.1. The relay of Sen Example 1.4: N = 400, l_c = 50 cm, g = 1 mm, $A_c = A_g$ = 15 cm², μ_r = 3000, i = 1 A. Find B_g and L, and the percentage error of the $\mu \to \infty$ approximation.

▼ **Solution**
$\mathcal{R}_c = 0.5/(3000\mu_0 \cdot 0.0015) = 8.842 \times 10^4$, $\mathcal{R}_g = 0.001/(\mu_0 \cdot 0.0015) = 5.305 \times 10^5$ A/Wb. $\Phi = 400/6.189 \times 10^5 = 6.463 \times 10^{-4}$ Wb, $B_g = 0.431$ T, $L = N^2/\mathcal{R} = 0.2585$ H (Sen: 258.5 mH). With $\mu \to \infty$: $L = 0.3016$ H, 16.7 % high — the 0.5 m iron path is not negligible against a 1 mm gap when μ_r is only 3000 (l_c/μ_r = 0.167 mm).

P13.2. In Example 13.5 (crane magnet, $\mu \to \infty$), what current would give the same 1.257 T if one gap is closed completely (part rests on one pole)?

▼ **Solution**
Only one 0.5 cm gap remains: $\mathcal{R}$ halves, so I halves to 10 A for the same flux. (In practice an iron part resting on a pole face still has a small residual gap from surface roughness.)

P13.3. A gapped inductor stores 2.7 mJ at saturation. If the gap is doubled (keeping N), by what factor do L, I_{sat} and W_{max} change (neglect the iron)?

▼ **Solution**
$\mathcal{R}$ doubles: L halves; $I_{sat} = B_{sat} A \mathcal{R}/N$ doubles; W_{max} doubles, because $\frac{1}{2}(L/2)(2I_{sat})^2 = 2 \times \frac{1}{2} L I_{sat}^2$. Energy capacity grows with gap volume, because at saturation the gap stores $B_{sat}^2/(2\mu_0)$ per unit volume.

P13.4 (concept, MT I Fall 2019 Q2(g)). Is an air gap desirable in a transformer core? Answer in terms of the equivalent circuit.

▼ **Solution**
No. A gap increases the core reluctance, which lowers the magnetizing inductance $L_m = N^2/\mathcal{R}$ (magnetizing reactance X_m in Lesson 18). The no-load magnetizing current rises, power factor at light load worsens, and leakage increases (Section 4.2), giving larger leakage reactances and worse regulation. Transformers therefore use gap-free (or interleaved-joint) cores.

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9. Summary

Relation	Meaning	Validity
$\mathcal{R}_g = g/(\mu_0 A_g)$	Gap reluctance	Uniform gap field
$l_{fe,equiv} = \mu_r g$	Iron length with the same reluctance as the gap	Same area
$B \approx \mu_0 Ni/g$, $L \approx \mu_0 N^2 A/g$	Gap-dominated circuit	$g \gg l_c/\mu_r$
$A_g \approx (a+g)(b+g)$	Fringing correction	$g \ll a, b$; empirical
$\Phi_\ell/\Phi_{tot} = \mathcal{R}_m/(\mathcal{R}_m + \mathcal{R}_\ell)$	Leakage share	Leakage path across the coil
$I_{sat} = B_{sat} A\mathcal{R}/N$	Saturation current	Uniform B

10. Vocabulary

English	Türkçe
air gap	hava aralığı
fringing (effect)	saçaklanma (etkisi)
leakage flux	kaçak akı
main / mutual flux	ana / ortak akı
effective area	etkin alan
gapped inductor	hava aralıklı endüktör
saturation current	doyma akımı
tolerance	tolerans

11. Sources

- LN3 viewer pp. 39–42 (slides 13–14), p. 62 (slide 24). LN4 viewer pp. 10–12 (slides 5–6). LN8 viewer pp. 21–26 (slides 9–12). Notes: LN03, LN04, LN08.
- Sen §1.1.5 pp. 5–7 (viewer 29–31), Fig. 1.10 fringing; Example 1.4 pp. 13–14; Problems 1.12–1.13 p. 35 (viewer 59).
- MT I Fall 2015 Q1 (pp. 1–2, no key supplied); MT I Fall 2016 Q1 (pp. 2–4, no key for Q1 supplied); MT I Fall 2017 Q1 with handwritten key (pp. 2–4); MT I Fall 2019 Q2(g) (p. 5).
- Verification: `lesson13_air_gaps.py`.

- More solved archive problems: Ch II–III archive collection §3 (MT I Fall 2019 Q2(e): ungapped inductor design).

Lesson 14 — Flux linkage, self- and mutual inductance, coupled circuits and magnetic energy

Chapter II: Magnetic Circuits · Week 5 · Lesson 14 of 31

Assistant-written teaching note. Teacher sources: LN4 slide 15; LN6 slides 3–9; LN7 slides 19–20 (preview); LN12 slides 13–15; LN14 slides 10–13. Book: Sen §1.1.6 (printed pp. 13–15, viewer 37–39), §3.2 field energy (pp. 94–99, viewer 118–123; used for the energy derivation). Exams: 2025 MT1 Q4(b); 2021 MT Q2(c,d); MT I Fall 2016 Q1(d), Fall 2017 Q1(d). Numbers: `lesson14_inductance_energy.py`.

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1. Place and goals

So far a magnetic circuit has been a calculation that turns current into flux. Circuits, however, are described by voltages and currents. **Inductance** (endüktans) is the bridge: it packs the whole magnetic circuit into one number that relates a coil's flux linkage to its current. **Mutual inductance** (karşılıklı endüktans) does the same for two coils sharing flux — the physics of every transformer (Chapter III). And because building up a magnetic field takes energy, this is also where we learn how much energy a magnetic circuit stores and *where* it stores it — the foundation of force and torque calculation in Chapter IV.

After this lesson you should be able to:

1. Define self-inductance from flux linkage, derive $L = N^2/\mathcal{R}$, and explain why $L \propto N^2$.
2. Define mutual inductance, compute it from a reluctance network, prove reciprocity, and define the coupling coefficient k.
3. Use the dot convention to write the terminal equations of coupled coils, and find the equivalent inductance of series and parallel connections.
4. Compute stored magnetic energy from $\int i\,d\lambda$, from $\frac{1}{2}Li^2$ when linear, and from energy density $B^2/(2\mu)$; know which formula holds when.

Prerequisites: Lessons 10–13. The coil-voltage relation $v = d\lambda/dt$ is used here in its simplest form and derived carefully in Lesson 15.

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2. Self-inductance

2.1 Definition

LN6 slide 4 defines inductance as "the flux linkage created per ampere":

$$L = \frac{d\lambda}{di} \quad \text{(incremental)}, \qquad L = \frac{\lambda}{i} \quad \text{(secant; equal to the incremental value for a linear coil)}.$$

For a linear coil (air core, or iron in its linear region, or a gap-dominated core) $\lambda = Li$ with constant L, and the two definitions coincide (LN6 slide 5). Units: Wb/A = henry (H).

2.2 Inductance from the magnetic circuit

For an N-turn coil on a core of reluctance $\mathcal{R}$ (as seen by the coil), with every turn linking the same flux:

$$\Phi = \frac{Ni}{\mathcal{R}}, \qquad \lambda = N\Phi = \frac{N^2}{\mathcal{R}}\, i \qquad \Longrightarrow \qquad \boxed{L = \frac{N^2}{\mathcal{R}} = \frac{N^2 \mu A}{l}} \tag{14.1}$$

(LN6 slide 6; Sen Eqs. 1.24–1.25). **Why N^2?** Doubling the turns doubles the MMF, so the flux doubles; and each unit of flux is now linked by twice as many turns, so flux linkage quadruples (LN6 slide 5's question; Example 14.1).

2.3 Voltage of an inductor

With the passive sign convention (current entering the + terminal) and a coil of negligible resistance, Faraday's law (Lesson 15) gives $v = d\lambda/dt$. For a linear coil, $\lambda = Li$ with L constant, so

$$v = N\frac{d\Phi}{dt} = N\frac{d\Phi}{di}\frac{di}{dt} = L\frac{di}{dt} \qquad \text{(LN6 slide 3)}.$$

If L depends on current (saturation) the correct form is $v = (d\lambda/di)(di/dt) = L_{inc}\, di/dt$; if L changes because something *moves* ($L(x)$), there is an extra term $i\,(dL/dx)(dx/dt)$ — the speed voltage of Chapter IV.

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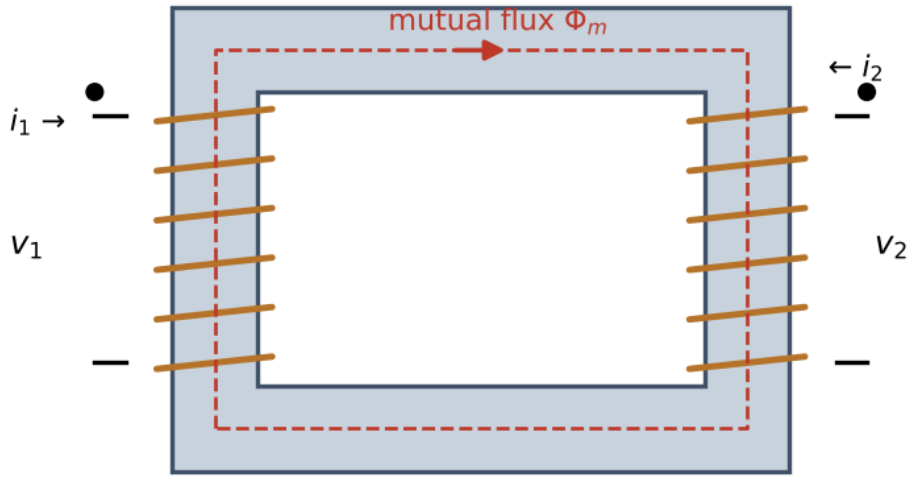
3. Mutual inductance

3.1 Definition

Two coils share part of their flux (LN6 slides 8–9, Figure 14.1). Let Φ_{21} be the flux produced by i_1 that links coil 2. The **mutual inductance** is

$$M_{21} = \frac{\lambda_{21}}{i_1} = \frac{N_2\Phi_{21}}{i_1}, \qquad M_{12} = \frac{N_1\Phi_{12}}{i_2}.$$

LN6 slide 9 writes the consequence $V_2 = M\, dI_1/dt$: a changing current in coil 1 induces a voltage in coil 2 even with no electrical connection — "the same logic" behind wireless chargers and transformers (LN7 slides 17–20).



Dots: currents entering both dotted terminals drive mutual flux the same way (aiding)

Figure 14.1. Two coils on a common core. Dots mark the terminals into which current produces mutual flux in the same direction.

3.2 Mutual inductance from a reluctance network

Treat coil 1 as the only source ($i_2 = 0$). Solve the network for the flux in the limb carrying coil 2, Φ_{21}; then $M = N_2\Phi_{21}/i_1$. For the common three-limb case with coil 1 and coil 2 on the outer legs and a third (centre) branch of reluctance $\mathcal{R}_c$:

$$\Phi_1 = \frac{N_1 i_1}{\mathcal{R}_1 + \mathcal{R}_2 \parallel \mathcal{R}_c}, \qquad \Phi_{21} = \Phi_1 \frac{\mathcal{R}_c}{\mathcal{R}_2 + \mathcal{R}_c}, \qquad M = \frac{N_1 N_2}{\mathcal{R}_{eq,1}} \cdot \frac{\mathcal{R}_c}{\mathcal{R}_2 + \mathcal{R}_c} \tag{14.2}$$

— exactly the expression in the 2025 Q4(b) key. Flux that returns through the centre limb links coil 1 but not coil 2: from coil 2's point of view it is *leakage*.

3.3 Reciprocity: $M_{12} = M_{21}$

Is the flux linkage of coil 2 per ampere of coil 1 always equal to that of coil 1 per ampere of coil 2? For linear magnetic circuits, yes. Algebraically, (14.2) and its mirror image give the same number for any network (Example 14.2 checks this numerically). Physically it follows from energy: establish i_1 first and then i_2, or the other way round; the stored energy must be the same final value, and that forces $M_{12} = M_{21}$ (Section 6.3). We therefore write simply M.

3.4 Coupling coefficient

$$k = \frac{|M|}{\sqrt{L_1 L_2}}, \qquad 0 \le k \le 1. \tag{14.3}$$

$k = 1$ means perfect coupling: all the flux of each coil links the other. Leakage makes $k < 1$. Typical values: power transformers 0.99+; gapped two-winding inductors 0.6–0.95; loosely coupled wireless-charging pads 0.1–0.5. The limit $k \le 1$ is a consequence of stored energy being non-negative (Section 6.3).

For two coils on a single core path with no leakage, $\mathcal{R}$ common: $L_1 = N_1^2/\mathcal{R}$, $L_2 = N_2^2/\mathcal{R}$, $M = N_1 N_2/\mathcal{R}$, so $M = \sqrt{L_1 L_2}$ and $k = 1$.

---

4. Coupled-circuit equations and the dot convention

4.1 Terminal equations

For two coils with resistances neglected, linear, both currents defined *into* the dotted terminals and voltages positive at the dots:

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}, \qquad v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}. \tag{14.4}$$

LN14 slide 10 asks "write down the voltage equation of inductor 2": that is the second line. With coil 2 open ($i_2 = 0$), $v_2 = M\, di_1/dt$ (LN6 slide 9).

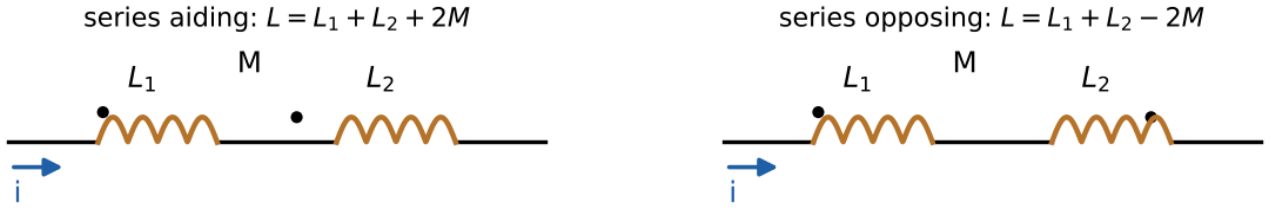
4.2 What the dots mean (and how to place them)

Dot convention (nokta kuralı): current entering a dotted terminal of one coil produces flux in the core in the same direction as current entering the dotted terminal of the other coil. If one current enters a dot and the other leaves a dot, the mutual term enters with a minus sign.

To place dots from a drawing: (1) pick a terminal on coil 1 and inject current there; (2) find the core flux direction with the right-hand rule; (3) on coil 2, find the terminal into which current would drive flux the *same* way; dot both chosen terminals. LN7 slide 8: “Beware of dots” — a dot error flips the sign of M and turns aiding into opposing.

4.3 Series and parallel connections

Series aiding (current enters both dots): $\lambda = (L_1 + M)i + (M + L_2)i$, so $L_{aid} = L_1 + L_2 + 2M$. **Series opposing**: $L_{opp} = L_1 + L_2 - 2M$ (Figure 14.2). Measuring both gives $M = (L_{aid} - L_{opp})/4$ — a standard laboratory method (Example 14.3).

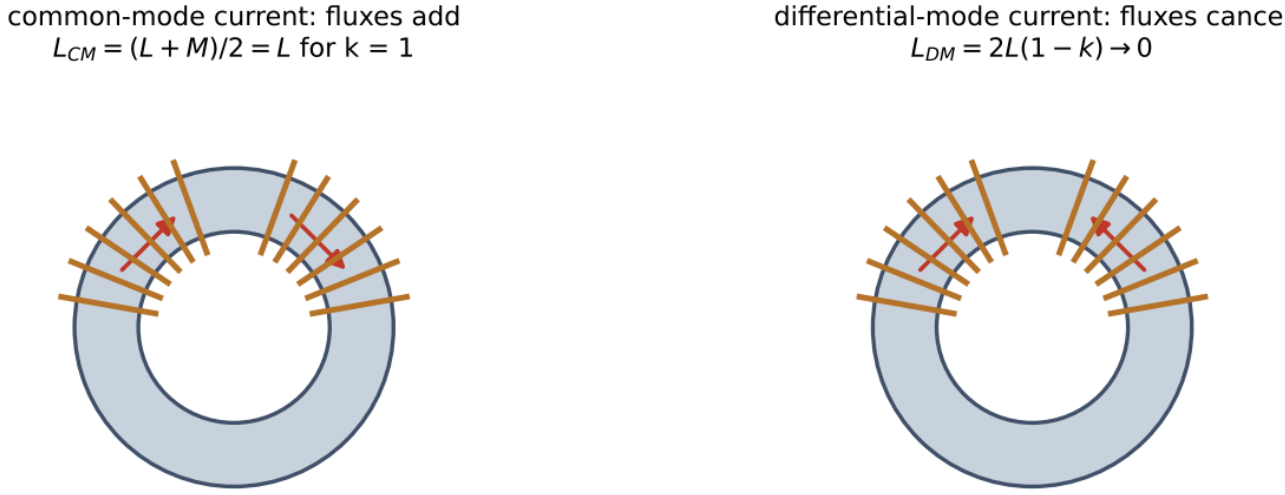


Parallel aiding (both coils between the same two nodes, currents into the dots): from $v = L_1\dot{i}_1 + M\dot{i}_2 = M\dot{i}_1 + L_2\dot{i}_2$ and $i = i_1 + i_2$,

$$L_{par} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \qquad (\text{for } L_1 = L_2 = L,\ M = kL:\ L_{par} = \tfrac{L(1+k)}{2}). \tag{14.5}$$

4.4 Application: the common-mode choke

A common-mode choke (ortak mod bobini) has two identical windings on one toroid. Common-mode noise currents flow in the *same* direction in both conductors; the load (differential) current flows out on one and back on the other. In the CM case the two windings act in parallel, aiding: $L_{CM} = L(1 + k)/2 \to L$ for $k = 1$. In the DM case the loop current passes through the windings in series, opposing: $L_{DM} = 2L(1 - k) \to 0$. The choke therefore blocks high-frequency CM noise while leaving the power current unaffected (2021 MT Q2(c,d); Figure 14.3). With realistic $k < 1$, the DM inductance is the small leakage inductance $2L(1 - k)$.



5. Energy stored in a magnetic field

5.1 Derivation from electrical input

Take a lossless coil (no resistance, no core loss) and nothing moving. The power entering its terminals is $p = vi = i\,d\lambda/dt$. Integrating from an unexcited state (λ = 0) to flux linkage λ:

$$W_f = \int p\,dt = \int_0^{\lambda} i(\lambda')\,d\lambda' \tag{14.6}$$

(LN12 slide 14; Sen Eq. 3.9). Geometrically, W_f is the area **between the λ–i curve and the λ axis** (Figure 14.4). The area between the curve and the i axis is the **coenergy** $W' = \int_0^i \lambda\,di$; together they fill the rectangle: $W + W' = \lambda i$ (LN12 slide 15). Coenergy is the tool for force calculation in Lesson 22.

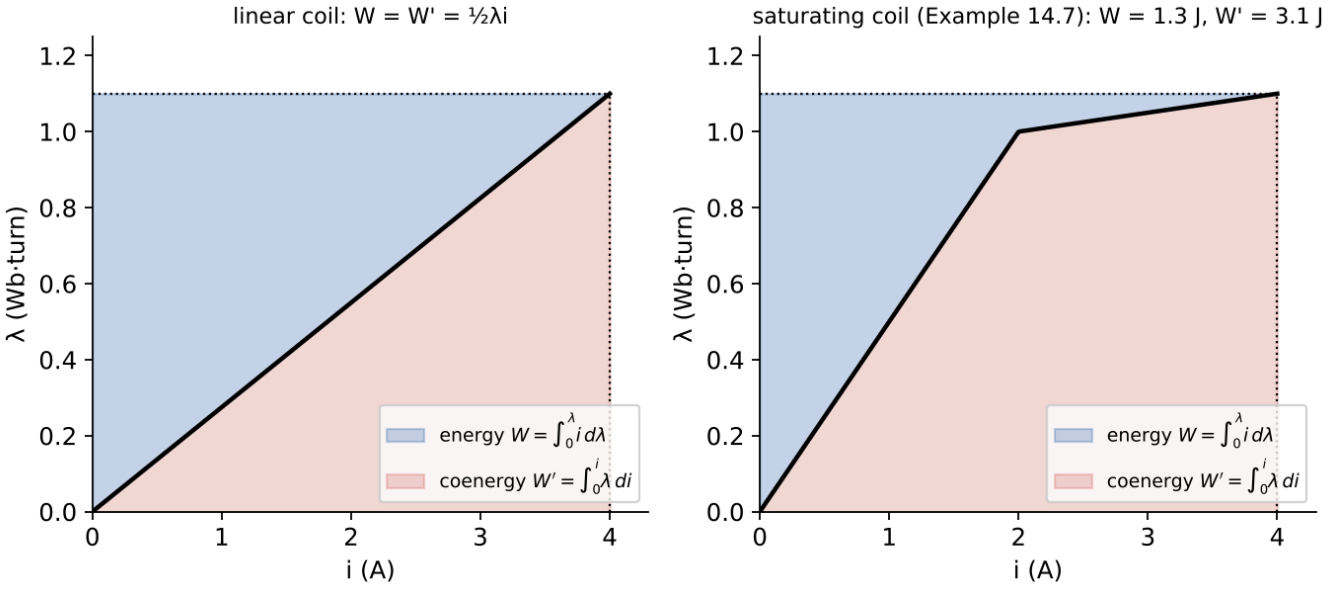


Figure 14.4. Energy (blue) and coenergy (red) areas for a linear coil (left) and for the saturating coil of Example 14.7 (right).

5.2 Linear case

If $\lambda = Li$ with constant L: $W_f = \int_0^{\lambda} (\lambda'/L)\,d\lambda' = \lambda^2/(2L)$, so

$$W_f = \tfrac{1}{2}Li^2 = \tfrac{1}{2}\lambda i = \frac{\lambda^2}{2L} \qquad \text{(only for linear coils; LN12 slide 15).} \tag{14.7}$$

In terms of the magnetic circuit, $W_f = \tfrac{1}{2}\mathcal{F}\Phi = \tfrac{1}{2}\mathcal{R}\Phi^2$, and for a network $W_f = \tfrac{1}{2}\sum_k \mathcal{R}_k \Phi_k^2$ — each reluctance stores energy in proportion to its reluctance for a given flux.

5.3 Energy density and *where* the energy is

Insert $\lambda = N\Phi = NBA$ and $i = Hl/N$ into (14.6) for a uniform section: $d\lambda = NA\,dB$ and

$$W_f = \int_0^{B} \frac{Hl}{N} NA\,dB' = (Al)\int_0^{B} H\,dB' \quad\Longrightarrow\quad w_f = \int_0^{B} H\,dB \;\; [\mathrm{J/m^3}] \tag{14.8}$$

(LN12 slide 14 “from the B–H curve”; Sen Eq. 1.29; Sen Eqs. 3.15–3.18 split the total into core and gap parts). For a linear material $w_f = \dfrac{B^2}{2\mu} = \tfrac{1}{2}BH = \tfrac{1}{2}\mu H^2$ (glossary: magnetic energy density, manyetik enerji yoğunluğu).

Because $w_f = B^2/(2\mu)$, **for the same flux density the air gap stores μ_r times more energy per cubic metre than the iron.** In Example 13.4 the gap holds 75 % of the energy of the inductor, in Example 12.7 83 %. With $\mu \to \infty$ the iron stores nothing (MT 2016 Q1(d), MT 2017 Q1(d)).

5.4 Two coupled coils

LN14 slide 11 warns: “the stored energy in coil 1 is $\tfrac{1}{2}L_1 i_1^2$” is **not correct** when another current is present, because $\lambda_1 = L_1 i_1 + M i_2$. Integrating $dW = i_1 d\lambda_1 + i_2 d\lambda_2$ along any path to (i_1, i_2) (LN14 slides 12–13):

$$W_f = \tfrac{1}{2}L_1 i_1^2 + \tfrac{1}{2}L_2 i_2^2 + M i_1 i_2 = \tfrac{1}{2}\begin{bmatrix} i_1 & i_2 \end{bmatrix}\begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}. \tag{14.9}$$

5.5 Nonlinear coils

With saturation, use the areas of (14.6). The formulas $\tfrac{1}{2}Li^2$ and $\tfrac{1}{2}\lambda i$ are then *wrong*: Example 14.7 shows $\tfrac{1}{2}\lambda i$ = 2.2 J against the true 1.3 J. Energy and coenergy are no longer equal.

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6. Two short proofs worth knowing

6.1 Why $L = N^2/\mathcal{R}$ also equals $2W_f/i^2$

From $W_f = \tfrac{1}{2}\mathcal{R}\Phi^2$ and $\Phi = Ni/\mathcal{R}$: $W_f = \tfrac{1}{2}N^2 i^2/\mathcal{R} = \tfrac{1}{2}Li^2$. Consistent definitions.

6.2 Reciprocity

Bring i_1 to its final value with $i_2 = 0$: energy $\tfrac{1}{2}L_1 i_1^2$. Then raise i_2 with i_1 fixed: coil 2 receives $\int i_2\, L_2 di_2 = \tfrac{1}{2}L_2 i_2^2$ and coil 1 receives $\int i_1\, d\lambda_1 = i_1 M_{12} i_2$. Total: $\tfrac{1}{2}L_1 i_1^2 + \tfrac{1}{2}L_2 i_2^2 + M_{12} i_1 i_2$. Doing it in the opposite order gives the same with M_{21}. The final stored energy depends only on the final state, so $M_{12} = M_{21}$.

6.3 Why $k \le 1$

W_f in (14.9) must be ≥ 0 for every (i_1, i_2) (a passive structure cannot deliver energy it never received). A quadratic form is non-negative for all inputs only if its matrix is positive semidefinite: $L_1 \ge 0$, $L_2 \ge 0$, $L_1 L_2 - M^2 \ge 0$. Hence $|M| \le \sqrt{L_1 L_2}$ and $k \le 1$.

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7. Methods and common mistakes

- **Find L:** reluctance seen by the coil → $L = N^2/\mathcal{R}_{seen}$. **Find M:** excite one coil, find the flux in the other coil’s limb, multiply by the other coil’s turns.
- **Sign of M** depends on reference directions (dots), not on physics: always state the dot positions or flux references before giving a signed M.
- **Energy:** use $\tfrac{1}{2}Li^2$ only if linear; otherwise integrate $i\,d\lambda$; in coupled coils include Mi_1i_2.
- **Do not double count**: $\tfrac{1}{2}\sum \mathcal{R}_k \Phi_k^2$ already includes every section.

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8. Worked examples

Example 14.1 — Doubling the turns (LN6 slide 5)

Given. A coil of 100 turns on a core whose reluctance is 2×10^5 A/Wb. **Find** L, and L with 200 turns in the same window. **Solution.** $L = 100^2/2 \times 10^5 = 0.05$ H; with 200 turns 0.2 H. **Interpretation.** ×4, because flux ×2 and turns linking it ×2. (If the window is full, doubling turns halves the wire area, so resistance rises by ×4 as well: inductance per ohm stays constant — a useful design fact.)

Example 14.2 — Self- and mutual inductances of the 2025 MT1 Q4 core, part (b)

Given. Reluctances of Example 12.8: $\mathcal{R}_1 = \mathcal{R}_2 = 1.061 \times 10^6$, centre branch $\mathcal{R}_c = \mathcal{R}_3 + \mathcal{R}_g = 0.211 \times 10^6 + 1.592 \times 10^6 = 1.803 \times 10^6$ A/Wb; $N_1 = 400$, $N_2 = 800$; linear core. **Find** L_1, L_2, M_{12}, M_{21} and k.

Solution. From coil 1: $\mathcal{R}_{eq,1} = \mathcal{R}_1 + \mathcal{R}_2 \parallel \mathcal{R}_c = 1.729 \times 10^6$ A/Wb ⇒ $L_1 = 400^2/\mathcal{R}_{eq,1} = 92.5$ mH. By symmetry $\mathcal{R}_{eq,2} = \mathcal{R}_{eq,1}$ ⇒ $L_2 = 800^2/\mathcal{R}_{eq,2} = 370.2$ mH. From (14.2): $M_{12} = \dfrac{400 \times 800}{1.729 \times 10^6} \cdot \dfrac{1.803}{1.061 + 1.803} = 116.5$ mH; the mirror calculation gives M_{21} = 116.5 mH (reciprocity). $k = 116.5/\sqrt{92.5 \times 370.2} = 0.63$.

Key values: 92.59, 370.37, 116.365 mH — the 0.1 % differences come from the key's centre-limb length (errata E8).

Sign. If both coils are referenced so that their MMFs push flux into the centre limb from the top (as drawn in Figure 12.4), coil 1's flux passes *down* the right leg, against coil 2's reference, and $M = -116.5$ mH. Referencing coil 2 the other way (dot on the other terminal) gives $M = +116.5$ mH. The exam asks only for the magnitude.

Checks. The code rebuilds λ_1 and λ_2 by nodal analysis of the network for unit currents and reproduces L_1, L_2 and $\pm M$; energy at $i_1 = 1.2$ A, $i_2 = 0.3$ A from (14.9), 0.1252 J, equals $\frac{1}{2}\sum \mathcal{R}_k \Phi_k^2$ from the network. **Interpretation.** k is only 0.63 because the gapped centre limb returns 37 % of coil 1's flux before it reaches coil 2. Part (d) of the exam removes this path ("ignore the centre leg"), making $k = 1$ (Lesson 15).

Extension. With coil 2 open and $i_1 = 1.2\sin(2\pi 50 t)$ A, $v_2 = M\, di_1/dt$ has peak $\omega M I_{1m} = 314.16 \times 0.1165 \times 1.2 = 43.9$ V.

Example 14.3 — Measuring M with series connections (original)

Given. Two coils measure $L_{aid} = 0.9$ H in series aiding and $L_{opp} = 0.3$ H in series opposing; coil 1 alone is 0.4 H. **Find** M, L_2, k. **Solution.** $M = (0.9 - 0.3)/4 = 0.15$ H; $L_2 = 0.9 - 0.4 - 2(0.15) = 0.2$ H (check: $0.4 + 0.2 - 0.3 = 0.3$ ✓); $k = 0.15/\sqrt{0.08} = 0.53$.

Example 14.4 — Common-mode choke (2021 MT Q2 Part II)

Given. (exams/archive/MT/2021_Midterm_Q2_solution.pdf p. 6.) Two windings with $L_1 = L_2 = 12.5$ mH on one toroid. **Find** (c) the equivalent inductance and impedance at 400 kHz for common-mode current; (d) the same for differential-mode current; comment.

Solution. (c) CM: windings in parallel, currents into the dots, perfect coupling ($M = L$): $L_{CM} = (L + M)/2 = 12.5$ mH; $|Z| = 2\pi(400 \times 10^3)(12.5 \times 10^{-3}) = 31.4$ kΩ. The key reaches the same value from the magnetic circuit: flux $2NI/\mathcal{R}$ from current $2I$, so $L = N^2/\mathcal{R}$.

d. DM: the current goes out through one winding and back through the other; MMFs cancel, $\Phi = 0$, $L_{DM} = 2L - 2M = 0$, $Z = 0$. **Extension (realistic coupling).** With $k = 0.99$: $L_{CM} =$ 12.44 mH (31.3 kΩ), $L_{DM} = 2L(1-k) = 0.25$ mH (628 Ω at 400 kHz — but only 0.08 Ω at 50 Hz, so the mains current is unaffected). **Interpretation.** The choke is a frequency-and-mode selective filter: high impedance to CM noise, almost none to the power current — and because DM MMFs cancel, the core does not saturate under load current.

Example 14.5 — Where the energy is (MT I Fall 2017 Q1(d))

Given. The gapped E-core of Example 13.6 at I = 1 A (L = 0.565 H, B_g = 0.377 T in each 25 cm² × 0.1 cm gap, iron $\mu \to \infty$). **Find** stored energy, energy density in the gaps and in iron. **Solution.** $W = \frac{1}{2}LI^2 = 0.283$ J. Gap: $w = B_g^2/(2\mu_0) = 5.65 \times 10^4$ J/m³; total gap volume $2 \times 25 \times 10^{-4} \times 10^{-3} = 5 \times 10^{-6}$ m³ ⇒ $W = 0.283$ J ✓. Iron: $w = 0$ for $\mu \to \infty$; with $\mu_r = 2000$ and the same B it would be 28 J/m³, 2000 times less.

Example 14.6 — Energy of the six-gap core (MT I Fall 2014 Q1(b) revisited)

From Example 12.7: steel 0.279 J, gaps 1.398 J, total 1.676 J $= \frac{1}{2}LI^2 = \frac{1}{2}(0.0335)(10)^2$ ✓. Even though the steel is near its knee, 83 % of the energy is in 1.8 mm of air.

Example 14.7 — Energy and coenergy of a saturating coil (original)

Given. A coil's λ–i characteristic is two straight segments: $\lambda = 0.5i$ for $i \le 2$ A (λ ≤ 1 Wb·t), then $\lambda = 1 + 0.05(i-2)$. **Find** W, W' at $i = 4$ A and compare with $\frac{1}{2}\lambda i$. **Solution.** At 4 A, $\lambda = 1.1$ Wb·t. Energy: $\int_0^1 \frac{\lambda}{0.5} d\lambda + \int_1^{1.1} \left[2 + \frac{\lambda - 1}{0.05}\right] d\lambda = 1.0 + (0.2 + 0.1) = 1.3$ J. Coenergy: $W' = \lambda i - W = 4.4 - 1.3 = 3.1$ J (direct integration of $\lambda\, di$ agrees). The linear formula would give $\frac{1}{2}\lambda i = 2.2$ J — wrong. **Interpretation.** In saturation, extra current adds little flux linkage and therefore little stored energy; coenergy grows quickly. Lesson 22 uses exactly this asymmetry.

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9. Optional practice

P14.1 (dots). In Figure 14.1, coil 1's current enters its upper terminal and produces clockwise flux. Coil 2 is wound so that current entering its *lower* terminal produces clockwise flux. Where is coil 2's dot, and what is the sign in $v_2 = L_2\, di_2/dt \;?\; M\, di_1/dt$ if i_2 is defined into the upper terminal?

▼ **Solution**
Coil 2's dot is on the lower terminal (it must correspond to coil 1's upper, dotted terminal). With i_2 defined into the undotted upper terminal, one current enters a dot and the other leaves a dot, so the mutual term is negative: $v_2 = L_2\, di_2/dt - M\, di_1/dt$ (with v_2 positive at the upper terminal).

P14.2. A 0.5 mm gap with area 4 cm² carries 0.8 T. How much energy does it store? How long would the iron path (μ_r = 3000, same area, same B) have to be to store the same energy?

▼ **Solution**
$w = 0.8^2/(2\mu_0) = 2.546 \times 10^5$ J/m³; volume 2×10^{-7} m³ ⇒ $W = 50.9$ mJ. Iron stores $w/3000$, so it would need 3000 times the gap length: 1.5 m.

P14.3. Show (14.5) and evaluate it for L_1 = 0.2 H, L_2 = 0.05 H, M = 0.08 H.

▼ **Solution**
$v = L_1\dot{i}_1 + M\dot{i}_2 = M\dot{i}_1 + L_2\dot{i}_2 \Rightarrow \dot{i}_1(L_1 - M) = \dot{i}_2(L_2 - M)$. With $\dot{i} = \dot{i}_1 + \dot{i}_2$: $\dot{i}_1 = \dot{i}(L_2 - M)/(L_1 + L_2 - 2M)$, $\dot{i}_2 = \dot{i}(L_1 - M)/(L_1 + L_2 - 2M)$, and $v = \dot{i}\,(L_1L_2 - M^2)/(L_1 + L_2 - 2M)$. Numbers: $(0.01 - 0.0064)/(0.09) = 0.04$ H. *(Code checks by solving the linear system.)*

P14.4. Two coils have $L_1 = 10$ mH, $L_2 = 40$ mH. A student reports $M = 25$ mH. Why must the measurement be wrong?

▼ **Solution**
$\sqrt{L_1L_2} = 20$ mH, so $M = 25$ mH would mean $k = 1.25 > 1$. Then W_f would be negative for $i_1 = 1$, $i_2 = -0.5$ A ($0.005 + 0.005 - 0.0125 < 0$): the coils would deliver energy without ever receiving it. Impossible for a passive structure.

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10. Summary

Relation	Validity
$L = \lambda/i = N^2/\mathcal{R}$; $L_{inc} = d\lambda/di$	Linear; incremental form general
$M = N_2\Phi_{21}/i_1 = M_{12}$	Linear (reciprocity)
$k = \lvert M\rvert/\sqrt{L_1L_2} \le 1$	Always
$v_1 = L_1\dot{i}_1 \pm M\dot{i}_2$	Linear, stationary; sign from dots
$L_{aid/opp} = L_1 + L_2 \pm 2M$	Series
$W_f = \int i\,d\lambda$; $W' = \int \lambda\,di$; $W + W' = \lambda i$	General (lossless)
$W_f = \frac{1}{2}Li^2$; $w_f = B^2/(2\mu)$	Linear
$W_f = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 + Mi_1i_2$	Linear coupled coils

11. Vocabulary

English	Türkçe
inductance / self-inductance	endüktans / öz endüktans
mutual inductance	karşılıklı endüktans
magnetic coupling; coupled circuits	manyetik kuplaj; kuplajlı devreler
coupling coefficient	kuplaj katsayısı
dot convention	nokta kuralı (nokta işareti)
series aiding / opposing	seri yardımcı / karşıt bağlantı
stored magnetic energy	depolanan manyetik enerji
magnetic energy density	manyetik enerji yoğunluğu
coenergy	koenerji (eş-enerji)
common-mode choke	ortak mod bobini

12. Sources

- LN6 viewer pp. 6–13 (slides 3–9): $V_L = L\,di/dt$, inductance definitions, $L = N^2/\mathcal{R}$, mutual inductance. LN4 p. 38 (slide 15). LN7 p. 15 (slide 8, dots). LN12 pp. 25–31 (slides 13–15, energy/coenergy). LN14 pp. 11–25 (slides 9–14, coupled energy, matrix form).

- Sen §1.1.6 pp. 13–15 (viewer 37–39), Example 1.4; §1.2.1 Eqs. 1.26–1.29 (energy per volume); §3.2 field energy pp. 94–99.
- 2025 MT1 Q4(b) p. 10 and key p. 1; 2021 MT Q2(c,d) p. 6 with key; MT I 2016 Q1(d) p. 3; MT I 2017 Q1(d) p. 2, key p. 3.
- Verification: `lesson14_inductance_energy.py`.

Lesson 15 — AC excitation: Faraday’s and Lenz’s laws, waveforms and saturation

Chapter II: Magnetic Circuits · Week 6 · Lesson 15 of 31

Assistant-written teaching note. Teacher sources: LN4 slides 11–17; LN5 slide 8 (Lenz); LN6 slides 10–14 (AC excitation, phasors). Book: Sen §1.3 Sinusoidal excitation and §1.3.1 Exciting current (printed pp. 21–25, viewer 45–49), Examples 1.6–1.7, Problem 1.20. Exams: 2025 MT1 Q4(c,d); 2022 MT1 Q2 (RCD); MT I Fall 2019 Q2; MT I Fall 2016 Q1 Part II(c). Numbers: `lesson15_ac_excitation.py`.

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1. Place and goals

Up to now currents were constant, fluxes were constant, and nothing was induced. Real transformers and inductors run on alternating voltage. Once the flux changes in time, **Faraday’s law** (Faraday’ın elektromanyetik indüksiyon yasası) produces a voltage in every turn it links. LN4 calls it “the most important equation for EE362” — it is equally the key to every transformer calculation in Chapter III.

After this lesson you should be able to:

1. State Faraday’s and Lenz’s laws, explain the minus sign, and write a coil’s terminal equation $v = Ri + d\lambda/dt$ with a clearly defined reference polarity.
2. Explain why, for a voltage-driven coil, **the voltage fixes the flux** and the core then demands whatever current is needed.
3. Derive $E_{rms} = 4.44 f N \Phi_{max}$ with its assumptions, and use it for design and for the effect of changing voltage or frequency.
4. Sketch and compute voltage, flux and current waveforms for linear and saturating cores, including non-sinusoidal (square-wave) voltages and harmonic content of the magnetizing current.
5. Analyse a coil coupled to a loaded second coil with phasors (preview of the transformer), and apply induction to sensing devices such as the RCD.

Prerequisites: Lessons 10–14; Chapter I sinusoids, RMS and phasors (Lessons 3–4).

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2. Faraday’s law and the sign question

2.1 Faraday’s law

For any closed path C bounding a surface S (LN4 slides 11–12; Maxwell’s third equation $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$):

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{A} = -\frac{d\Phi}{dt}. \tag{15.1}$$

The left side is the **induced EMF** (indüklenen elektromotor kuvvet) around the path. The direction of traversal and the positive side of S are related by the right-hand rule, exactly as in Ampère’s law.

LN4 slide 13 lists four ways to make $d\Phi/dt \neq 0$: time variation of the flux (AC excitation — this lesson and Chapter III); movement of a conductor in a stationary field; rotation (machines — Chapter IV); and deformation of the coil (changing area).

2.2 Lenz's law: the meaning of the minus sign

LN5 slide 8: "An induced EMF always gives rise to a current whose magnetic field opposes the original change in magnetic flux" (Lenz yasası). If the flux through a closed loop increases, the induced current circulates so that its own flux points against the increase. Without the minus sign, the induced current would reinforce the change, which would reinforce the current — energy from nothing. Lenz's law is energy conservation applied to induction. Eddy currents (Lesson 16) and the "magnet falling slowly through a copper tube" demonstration (LN5 slide 9) are direct consequences.

2.3 From field EMF to the terminal voltage of a coil

A coil has two terminals. Define:

- the current i entering the terminal marked +;
- the flux linkage λ positive in the direction that a positive i would produce (right-hand rule);
- the terminal voltage v = potential of + terminal minus potential of − terminal (**passive sign convention**, the same as for any load in Chapter I).

Integrate (15.1) around a path that goes through the wire (inside a perfect conductor $E = 0$, apart from the resistive drop) and returns through the air between the terminals. Carefully keeping signs gives the circuit law

$$\boxed{v = Ri + \frac{d\lambda}{dt}}, \qquad \lambda = N\Phi \text{ if all turns link the same flux.} \tag{15.2}$$

This is LN4 slide 14's $V_{coil} = N\,d\Phi/dt = d\lambda/dt$ (with $R = 0$) and Sen's Eq. 1.26 $e = N\,d\Phi/dt$. The slide-12 form $e = -d\Phi/dt$ describes the *EMF around the loop* in the direction of the current reference; the terminal voltage of a load-convention coil is the negative of that EMF, hence the plus sign. Both slides are correct; they describe different quantities.

Sanity check with Lenz. If you push increasing current into the + terminal, λ increases, so $v = d\lambda/dt > 0$: the + terminal is at the higher potential, i.e. the coil *opposes* the increasing current — exactly Lenz. A linear coil gives $v = L\,di/dt$: an inductor (Lesson 14).

"KVL out, Faraday in" (LN4 slides 16–17). KVL assumes $\oint \mathbf{E}\cdot d\mathbf{l} = 0$ around every loop. When the loop links changing flux, that integral is $-d\Phi/dt \neq 0$, so KVL fails *in the field*. Lumped circuits rescue KVL by confining the changing flux inside the coils and representing it by the terminal voltage $d\lambda/dt$. The circuit you draw is then consistent, provided no changing flux links your wiring loop.

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3. Sinusoidal excitation

3.1 The voltage determines the flux

Apply a voltage $v(t)$ to a coil with negligible resistance and core loss. From (15.2), $v = N\,d\Phi/dt$, hence

$$\Phi(t) = \frac{1}{N}\int v\,dt \qquad \text{(plus a constant that is zero in periodic steady state).} \tag{15.3}$$

This is a central fact for transformers: **the applied voltage and frequency set the flux; the core material and geometry then set how much current is needed to produce that flux.** A better core does not change the flux; it reduces the magnetizing current.

3.2 Derivation of the 4.44 formula

Let the flux be sinusoidal, $\Phi(t) = \Phi_{max}\sin\omega t$ (LN6 slide 10; Sen Eq. 1.37). Then

$$v(t) = N\frac{d\Phi}{dt} = N\omega\Phi_{max}\cos\omega t = E_{max}\cos\omega t, \qquad E_{max} = 2\pi f N \Phi_{max}.$$

Dividing the peak by $\sqrt{2}$ (valid for a sinusoid only):

$$\boxed{E_{rms} = \frac{2\pi}{\sqrt{2}} f N \Phi_{max} = 4.44\, f N \Phi_{max} = 4.44\, f N B_{max} A} \tag{15.4}$$

(Sen Eq. 1.40; $\sqrt{2}\pi = 4.4429$, so "4.44" is accurate to 0.07 %). Note the mixture: RMS voltage, **peak** flux. The voltage is a cosine when the flux is a sine — the voltage **leads** the flux by 90° (Figure 15.1).

Assumptions: sinusoidal flux (hence sinusoidal voltage), every turn links the same flux, winding resistance and leakage neglected (otherwise E is the internal EMF, not the terminal voltage).

3.3 Consequences: voltage, frequency and flux density

From (15.4), $B_{max} = E_{rms}/(4.44 f N A)$:

- **At fixed frequency, $B_{max} \propto V$.** An overvoltage pushes the core toward saturation ("over-excitation", LN6 slide 11).
- **At fixed voltage, $B_{max} \propto 1/f$.** A 50 Hz transformer on 60 Hz runs at 5/6 of its design flux density (fine); on 30 Hz it would need 5/3 of it (saturation) — MT I Fall 2015 Q3(c) asks exactly this. Motor drives keep V/f constant for this reason.
- **Design:** for a given voltage and frequency, $NA = E/(4.44 f B_{max})$; at higher frequency a smaller core suffices — why 400 Hz aircraft and kHz–MHz switch-mode transformers are small (MT I Fall 2014 Q3 compares 50 Hz and 500 Hz designs).

3.4 Waveforms and phasors of a linear inductor

If the core is linear, $i = \lambda/L = N\Phi/L = \mathcal{R}\Phi/N$, so i is a sinusoid **in phase with $\boldsymbol{\Phi}$** and lagging v by 90° (LN6 slide 10 waveform; slide 14 phasor diagram: "in ideal inductor current lags voltage by 90°"). In phasor form (Chapter I, Lesson 4), $\mathbf{V} = j\omega L\mathbf{I} = jX_L\mathbf{I}$ with **inductive reactance** (endüktif reaktans) $X_L = \omega L$. No average power is absorbed: $p = vi$ averages to zero because v and i are in quadrature.

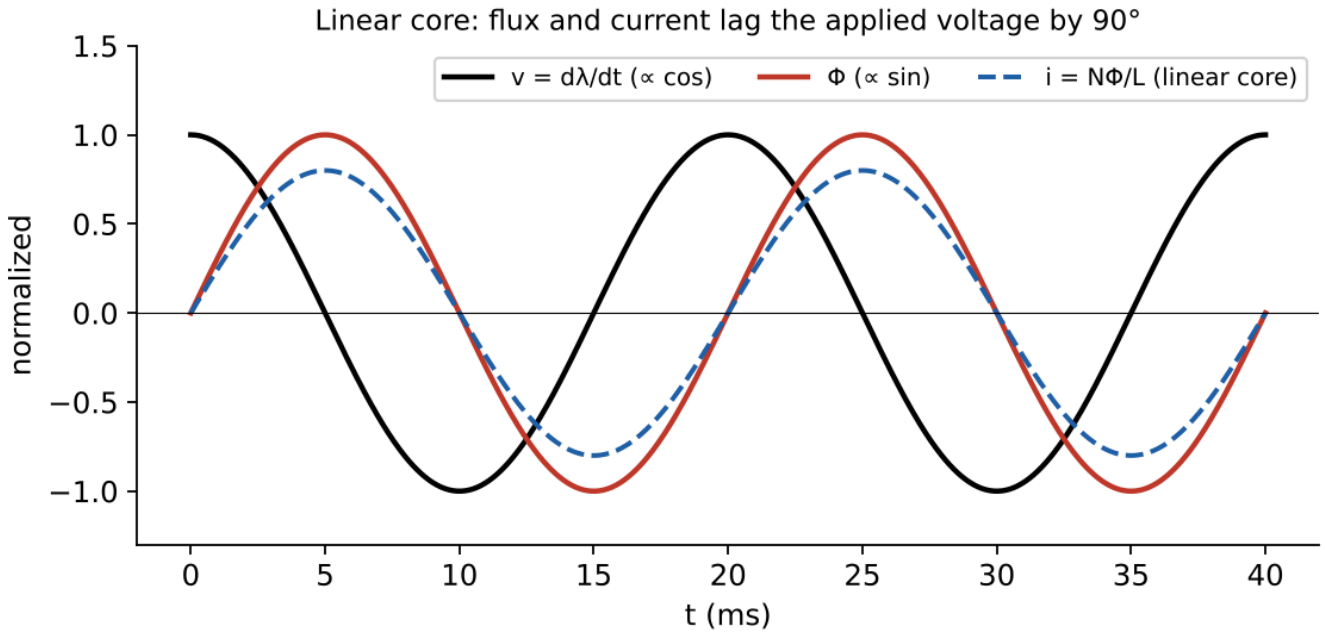


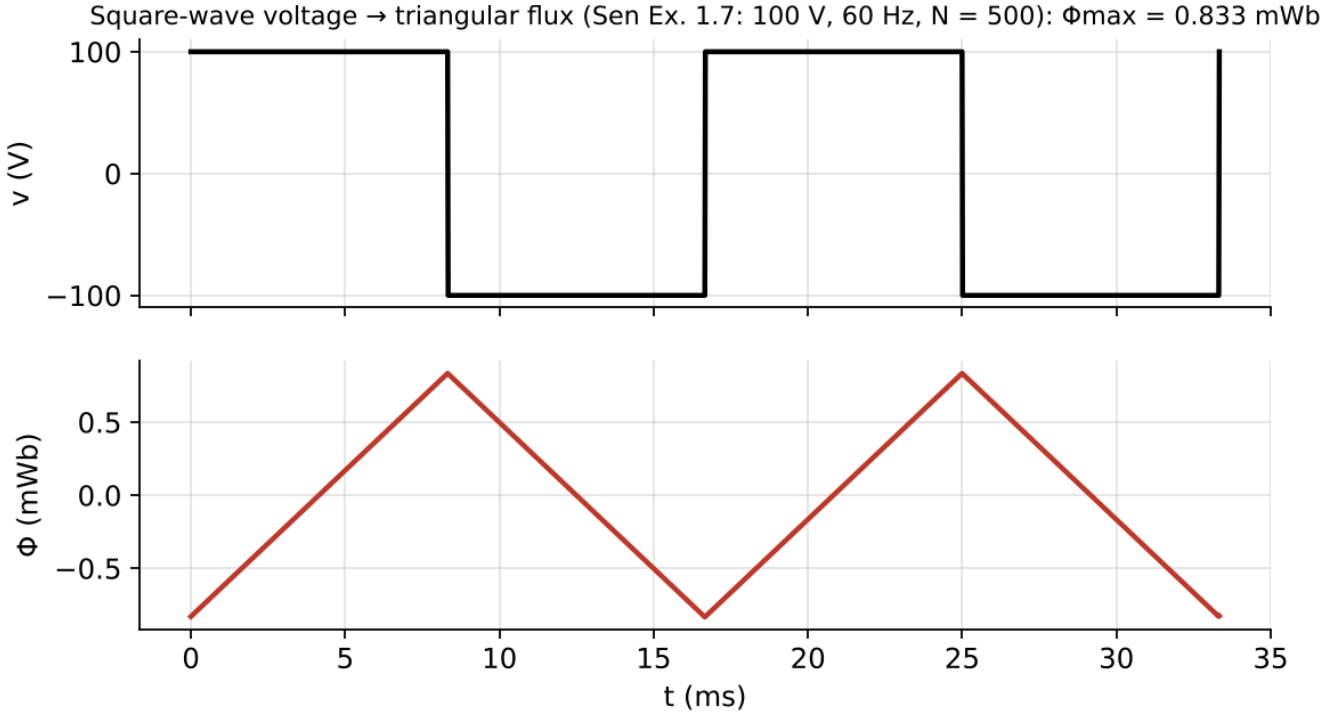
Figure 15.1. Linear core: Φ and i are sinusoids in phase; the voltage $v = d\lambda/dt$ leads them by 90°. Redrawn from the idea of LN6 slide 10.

3.5 Non-sinusoidal voltages: volt-seconds

Equation (15.3) works for any waveform. Over a half-period in which the flux swings from $-\Phi_{max}$ to $+\Phi_{max}$:

$$N(2\Phi_{max}) = \int_{\text{half cycle}} v\,dt \qquad \text{(volt-second area, Sen Eq. 1.42).} \tag{15.5}$$

A **square-wave** voltage of amplitude E gives a **triangular** flux with $\Phi_{max} = E\,(T/2)/(2N)$ (Figure 15.2, Sen Example 1.7). The same volt-second idea governs inductors in switching converters (Lessons 28–29).



4. Saturation: the magnetizing current is not sinusoidal

4.1 Voltage-driven core (the usual case)

Suppose the voltage, hence the flux, is sinusoidal, but the core's Φ–i curve (the B–H curve rescaled by A and l/N) saturates. Then the current at each instant is read from the curve: $i(t) = \frac{l}{N} H(B_{max} \sin \omega t)$. Near the flux peaks the curve is flat, so small extra flux needs a lot of extra current: the current becomes **peaky** (LN6 slides 11–12; Sen Fig. 1.18; Figure 15.3).

Properties of this **magnetizing current** (mıknatıslanma akımı) when hysteresis is neglected:

- It is **in phase with the flux** (zero crossings together) and therefore its fundamental lags the voltage by 90°: no average power is absorbed (Sen §1.3.1).
- It has **half-wave symmetry** ($i(t + T/2) = -i(t)$) because the B–H curve is odd, so it contains **only odd harmonics** (3rd, 5th, 7th…). The 3rd harmonic is the largest.
- Its crest factor (peak/RMS) exceeds $\sqrt{2}$ and grows quickly with B_{max} (Example 15.6: from 1.51 at 0.9 T to 2.29 at 1.3 T).
- A modest overvoltage (10–20 %) can multiply the magnetizing current several times — why transformer over-excitation must be avoided.

With **hysteresis**, the multivalued loop shifts the current zero crossings ahead of the flux, so the current gains a component in phase with the voltage: the **core-loss current**. Lesson 16 completes this picture (Sen Fig. 1.19; LN6 slide 13).

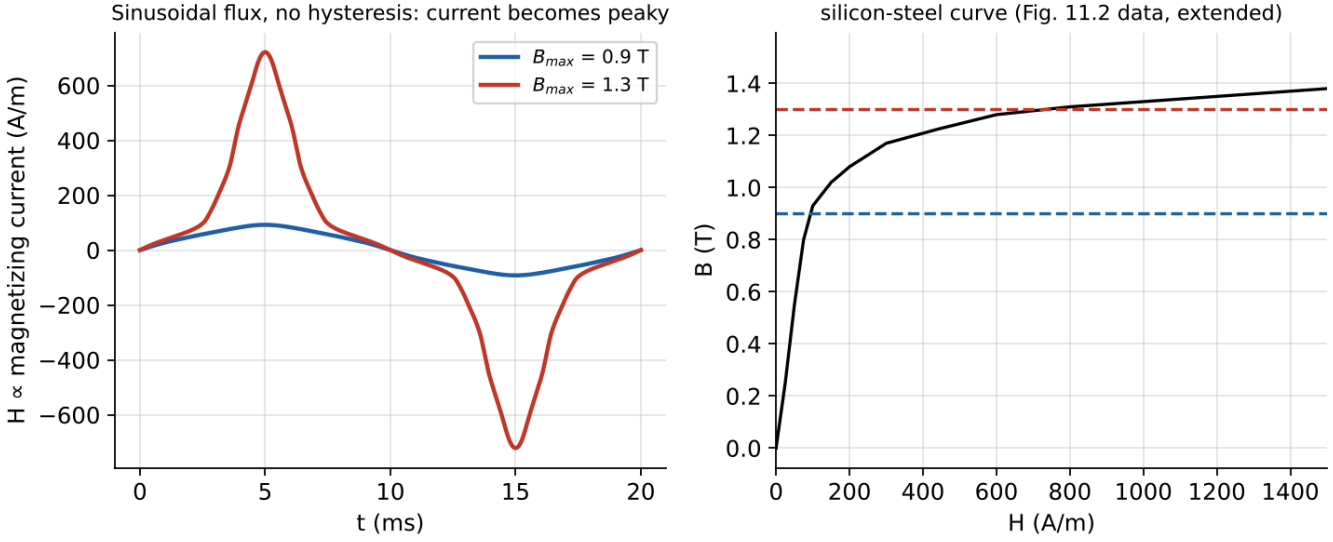


Figure 15.3. Sinusoidal flux in a silicon-steel core (curve of Figure 11.2, extended above 1.33 T). At B_{max} = 0.9 T the current is nearly sinusoidal; at 1.3 T it is strongly peaked.

4.2 Current-driven core

If instead a *current source* forces a sinusoidal current (e.g. a series reactor in a line, or the crane magnet of MT 2016 Q1 Part II(c) "excited with an AC current of 20 A rms"), the roles swap: H is sinusoidal, B is flattened at its peaks, and the induced voltage $v = NA\,dB/dt$ becomes peaky with sharp spikes when B leaves saturation. Either way, saturation generates harmonics.

5. Two coils under AC: the loaded secondary (transformer preview)

Put a second coil (N_2) on the same core, connect a resistor R_L across it, and drive coil 1 with a sinusoidal current. Three equations, all with phasors (peak or RMS consistently):

1. Ampère around the core (series core of reluctance $\mathcal{R}_s$, perfect coupling): $N_1\mathbf{I}_1 - N_2\mathbf{I}_2 = \mathcal{R}_s\boldsymbol{\Phi}$, with $\mathbf{I}_2$ defined as leaving the dotted end (it opposes the flux, Lenz).
2. Faraday for coil 2: $\mathbf{V}_2 = j\omega N_2\boldsymbol{\Phi}$.
3. Load: $\mathbf{I}_2 = \mathbf{V}_2/R_L$.

Eliminating: $N_1\mathbf{I}_1 = \boldsymbol{\Phi}\left(\mathcal{R}_s + j\omega N_2^2/R_L\right)$. The secondary MMF $N_2\mathbf{I}_2$ is **in quadrature** with the magnetizing MMF $\mathcal{R}_s\boldsymbol{\Phi}$ (because $\mathbf{V}_2$, and hence $\mathbf{I}_2$, leads $\boldsymbol{\Phi}$ by 90°). Two limits:

- $\mathcal{R}_s \to 0$ (ideal core): $N_1\mathbf{I}_1 = N_2\mathbf{I}_2$ and $\mathbf{V}_2 = R_L(N_1/N_2)\mathbf{I}_1$ — the **ideal transformer** of Lesson 17 (MMF balance, LN7 slide 6).
- $R_L \to \infty$ (open secondary): $\boldsymbol{\Phi} = N_1\mathbf{I}_1/\mathcal{R}_s$, the inductor of Section 3.

Peak magnitudes must therefore **not** be added or subtracted algebraically: $|N_1I_1 - N_2I_2| \neq N_1|I_1| - N_2|I_2|$ unless the currents are in phase. Example 15.4 shows what goes wrong otherwise.

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6. Methods and common mistakes

Task	Method
Flux from applied sinusoidal voltage	$\Phi_{max} = V_{rms}/(4.44fN)$
Flux from arbitrary periodic voltage	$\Phi = \frac{1}{N}\int v\,dt$; volt-seconds over a half-cycle = $2N\Phi_{max}$
Current from flux, linear core	$i = \mathcal{R}\Phi/N$ (sinusoid in phase with Φ)
Current from flux, saturating core	read $H(B(t))$ point by point; peak current from the curve at B_{max}
Induced voltage in a sense coil	$v = N_s\,d\Phi/dt$; RMS $= 4.44fN_s\Phi_{max}$
Coupled coils with load	phasors: Ampère (MMFs as phasors) + Faraday + load law

Mistakes to avoid. Using 4.44 with an RMS flux or a peak voltage; forgetting $\omega = 2\pi f$; applying 4.44 to square waves; adding peak MMFs of quadrature currents; assuming a better core reduces flux (it reduces current); forgetting that DC gives no induced voltage in steady state (RCD question (e)).

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7. Worked examples

Example 15.1 — Coil on a 120 V, 60 Hz supply (Sen Example 1.6)

Given. $N = 200$, core length 100 cm, area 20 cm², μ_r = 2500 (linear). **Find** expressions for (a) $B(t)$, (b) $i(t)$.

Solution. (a) $\Phi_{max} = 120/(4.44 \times 200 \times 60) = 2.252 \times 10^{-3}$ Wb; $B_{max} = \Phi_{max}/A = 1.126$ T; taking the flux as the sine reference, $B = 1.126\sin(2\pi 60t)$ T and the applied voltage is $120\sqrt{2}\cos(2\pi 60t)$ V. (b) $H_{max} = B_{max}/\mu = 1.126/(2500\mu_0) = 358.5$ A/m; $i_{max} = H_{max}l/N = 358.5 \times 1.0/200 = 1.79$ A; $i =$

$1.79\sin(2\pi 60t)$ A, in phase with B. **Check.** $L = N^2\mu A/l = 0.2513$ H; $V\sqrt{2}/(\omega L) = 1.79$ A ✓ (the 4.44 rounding changes the third significant figure only). **Interpretation.** 1.13 T is a realistic transformer flux density; with real steel the current would be slightly peaky.

Example 15.2 — Square-wave excitation (Sen Example 1.7)

Given. Square wave ±100 V, 60 Hz, on a 500-turn coil, core area 0.001 m², no resistance. **Find** (a) Φ_{max} and the waveforms; (b) the largest E for $B_{max} \le 1.2$ T.

Solution. (a) Over a half-period of 1/120 s the flux swings by $2\Phi_{max}$: $500(2\Phi_{max}) = 100 \times \frac{1}{120} \Rightarrow \Phi_{max} = 0.833 \times 10^{-3}$ Wb. The flux rises linearly during the positive half-cycle and falls linearly during the negative one: a triangle (Figure 15.2). (b) $\Phi_{max} = 1.2 \times 0.001 = 1.2 \times 10^{-3}$ Wb ⇒ $E = 120 \times 500 \times 2 \times 1.2 \times 10^{-3} = 144$ V. **Check.** The code integrates the square wave numerically and recovers 0.833 mWb.

Example 15.3 — Flux linkage and voltage of coil 1 (2025 MT1 Q4(c))

Given. The E-core of Examples 12.8/14.2, coil 2 open, $i_1 = 1.2\sin(2\pi 50t)$ A. **Find** the flux linkage of coil 1 and its induced voltage.

Solution. $\Phi_{1,max} = N_1 I_{1,max}/\mathcal{R}_{eq} = 480/1.729 \times 10^6 = 2.776 \times 10^{-4}$ Wb; $\lambda_{1,max} = N_1\Phi_{1,max} = 0.111$ Wb·turn, so $\lambda_1 = 0.111\sin(2\pi 50t)$ Wb·t. $v_1 = d\lambda_1/dt = 34.9\cos(2\pi 50t)$ V, i.e. **24.7 V rms** (key: 24.69 V via 4.44 with rounded flux). **Check.** $\omega L_1 I_{rms} = 314.16 \times 0.0925 \times 0.8485 = 24.67$ V ✓.

Example 15.4 — Loaded second coil with the centre leg removed (2025 MT1 Q4(d))

Given. Same core but with the centre leg ignored (infinite gap reluctance), so the two outer paths form a series core $\mathcal{R}_s = 2\mathcal{R}_1 = 2.122 \times 10^6$ A/Wb. Coil 2 ($N_2 = 800$) feeds $R_L = 20\ \Omega$; $i_1 = 1.2\sin(2\pi 50t)$ A; ideal coupling. **Find** the voltage and current of coil 2.

Solution (phasors, peak values, $\mathbf{I}_1 = 1.2\angle 0^\circ$). From Section 5:

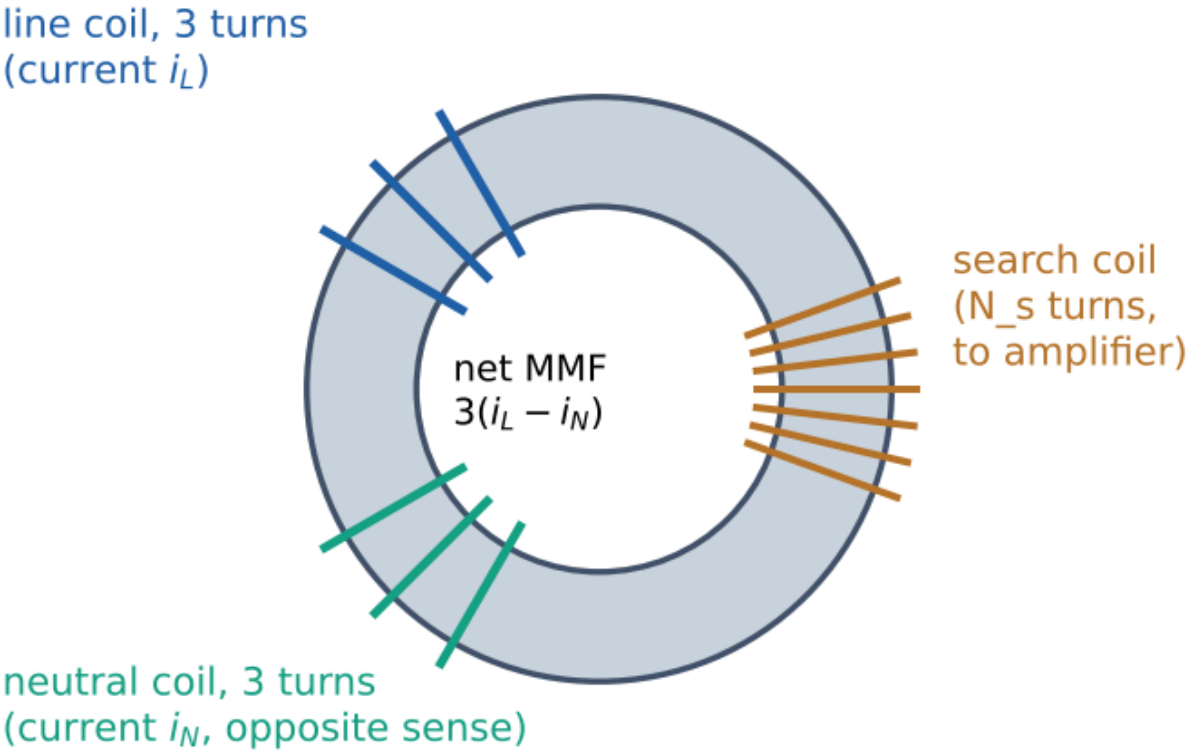
$$\boldsymbol{\Phi} = \frac{N_1\mathbf{I}_1}{\mathcal{R}_s + j\omega N_2^2/R_L} = \frac{480}{2.122 \times 10^6 + j\,1.005 \times 10^7} = 4.67 \times 10^{-5}\angle -78.1^\circ \text{ Wb.}$$

$\mathbf{V}_2 = j\omega N_2\boldsymbol{\Phi} = 11.74\angle 11.9^\circ$ V peak; $\mathbf{I}_2 = \mathbf{V}_2/R_L = 0.587\angle 11.9^\circ$ A peak (8.30 V and 0.415 A rms). **Checks.** (i) A time-domain simulation of $d\Phi/dt = R_L(N_1 i_1 - \mathcal{R}_s\Phi)/N_2^2$ reaches a steady-state peak i_2 = 0.587 A. (ii) Power into coil 1, $\frac{1}{2}\mathrm{Re}\{j\omega N_1\boldsymbol{\Phi}\,\mathbf{I}_1^*\}$, equals $\frac{1}{2}|I_2|^2 R_L$ = 3.45 W (lossless core). (iii) Ideal-transformer limit: 0.6 A, 12 V.

About the key. The handwritten key writes $(N_1 I_{1m} - N_2 I_{2m})/(2\mathcal{R}_1) = \Phi_m$ and $\Phi_m = R_L I_{2m}/(4.44\sqrt{2}N_2 f)$, giving I_{2m} = 0.495 ≈ 0.5 A and 10 V. That subtracts *peak magnitudes* of $N_1 i_1$ and $N_2 i_2$ as if they were in phase; they are not (Section 5). The phasor answer is 0.587 A and 11.7 V peak (errata E9). The size of the discrepancy reflects how far this core is from ideal: $\omega N_2^2/R_L$ is only 4.7 times $\mathcal{R}_s$.

Example 15.5 — Residual current device (2022 MT1 Q2)

Given. (exams/archive/20221/EE361 MT1_2022f_Q2.pdf pp. 1–3.) Toroid OD 15 mm, ID 5 mm, height 20 mm, μ_r = 10 000; line and neutral each pass 3 turns in opposite senses (Figure 15.4); a search coil drives an amplifier (negligible current). **Find** (a) the core flux in normal operation; (b) the peak core flux density for a 30 mA rms earth-leakage current; (c) the minimum search-coil turns to induce 100 mV rms at 30 mA, 50 Hz; (d) whether the RCD protects someone touching line and neutral; (e) whether it detects DC faults.



Solution. (a) Net MMF $3i_L - 3i_N = 0$ when all current returns through the neutral, so $H = 0$ and $B = 0$.

b. With 30 mA rms leaking to earth, the unbalance is Δi; mean path $l = \pi \times 10$ mm. $H_{peak} = 3(0.030\sqrt{2})/(\pi \times 0.01) = 4.05$ A/m; $B_{peak} = \mu_r \mu_0 H = 10^4(4\pi \times 10^{-7})(4.05) = 50.9$ mT.

c. Core area $A = \frac{15-5}{2} \times 20 = 100$ mm². $V_{rms} = 4.44 f N_s B_{peak} A \Rightarrow N_s = \dfrac{0.1}{4.44(50)(0.0509)(10^{-4})} = 88.4$ ⇒ **89 turns** (round up so at least 100 mV).

d. A person between line and neutral is just another load: line and neutral currents remain equal, net MMF zero, nothing is detected. The RCD protects only against current that returns *outside* the core (to earth).

e. In steady DC, $d\Phi/dt = 0$, so no voltage is induced. A *sudden* DC fault produces a transient flux change and a pulse in the search coil, so detection is possible but depends on the fault waveform (the key's answer); standard "type AC" RCDs are not specified for smooth DC leakage (special types exist). **Interpretation.** Faraday's law turns a tiny current imbalance into a measurable voltage thanks to the high-μ core and many search turns.

Example 15.6 — Magnetizing current harmonics in a saturating core (original)

Given. Silicon-steel core, l = 0.5 m, A = 10 cm², N = 200, 50 Hz sinusoidal voltage. Compare B_{max} = 0.9 T and 1.3 T (curve of Figure 11.2, extended above 1.33 T by a smooth monotone fit).

Solution. Required voltages: $V = 4.44 f N B_{max} A$ = 40.0 V and 57.8 V rms. For each B_{max}, compute $i(t) = (l/N)H(B_{max} \sin \omega t)$ and its Fourier series:

B_{max}	V_{rms}	i_{peak}	i_{rms}	I_1 **(peak)**	I_3/I_1	I_5/I_1	**crest factor**
0.9 T	40.0 V	0.230 A	0.153 A	0.216 A	0.03	0.03	1.51
1.3 T	57.8 V	1.80 A	0.788 A	0.982 A	0.48	0.23	2.29

Even harmonics are zero (checked numerically). **Interpretation.** A 44 % increase in voltage multiplies the RMS magnetizing current by 5 and the peak by 8, with a 3rd harmonic of nearly half the fundamental. (The absolute numbers depend on the curve read from Sen Fig. 1.7; the trend does not.)

Example 15.7 — Saturating inductor on a sinusoidal supply (MT I Fall 2019 Q2)

Given. (exams/archive/MT/MT I - Fall2019.pdf pp. 2–3.) Core with piecewise-linear B–H: straight from the origin to (250 A/m, 1.6 T), then slope μ_0 ("perfectly saturated"); $A_c = A_g$ = 400 mm², mean path 220 mm including a 2 mm gap, N = 200. **Find** (a) the largest peak current without saturating the core; (b) the Φ–i characteristic and slopes; (c) incremental inductance in both regions; (d) $\Phi(t)$ and $i(t)$ for $v = 41\sin(2\pi 50 t)$ V.

Solution. (a) At the knee $B = 1.6$ T in core and gap: $Ni = H_c l_c + (B/\mu_0) l_g = 250(0.218) + (1.6/\mu_0)(0.002) = 54.5 + 2546.5 \Rightarrow i_{sat} = 13.0$ A.

b. $\Phi_{sat} = 1.6 \times 400 \times 10^{-6} = 0.64$ mWb. Unsaturated slope $\Phi_{sat}/i_{sat} = 4.92 \times 10^{-5}$ Wb/A. Saturated: both core and gap behave like air for increments, $d\Phi/di = \mu_0 N A / l_{mean} = 4.57 \times 10^{-7}$ Wb/A.

c. $L_1 = N \times 4.92 \times 10^{-5} = 9.84$ mH; $L_2 = N \times 4.57 \times 10^{-7} = 91.4$ µH (the key prints 0.912 µH — a factor-100 slip, errata E13).

d. $\Phi = -\dfrac{41}{N\omega}\cos\omega t$, so $\Phi_{peak} = 41/(200 \times 100\pi) = 0.6525$ mWb — just above the knee. The current follows the flux (no hysteresis) on the first slope up to 13.0 A, then rises steeply: $i_{peak} = 13.0 + (0.6525 - 0.64) \times 10^{-3}/4.57 \times 10^{-7} = 40.4$ A. The core is saturated for about 22° around each flux peak, where the current forms a narrow spike. The key's 34.9 A comes from rounding Φ_{peak} to 0.65 mWb (errata E14). **Interpretation.** A 2 % flux excess over the knee triples the peak current. In saturation, small rounding errors in flux become large errors in current — compute from exact values.

Example 15.8 — Voltage, frequency and flux density

Given. A core is designed for B_{max} = 1.2 T at 50 Hz and rated voltage. **Find** B_{max} at rated voltage and (a) 60 Hz; (b) 30 Hz. **Solution.** $B_{max} \propto V/f$: (a) $1.2 \times 50/60 = 1.0$ T; (b) $1.2 \times 50/30 = 2.0$ T — beyond saturation of electrical steel; the magnetizing current would be very large and distorted. Reduce the voltage to 60 % (keep V/f constant) to operate at 30 Hz.

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8. Optional practice

P15.1 (Sen Problem 1.20). A two-winding laminated core: N_1 = 200 driven so that $B = 1.2\sin 377t$ T; N_2 = 400 open; stacking factor 0.95; gross area 25 cm²; μ_r = 10 000; l_c = 90 cm. Find (a) E_1 rms; (b) the current; (c) E_2 rms.

▼ **Solution**

Net iron area $0.95 \times 25 = 23.75$ cm² (flux density refers to the iron). (a) $E_1 = \omega N_1 B_{max} A_{net}/\sqrt{2} = 377(200)(1.2)(2.375 \times 10^{-3})/\sqrt{2} = 152$ V. (b) $H_{max} = 1.2/(10^4\mu_0) = 95.5$ A/m; $i_{max} = H_{max} l_c / N_1 = 0.430$ A (0.304 A rms), in phase with B. (c) Same flux links coil 2: $E_2 = E_1 N_2/N_1 = 304$ V.

P15.2. A 1.2 T, 50 Hz design is fed from a 60 Hz supply at 10 % overvoltage. New B_{max}?

▼ **Solution**

$B_{max} = 1.2 \times 1.1 \times 50/60 = 1.10$ T — the overvoltage is more than compensated by the higher frequency.

P15.3. In the RCD of Example 15.5, what search-coil RMS voltage results from a 250 mA earth fault (assume the core stays linear)?

▼ Solution
Linear scaling: $V = 0.1 \times (250/30) \times (89/88.4) = 0.84$ V rms. (A 30 mA trip setting thus gives plenty of signal for larger faults.)

P15.4 (concept). Why does a transformer energized at no load sometimes draw a current many times its rated current for a few cycles (inrush), even though its steady-state magnetizing current is tiny? Use (15.3).

▼ Solution
$\Phi = \frac{1}{N}\int v\,dt$ includes the initial condition. If the transformer is switched on at a voltage zero with no residual flux, the flux starts at zero but must follow $\Phi_{max}(1 - \cos\omega t)$, reaching $2\Phi_{max}$ (plus any residual flux) after half a cycle — far into saturation, where the current is enormous. Resistance gradually removes the DC offset. (Sen §9.4 treats this; a supplement beyond the syllabus.)

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9. Summary

Relation	Validity
$\oint \mathbf{E}\cdot d\mathbf{l} = -d\Phi/dt$	Always (Faraday)
$v = Ri + d\lambda/dt$	Coil, passive sign convention
$\Phi = \frac{1}{N}\int v\,dt$	Lossless coil, periodic steady state
$E_{rms} = 4.44 f N \Phi_{max} = 4.44 f N B_{max} A$	Sinusoidal flux
$2N\Phi_{max} = \int_{T/2} v\,dt$	Any symmetric periodic voltage
$\mathbf{V} = jX_L\mathbf{I}$, $X_L = \omega L$	Linear inductor, phasors
Magnetizing current: in phase with Φ, odd harmonics	Saturation, no hysteresis

10. Vocabulary

English	Türkçe
electromagnetic induction	elektromanyetik indüksiyon
Faraday’s law; Lenz’s law	Faraday yasası; Lenz yasası
induced EMF / induced voltage	indüklenen EMK / indüklenen gerilim
AC excitation	alternatif akım uyartımı
time-varying magnetic field	zamanla değişen manyetik alan
magnetizing (exciting) current	mıknatıslanma (uyartım) akımı
inductive reactance	endüktif reaktans
over-excitation	aşırı uyartım
harmonic	harmonik

English	Türkçe
residual current device (RCD)	kaçak akım koruma rölesi (KAKR)

11. Sources

- LN4 viewer pp. 29–40 (slides 11–17). LN5 viewer p. 11 (slide 8, Lenz), p. 12 (slide 9). LN6 viewer pp. 16–20 (slides 10–14). Notes in `../full-semester/source-notes/`.
- Sen §1.3 pp. 21–23 (viewer 45–47), Eqs. 1.37–1.42, Examples 1.6–1.7; §1.3.1 pp. 24–25 (viewer 48–49), Figs. 1.18–1.19; Problem 1.20 p. 37 (viewer 61).
- 2025 MT1 Q4(c,d): question pp. 10–11; key p. 2 (errata E9). 2022 MT1 Q2: pp. 1–3 with key. MT I Fall 2019 Q2: pp. 2–3 with key (errata E13–E14). MT I Fall 2016 Q1 Part II(c) p. 4; MT I Fall 2015 Q3(c) p. 4; MT I Fall 2014 Q3 pp. 5–7.
- Verification: `lesson15_ac_excitation.py` (includes an ODE simulation and FFT harmonic analysis).
- More solved archive problems: Ch II–III archive collection §1–2, §7.1 (MT I Fall 2015 Q2 trapezoidal flux, Q3(a)–(c) frequency and saturation, V/f recitation example).

Lesson 16 — Hysteresis, eddy currents, core losses, lamination, skin effect and inductors

Chapter II: Magnetic Circuits · Week 6 · Lesson 16 of 31

Assistant-written teaching note. Teacher sources: LN5 (all 27 slides); LN8 slides 2–5 and 13–17 (core losses, excitation current); LN6 slides 2, 7, 13 (inductors, hysteresis waveform). Book: Sen §1.2 Hysteresis, §1.2.1–1.2.3 losses (printed pp. 15–21, viewer 39–45), §1.3.1 with hysteresis (pp. 24–25), Problems 1.17–1.18. Exams: 2025 MT1 Q4(e); MT I Fall 2014 Q3 (frequency scaling). The single- and three-phase inductor row of the syllabus has no dedicated slide in the supplied pack; it is covered here from LN6, Chapter I three-phase theory and standard construction, labelled as such. Numbers: `lesson16_core_losses.py`.

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1. Place and goals

Lessons 10–15 treated the core as lossless. Real cores get warm. This lesson explains the two physical mechanisms behind **core loss** (nüve kaybı, demir kaybı) — **hysteresis** and **eddy currents** — how they scale with frequency and flux density, why cores are laminated, and how the loss shows up in the exciting current. It also covers the related **skin effect** in conductors (LN5's second half) and closes Chapter II with single- and three-phase inductors. The core-loss resistance R_c introduced here becomes a branch of the transformer equivalent circuit (Lesson 18) and its value is measured by the open-circuit test (Lesson 19).

After this lesson you should be able to:

1. Describe a hysteresis loop (remanence, coercivity, saturation), prove that the loop area is the energy lost per cycle per unit volume, and compute hysteresis loss.
2. Explain eddy currents from Faraday's law, derive the lamination eddy-loss formula, and explain why lamination and silicon alloying reduce it.
3. Use the scaling laws $P_h \propto f B_{max}^n$ and $P_e \propto f^2 B_{max}^2$ for changes of voltage and frequency, and separate the two losses from measurements.
4. Explain how hysteresis distorts the exciting current and how core loss is represented by R_c in parallel with the magnetizing reactance.
5. Compute skin depth, explain why AC resistance exceeds DC resistance, and describe remedies (bundled, foil, hollow, stranded conductors).
6. Describe single- and three-phase inductors and calculate their basic quantities.

Prerequisites: Lessons 11, 14, 15.

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2. Hysteresis

2.1 The loop

Magnetize an unmagnetized core slowly with increasing current: B follows the initial magnetization curve (0 → a in Figure 16.1; LN5 slide 5, Sen Fig. 1.12). Reduce the current to zero: B does not return along the same curve; it stays at the **remanence** B_r (kalıcı mıknatıslanma, point b). A reverse field $-H_c$, the **coercivity** (koersivite, point c), is needed to

bring B to zero. Continuing to $-H_{max}$ gives negative saturation (d), and returning through e and f closes the **hysteresis loop** (histerezis döngüsü). After a few cycles the loop repeats exactly. "Hysteresis" means *lagging behind*: throughout the cycle B lags H (Sen §1.2).

Smaller excitation gives smaller loops nested inside (LN5 slide 7 shows loops for B_{max} = 0.3…1.7 T). The locus of the loop tips is the normal magnetization curve used in Lessons 11–15.

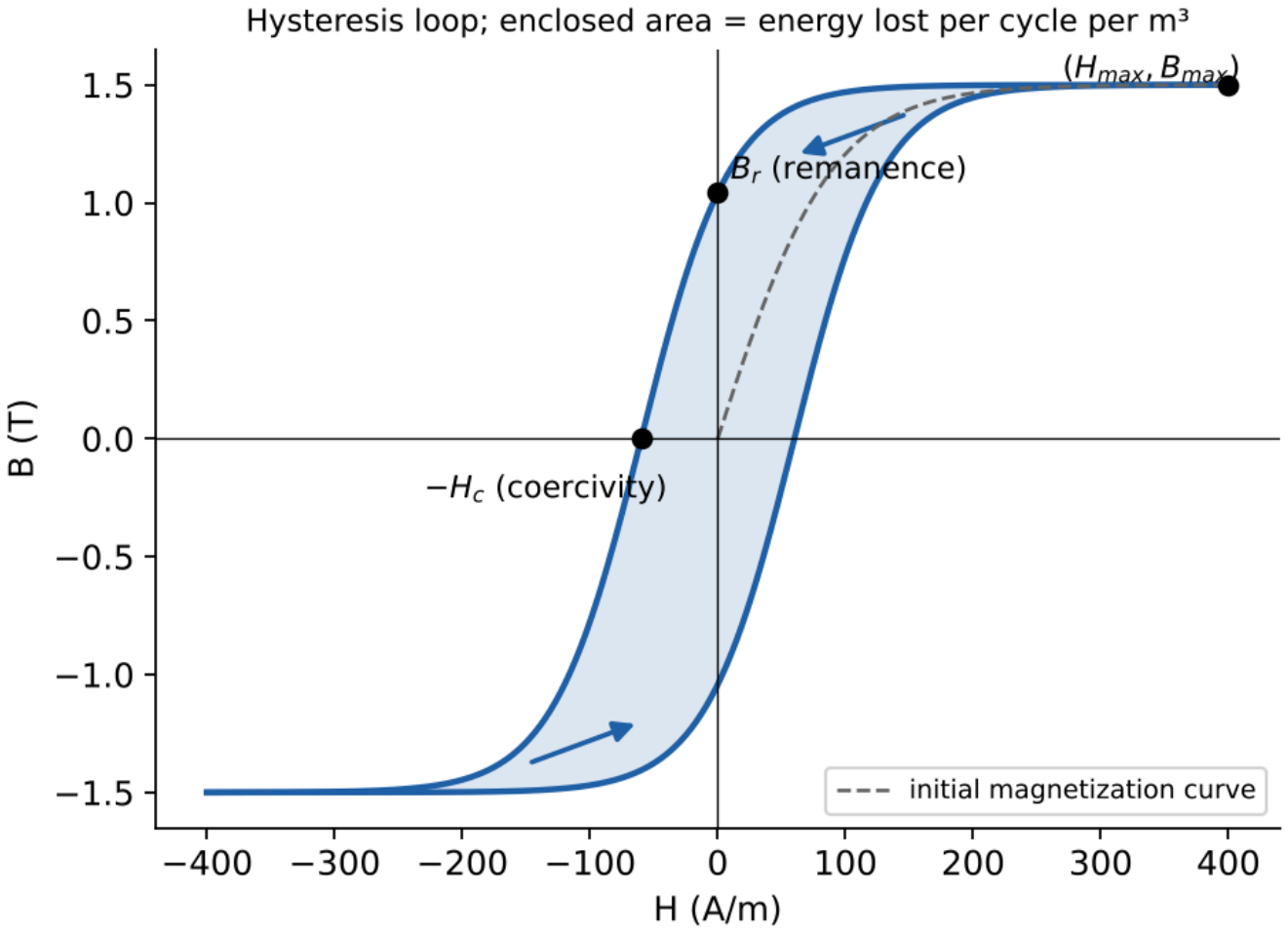


Figure 16.1. Hysteresis loop (assistant-generated model shape) with remanence B_r, coercivity H_c and tip (H_{max}, B_{max}). The shaded area is the energy lost per cycle per m³.

2.2 Loop area = energy lost

From Lesson 14, the energy delivered to a coil–core assembly (no winding resistance) between two instants is $W = \int i\, d\lambda$. With $\lambda = NAB$ and $i = Hl/N$:

$$W = \int \frac{Hl}{N} NA\, dB = (Al) \int H\, dB = V_{core} \int H\, dB \qquad \text{(Sen Eq. 1.29)}.$$

Around one complete cycle the core returns to the same magnetic state, so any *stored* energy is the same at start and end; whatever net energy was delivered has been turned into heat:

$$W_{cycle} = V_{core} \oint H\, dB = V_{core} \times (\text{loop area}), \qquad P_h = V_{core}\, f \oint H\, dB \tag{16.1}$$

(Sen Eqs. 1.30–1.31). LN5 slide 6 shows the same idea graphically: going up from P1 to P2 the source supplies the area between the rising branch and the B axis; coming back from P2 to P3 the field returns a smaller area; the difference is lost. Physically, the loss is the irreversible work

of moving domain walls past pinning sites in the crystal.

Consequences: at fixed B_{max}, energy per cycle is fixed, so **hysteresis loss ∝ frequency** (LN5 slides 6 and 25). A narrow loop (soft material, Lesson 11) means low loss.

2.3 Steinmetz's empirical law

The loop area grows with B_{max}. Steinmetz found experimentally that loop area $\approx K B_{max}^{n}$ with n between about 1.5 and 2.5 for machine steels, so

$$P_h = K_h f B_{max}^{n} \qquad \text{(Sen Eq. 1.33)}, \tag{16.2}$$

with K_h depending on material and core volume. It is an empirical fit, valid over the range of B_{max} and f for which it was measured.

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3. Eddy currents

3.1 Origin

Iron is a conductor. A changing flux inside a solid piece of iron induces EMFs around closed paths in the iron (Faraday), and those EMFs drive **eddy currents** (girdap akımları, Fuko akımları) around the flux (LN5 slide 9; Sen Fig. 1.15). By Lenz's law the eddy currents oppose the change of flux, and their i^2R heating is the **eddy-current loss**. Applications use them deliberately: induction cooking and heating, eddy-current brakes (an elliptical trainer, LN8 slide 4), eddy-current separators that kick aluminium out of a waste stream, and the slow fall of a magnet in a copper tube (LN5 slides 9, 26).

3.2 Why lamination works: derivation for a thin sheet

Consider one lamination of thickness d (in x, from $-d/2$ to $d/2$), much larger in the other two directions, carrying a uniform sinusoidal flux density $B(t) = B_{max}\sin\omega t$ along its length. Assume the sheet is thin compared with the skin depth (Section 6), so the eddy currents do not distort B.

1. **EMF.** Take a rectangular loop in the sheet extending from $-x$ to $+x$, of length ℓ (much longer than $2x$). Faraday: the EMF around it is $(2x\ell)\,dB/dt$; almost all of the loop length is the two long sides, so the electric field along each long side is $E(x) = x\,dB/dt$.
2. **Current density.** $J(x) = \sigma E(x) = \sigma x\,dB/dt$ (σ = 1/ρ, resistivity ρ).
3. **Power per unit volume** averaged over the thickness: $\dfrac{1}{d}\displaystyle\int_{-d/2}^{d/2}\frac{J^2}{\sigma}dx = \frac{\sigma}{d}\left(\frac{dB}{dt}\right)^2\int_{-d/2}^{d/2}x^2dx = \frac{\sigma d^2}{12}\left(\frac{dB}{dt}\right)^2$.
4. **Time average:** $\langle (dB/dt)^2\rangle = \omega^2 B_{max}^2/2$, so

$$p_e = \frac{\sigma\omega^2 B_{max}^2 d^2}{24} = \frac{\pi^2 f^2 B_{max}^2 d^2}{6\rho} \quad [\mathrm{W/m^3}], \qquad P_e = V_{core}\,p_e. \tag{16.3}$$

This is LN5 slide 24's formula $P = V\pi^2 B_{peak}^2 d^2 f^2/(k\rho)$ with $k = 6$ for a thin sheet and sinusoidal flux (the slide leaves k undefined; other geometries and waveforms give other constants). Sen writes the same scaling compactly as

$$P_e = K_e B_{max}^2 f^2 \qquad \text{(Sen Eq. 1.34)}. \tag{16.4}$$

What (16.3) teaches:

- ∝ **f²**: doubling frequency at constant B_{max} quadruples eddy loss (LN5 slide 25, LN8 slide 3).
- ∝ **d²**: cutting a core into sheets n times thinner reduces eddy loss n^2 times. Laminations are 0.23–0.5 mm in power transformers and 0.5–5 mm in machines (Sen quotes 0.5–5 mm for machines and 0.01–5 mm for electronic devices). The sheets are insulated from each other and stacked *in the plane of the flux*, so eddy loops are confined to thin strips (LN8 slides 5, 15–16; Figure 16.2).
- ∝ **1/ρ**: adding a few percent silicon raises steel's resistivity several times (Sen §1.2.2: "addition of a few percent of silicon (say 4 %)"). Ferrites, being nearly insulators, have negligible eddy loss and are used at high frequency.
- **At constant voltage** ($B_{max}f$ fixed by $V = 4.44fNB_{max}A$), $P_e \propto (B_{max}f)^2 \propto V^2$ — independent of frequency! This is the trick behind Sen Problem 1.17 (Example 16.2).

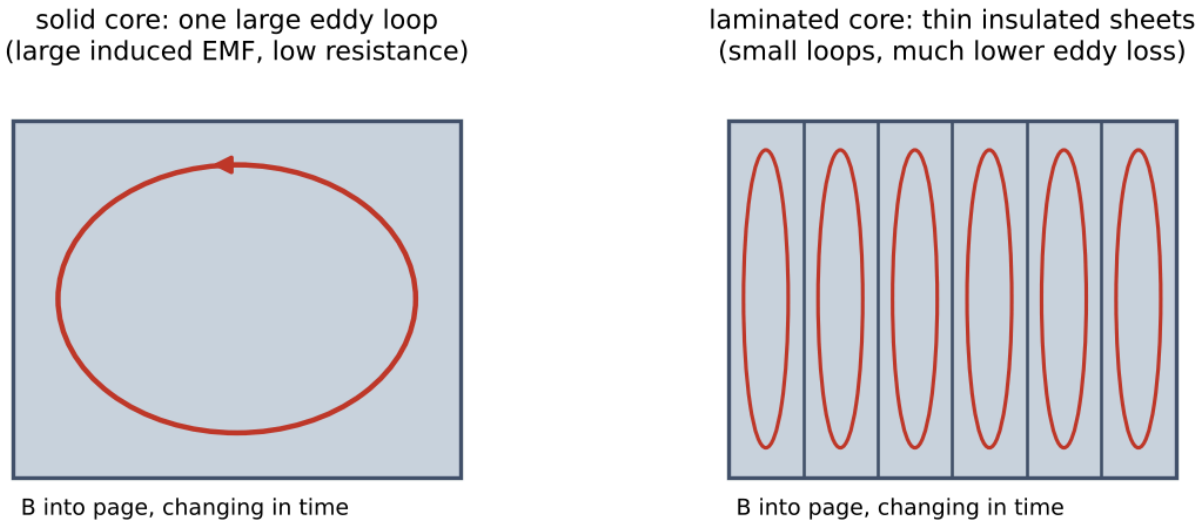


Figure 16.2. Solid core: one large eddy loop. Laminated core: thin insulated sheets confine eddy currents to narrow loops with much higher resistance. After LN8 slide 5 and Sen Fig. 1.15.

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4. Core loss as a whole

4.1 Total and measurement

$$P_c = P_h + P_e = K_h f B_{max}^n + K_e f^2 B_{max}^2 \qquad \text{(Sen Eq. 1.35)}. \tag{16.5}$$

If the current varies slowly the eddy currents vanish and the B–H loop is the **static** loop. At power frequency eddy currents widen it into a **dynamic** loop (Sen Fig. 1.16), whose area times $V_{core}f$ gives the total core loss (Sen Eq. 1.36). A wattmeter on the exciting winding measures the total; separating the two parts needs tests at two frequencies at the same B_{max} (Example 16.3). For machine and transformer analysis the total is what matters (Sen: "it is not necessary to know the losses separately").

Core loss depends on B_{max} and f, i.e. on the **applied voltage** and frequency, but hardly on the load current. That is why transformer core loss is called a **constant loss** (LN10 slide 12) and is measured at rated voltage in the open-circuit test.

4.2 Effect on the exciting current and the equivalent circuit

With hysteresis, the multivalued Φ–i loop makes the exciting current non-symmetric relative to the voltage (Sen Fig. 1.19; LN6 slide 13). Figure 16.3 shows that the current's zero crossings move *ahead* of the flux zero crossings. Decompose the current (fundamental components):

- a **core-loss component** i_c in phase with the voltage — it carries the average power P_c;
- a **magnetizing component** i_m in phase with the flux (lagging the voltage by 90°) — identical to the no-hysteresis current of Lesson 15.

$$i_\varphi = i_c + i_m \qquad \text{(LN8 slide 17: } I_e = I_c + I_m\text{)}.$$

So the coil behaves like a resistance R_c in parallel with a magnetizing reactance X_m (Sen Fig. 1.19c):

$$R_c = \frac{E^2}{P_c}, \qquad X_m = \frac{E}{I_m}. \tag{16.6}$$

These are the two shunt elements of the transformer model (LN8 slides 7, 18–19). Example: the LN9 transformer's 30 W core loss at 230 V gives $R_c = 230^2/30 = 1763\ \Omega$ (Lesson 19 completes it).

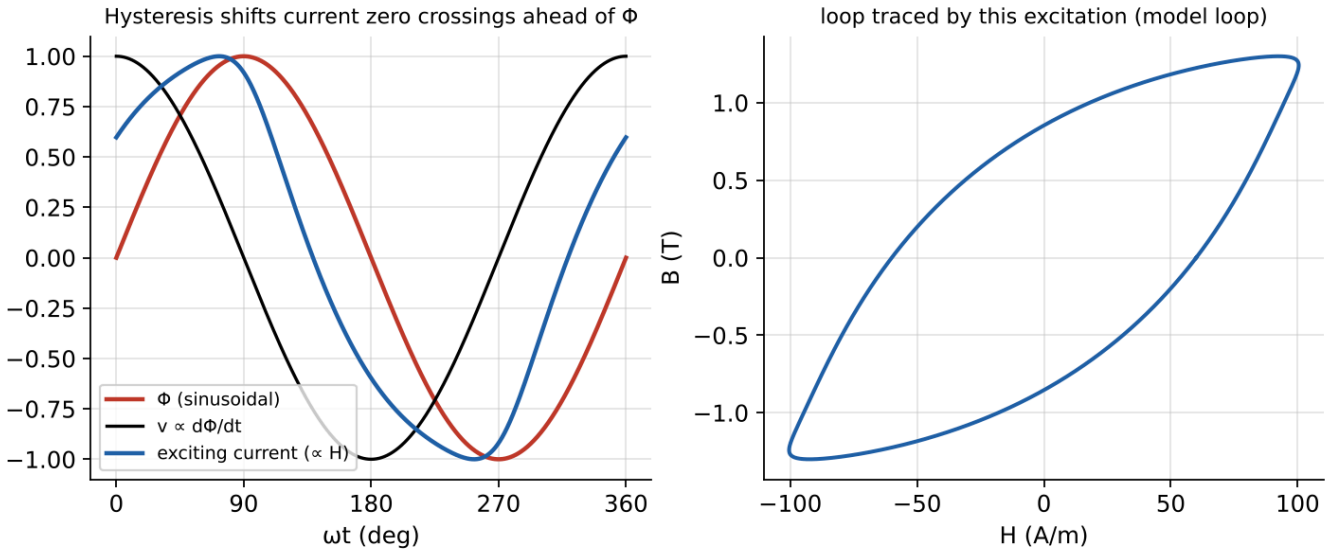


Figure 16.3. Sinusoidal flux with a model hysteresis loop: the exciting current (blue) crosses zero before the flux (red), so it contains a component in phase with the voltage (black). Assistant model; compare LN6 slide 13 and Sen Fig. 1.19.

4.3 Frequency, voltage and flux trade-offs

Combining (15.4) with (16.5):

Change (others fixed)	B_{max}	Hysteresis loss	Eddy loss
Voltage ×k, same f	×k	$\times k^n$	$\times k^2$
Frequency ×k, same V	×1/k	$\times k \cdot k^{-n} = k^{1-n}$	×1
Frequency ×k, same B_{max} (V ×k)	×1	×k	$\times k^2$

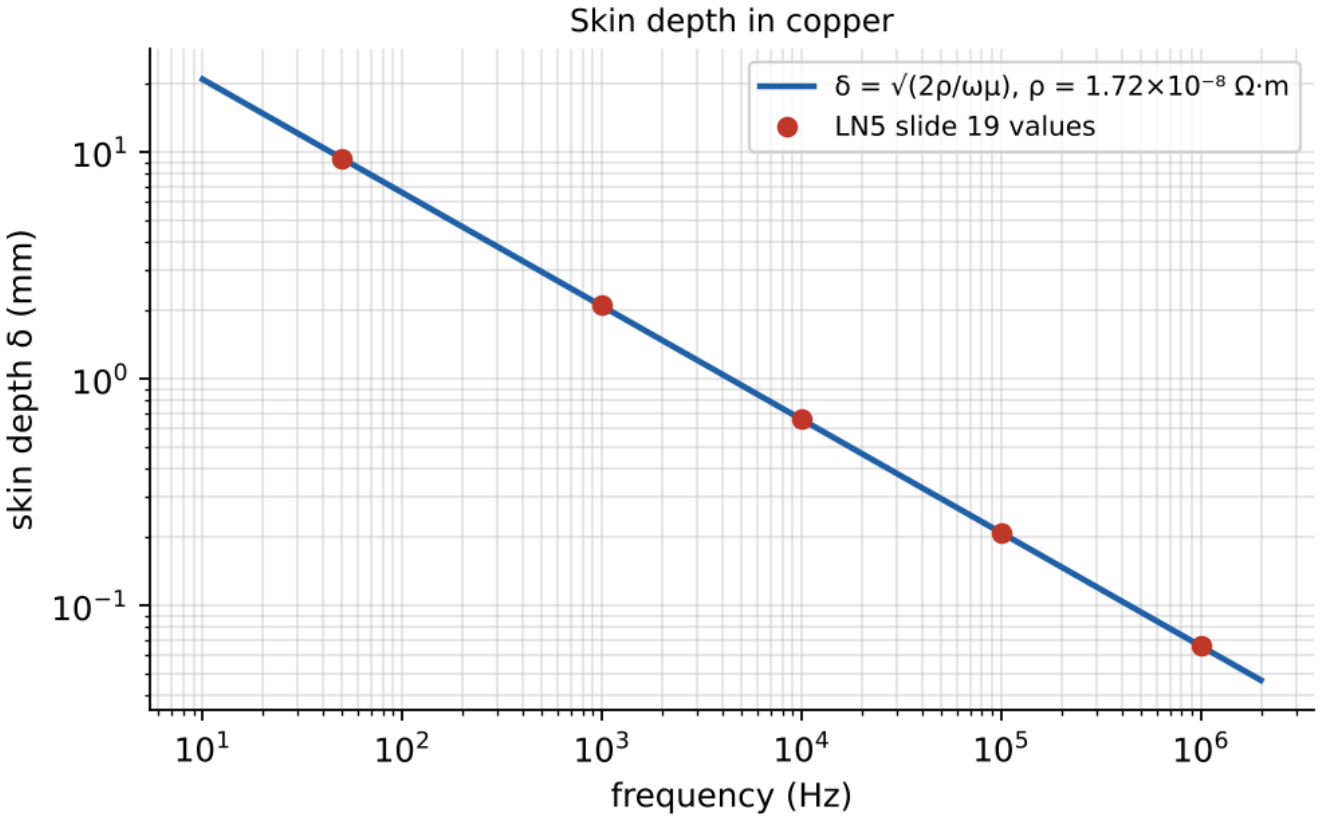
Designers raise frequency to shrink cores (fewer turns × area for the same voltage, Lesson 15) but must then use thinner laminations or ferrites to control the f^2 eddy loss — the trade-off examined in MT I Fall 2014 Q3 (50 Hz vs 500 Hz).

5. Skin effect and AC resistance (LN5 slides 10–23)

Eddy currents also arise inside a current-carrying conductor. The conductor's own alternating field induces circulating currents that oppose the current near the centre and reinforce it near the surface (LN5 slide 17). Current crowds into a surface layer of characteristic thickness, the **skin depth** (deri kalınlığı):

$$J(d) = J_S e^{-d/\delta}, \qquad \delta = \sqrt{\frac{2\rho}{\omega \mu_r \mu_0}} \tag{16.7}$$

(LN5 slide 18, where the symbol σ is used for δ — errata E3). For copper ($\rho = 1.72 \times 10^{-8}$ Ω·m): 9.3 mm at 50 Hz, 2.1 mm at 1 kHz, 0.66 mm at 10 kHz, 0.21 mm at 100 kHz, 66 µm at 1 MHz (LN5 slide 19; Figure 16.4).



Consequences:

- **AC resistance > DC resistance** (LN5 slide 15): less of the cross-section carries current. For a 30 mm diameter copper conductor at 50 Hz, the exact Bessel-function solution gives R_{ac}/R_{dc} = 1.13; for 20 mm, 1.03 — consistent with LN5's rule of thumb $R_{ac} \approx 1.1 R_{dc}$ for large power-system conductors (slide 23).
- **Remedies** (LN5 slides 20–22): bundled conductors on transmission lines (also reduce corona), foil conductors, hollow (water-cooled) conductors in induction heaters, and in electronics stranded insulated "Litz" wire.
- In **iron**, $\mu_r \sim 10^3$ makes δ small even at 50 Hz (≈1.6 mm for μ_r = 1000, ρ = 0.5 µΩ·m). Laminations must be thinner than δ for (16.3) to hold and for flux to penetrate the whole sheet — another reason for 0.3–0.5 mm sheets.

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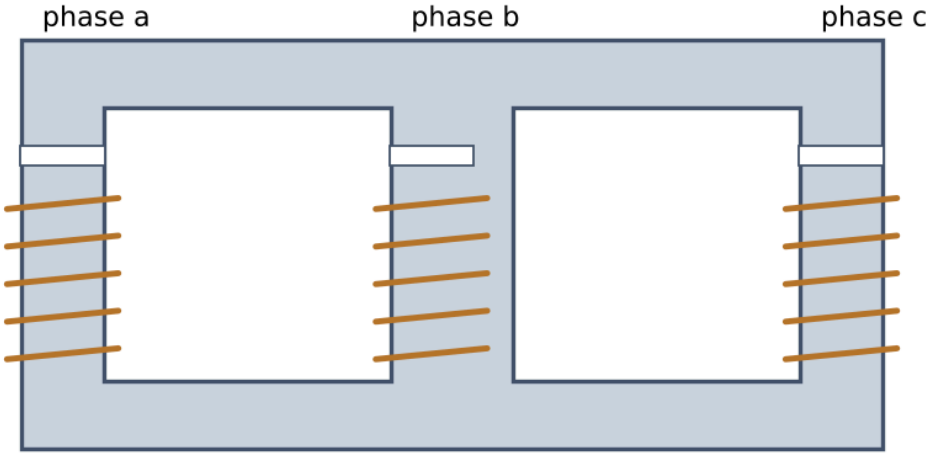
6. Single- and three-phase inductors

Single-phase inductors (tek fazlı endüktör) are coils on (usually gapped) cores: filter chokes, converter inductors (Lessons 28–29), current-limiting reactors. Their design combines everything in Chapter II: $L = N^2/\mathcal{R}$ with a gap for stability (Lesson 13), B_{max} below saturation at peak current, copper loss I^2R (with R_{ac} at high frequency) and core loss (this lesson). LN6 slides 2 and 7 show typical toroidal and bobbin inductors.

Three-phase inductors (üç fazlı endüktör) are either three single-phase units or one **three-limb core** with a winding on each limb (Figure 16.5). Because balanced phase fluxes sum to zero at every instant ($\Phi_a + \Phi_b + \Phi_c = 0$), each limb's flux returns through the other two limbs — no separate return limb is needed, saving iron (the same argument as for three-phase transformers, LN11 slide 8). Each phase behaves as an inductance L (reactance $X = \omega L$) in its own line. Uses:

- **series (line) reactors** limit fault current and smooth current drawn by drives and rectifiers;
- **shunt reactors** absorb reactive power on lightly loaded cables and lines — the inductive counterpart of the capacitor banks of Chapter I (the official Chapter 1 sample Q1 already needed reactors to correct leading loads).

Per-phase analysis uses the Chapter I per-phase-Y method: for a Y-connected shunt reactor $Q_{3\phi} = 3V_{ph}^2/X = V_{LL}^2/X$.



three-limb core, one winding per phase, gaps in each limb; balanced fluxes sum to zero at the yokes

Figure 16.5. Three-phase gapped inductor on a three-limb core (assistant schematic).

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7. Methods and common mistakes

- **Hysteresis loss from a loop:** area in J/m³ × core volume × frequency. Units: (A/m)(T) = (A/m)(V·s/m²) = J/m³.
- **Scaling problems:** first find how B_{max} changes (from V/f), then apply $P_h \propto fB^n$, $P_e \propto f^2B^2$.
- **Loss separation:** at constant B_{max}, $P_c/f = K_h' + K_e'f$ is a straight line in f.
- Do not apply the thin-lamination formula to thick solid iron (skin effect invalidates it).
- Core loss depends on voltage (flux), not on load current; copper loss depends on current.

- Skin depth uses resistivity and *relative permeability* of the conductor; copper and aluminium have $\mu_r = 1$.

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8. Worked examples

Example 16.1 — Hysteresis loss from a rectangular loop (original variant of Sen Problem 1.18)

Given. A toroid, mean length 15 cm, area 10 cm², 300 turns, fed from 100 V, 400 Hz. Its B–H loop is approximately rectangular with coercivity H_c = 50 A/m (value assumed here; Sen's figure value is not reproduced). **Find** B_{max} and the hysteresis loss.

Solution. $B_{max} = V/(4.44fNA) = 100/(4.44 \times 400 \times 300 \times 10^{-3}) = 0.188$ T. A rectangular loop of width $2H_c$ and height $2B_{max}$ has area $4H_cB_{max} = 4(50)(0.188) = 37.5$ J/m³. $P_h = V_{core} f \times$ area $= (0.15 \times 10^{-3})(400)(37.5) = 2.25$ W. **Check.** Numerical $\oint H\,dB$ around the rectangle equals $4H_cB_{max}$. **Interpretation.** With a rectangular loop the current is a square wave of amplitude $H_c l/N$ = 25 mA switching when the flux reverses — the "deltamax switch" behaviour mentioned by Sen (§1.2).

Example 16.2 — Same coil on 100 V/50 Hz and 110 V/60 Hz (Sen Problem 1.17)

Given. Compare hysteresis and eddy losses; take $n = 2$. **Solution.** $B_{max} \propto V/f$: ratio $(110/60)/(100/50) = 0.917$. Hysteresis $\propto fB_{max}^2$: $1.2 \times 0.917^2 = 1.008$ (+0.8 %). Eddy $\propto f^2B_{max}^2 = (V\text{-ratio})^2 = 1.1^2 = 1.21$ (+21 %). **Interpretation.** Raising V and f in the same proportion keeps the flux, but here V rose less than f, so the flux fell; hysteresis barely changed while eddy loss followed the voltage squared.

Example 16.3 — Separating hysteresis and eddy losses from a two-frequency test (original)

Given. At the same B_{max}, a core loses 100 W at 50 Hz and 130 W at 60 Hz. **Find** both components at 50 Hz, and the loss at 400 Hz with the same B_{max}. **Solution.** Write $P_c = af + bf^2$: $50a + 2500b = 100$, $60a + 3600b = 130 \Rightarrow a = 1.167$ W/Hz, $b = 0.01667$ W/Hz². At 50 Hz: $P_h = 58.3$ W, $P_e = 41.7$ W. At 400 Hz: $P_c = 467 + 2667 = 3133$ W — eddy loss dominates. **Interpretation.** At 400 Hz this steel would need much thinner laminations (eddy ∝ d²).

Example 16.4 — Eddy loss in a lamination and the case for laminating

Given. Silicon steel, $\rho = 0.5$ μΩ·m, density 7650 kg/m³, B_{max} = 1.5 T, 50 Hz, lamination 0.35 mm. **Find** eddy loss per m³ and per kg; compare a 10 mm solid block. **Solution.** (16.3): $p_e = \pi^2(50)^2(1.5)^2(0.35 \times 10^{-3})^2/(6 \times 0.5 \times 10^{-6}) = 2.27$ kW/m³ = 0.30 W/kg. Formally a 10 mm block would lose $(10/0.35)^2 = 816$ times more — but the formula is then invalid because δ in this steel at 50 Hz is only ≈1.6 mm (μ_r = 1000): flux and eddy currents crowd at the surface. The qualitative conclusion — solid iron cores are hopelessly lossy at AC — stands. **Check.** Numerical integration of J^2/σ over thickness and time reproduces (16.3) to 10^{-6}. **Interpretation.** 0.3 W/kg is the right order for "classical" eddy loss in 0.35 mm transformer steel; measured total losses (≈1 W/kg at 1.5 T) also include hysteresis and excess losses.

Example 16.5 — Skin effect in copper conductors (LN5 slides 18–23)

Given. Copper, $\rho = 1.72 \times 10^{-8}$ Ω·m. **Find** δ at 50 Hz, 1 kHz, 100 kHz; and R_{ac}/R_{dc} at 50 Hz for round conductors of 10, 20 and 30 mm diameter. **Solution.** δ = 9.33 mm, 2.09 mm, 0.209 mm. From the exact solution of the diffusion equation in a round wire, $Z/R_{dc} = \frac{1}{2}ka\,J_0(ka)/J_1(ka)$ with $k = (1-j)/\delta$: R_{ac}/R_{dc} = 1.002, 1.027, 1.125. **Interpretation.** Below a diameter of about 2δ the skin effect is negligible; large busbars and conductors at 50 Hz see the ~10 % increase quoted by LN5. In a 100 kHz converter, wire thicker than about 0.4 mm is wasteful — hence Litz wire and foil windings.

Example 16.6 — Three-phase reactors (original)

Given. (a) A series line reactor of 2 mH per phase carries 30 A rms on a 400 V, 50 Hz system. (b) A Y-connected shunt reactor of 100 mH per phase is connected to the same bus. **Find** (a) the reactance, voltage drop and reactive power; (b) the phase current and reactive power absorbed. **Solution.** (a) $X = 2\pi 50(0.002) = 0.628\ \Omega$; drop $= IX = 18.8$ V (8.2 % of the 231 V phase voltage); $Q = 3I^2X = 1.70$ kvar. (b) $X = 31.4\ \Omega$; $I = 231/31.4 = 7.35$ A; $Q = V_{LL}^2/X = 5.09$ kvar. **Interpretation.** Reactors are the inductive counterpart of the capacitor banks of Chapter I: a shunt reactor *absorbs* vars and raises (corrects) a leading power factor; a series reactor limits current at the cost of voltage drop.

Example 16.7 — Leakage and saturation effects on inductances (2025 MT1 Q4(e))

Question. Discuss how leakage flux and nonlinear core behaviour (near saturation) would affect the inductances of the two-coil E-core. **Answer.** *Leakage* (flux of one coil that does not reach the other) reduces the mutual inductance M and the coupling coefficient k; the self-inductance of each coil still includes its leakage flux (it links its own coil) — so L_1, L_2 are essentially unchanged while M falls. *Saturation* lowers the permeability of the iron, raising the iron reluctances; since the gap already dominates the centre branch but not the long outer paths (which carry 1.85 T in Example 12.8 — far beyond real ferrite), the self- and mutual inductances fall and become current-dependent (secant ≠ incremental, Lesson 11). The key's one-line answer ("leakage → lower mutual; non-linear → lower self-inductance") is correct; saturation lowers M too.

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9. Optional practice

P16.1. A transformer core loses 400 W at rated 50 Hz voltage, of which 60 % is hysteresis (n = 2). Estimate the core loss at 50 Hz and 110 % voltage.

▼ **Solution**
B_{max} ×1.1. Hysteresis: $240 \times 1.1^2 = 290$ W; eddy: $160 \times 1.1^2 = 194$ W; total 484 W (+21 %). With n = 2 both components scale as V^2 at fixed frequency.

P16.2. By what factor must the lamination thickness change to keep eddy loss unchanged when a design moves from 50 Hz to 400 Hz at the same B_{max}?

▼ **Solution**
$p_e \propto f^2 d^2$: keep fd constant, so d must be 8 times thinner (e.g. 0.35 mm → 0.044 mm, foil-like) — or switch to a ferrite or amorphous/nanocrystalline material.

P16.3. Find the skin depth in aluminium ($\rho = 2.82 \times 10^{-8}$ Ω·m) at 50 Hz and comment on why large aluminium conductors are stranded around a steel core.

▼ **Solution**
δ = √(2ρ/(ωμ0)) = √(2×2.82×10⁻⁸/(314.16×4π×10⁻⁷)) = 11.95 mm. Conductors of a few cm diameter would carry little current in their centre; ACSR conductors put the steel strength member in the centre, where it carries little current anyway, and the stranded aluminium outside.

P16.4 (concept, MT I Fall 2014 Q3(b)). A 50 Hz transformer is redesigned for 500 Hz with the same voltages, power, turns, core material, lamination and B_{max}. How do core area, maximum flux, hysteresis loss and eddy loss change?

▼ **Solution**
$V = 4.44 f N B_{max} A$ with V, N, B_{max} fixed ⇒ $A \propto 1/f$: area ÷10, so maximum flux $B_{max}A$ ÷10 and the core volume (hence weight) falls roughly ×10 if the path length is similar. Per unit volume, hysteresis loss ∝ f (×10) and eddy loss ∝ f² (×100) at the same lamination thickness; per transformer, hysteresis ≈ unchanged (×10 per volume, ÷10 volume) while eddy loss rises ≈ ×10 — so in practice thinner laminations or a different material are required.

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10. Summary

Relation	Validity
$P_h = V_{core} f \oint H\, dB \approx K_h f B_{max}^{n}$	Steady AC; Steinmetz $n \approx 1.5$–2.5 (empirical)
$p_e = \pi^2 f^2 B_{max}^2 d^2/(6\rho)$; $P_e = K_e f^2 B_{max}^2$	Thin laminations ($d \ll \delta$), sinusoidal B
$P_c = P_h + P_e$; $R_c = E^2/P_c$	Constant loss at fixed V, f
$i_\varphi = i_c + i_m$	Exciting current components
$\delta = \sqrt{2\rho/(\omega\mu_r\mu_0)}$	Good conductor, sinusoidal
$\Phi_a + \Phi_b + \Phi_c = 0$	Balanced three-limb core

11. Vocabulary

English	Türkçe
magnetic hysteresis; hysteresis loop	manyetik histerezis; histerezis döngüsü
remanence; coercivity	kalıcı mıknatıslanma; koersivite
hysteresis loss	histerezis kaybı
eddy current; eddy-current loss	girdap (Fuko) akımı; girdap akımı kaybı
core loss / iron loss	nüve kaybı / demir kaybı
core lamination; laminated core	nüve laminasyonu; lamine nüve
electrical steel	elektrik çeliği
skin effect; skin depth	deri etkisi; deri kalınlığı
AC resistance	AA direnci
single-phase / three-phase inductor	tek fazlı / üç fazlı endüktör
series reactor; shunt reactor	seri reaktör; şönt reaktör

12. Sources

- LN5 viewer pp. 2–37 (all slides): saturation, residual magnetism, hysteresis loops and loss $\propto$ f, Lenz, eddy currents, skin depth and copper values, bundled/foil/hollow conductors, $R_{ac} \approx 1.1 R_{dc}$, eddy formula, loss scaling, applications. Notes.
- LN8 viewer pp. 2–8, 27–32 (slides 2–5, 13–17). LN6 viewer pp. 3, 11, 19 (slides 2, 7, 13).
- Sen §1.2–1.2.3 pp. 15–21 (viewer 39–45), Eqs. 1.26–1.36, Figs. 1.12–1.16; §1.3.1 with hysteresis p. 24–25, Fig. 1.19; Problems 1.17–1.18 pp. 36–37 (viewer 60–61).
- 2025 MT1 Q4(e) p. 11 and key p. 2; MT I Fall 2014 Q3 pp. 5–7.
- Verification: `lesson16_core_losses.py` (loop integration, lamination integral check, Bessel-function skin effect).
- More solved archive problems: Ch II–III archive collection §1(f) (core loss from a hysteresis loop, MT I Fall 2015 Q2).